Interaction Field Theory

We will discuss: (a) ϕ^4 theory : $\mathcal{L}_{\phi^4} = \frac{1}{2}(\partial^\mu\phi)(\partial_\mu\phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 \quad \phi\in\mathbb{R}$.

Coupling constant

Interaction

(b) Yukawa theory: $\mathcal{L}_{Yukawa} = \mathcal{L}_S + \mathcal{L}_D + \mathcal{L}_Y$

with
$$\begin{cases} \mathcal{L}_S = \frac{1}{2}\partial^\mu\phi\,\partial_\mu\phi - \frac{1}{2}m_\phi^2\phi \\ \mathcal{L}_D = \bar{\psi}(i\not{\partial} - m_\psi)\psi \\ \mathcal{L}_Y = \cancel{-g\phi\bar{\psi}\psi} \\ \quad -g\phi\bar{\psi}\psi \end{cases}$$

← Interaction

Coupling constant.

(c) Scalar-QED: $\mathcal{L}_{S\text{-}QED} = \mathcal{L}_{CKG} + \mathcal{L}_{Max} + \mathcal{L}_{int}$

with
$$\begin{cases} \mathcal{L}_{CKG} = \frac{1}{2}\partial^\mu\phi\,\partial_\mu\phi^* - \frac{1}{2}m_\phi^2\phi\phi^* \\ \mathcal{L}_{Max} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} \\ \mathcal{L}_{int} = -ieA_\mu(\phi^*\partial^\mu\phi - (\partial^\mu\phi^*)\phi) + e^2|\phi|^2A_\mu A^\mu \end{cases}$$

(d) Spinor-QED: $\mathcal{L}_{QED} = \mathcal{L}_{Dirac} + \mathcal{L}_{Max} + \mathcal{L}_{int}$

Charge, take the role as coupling constant.

with
$$\begin{cases} \mathcal{L}_{Dirac} = \bar{\psi}(i\not{\partial} - m)\psi \\ \mathcal{L}_{Max} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} \\ \mathcal{L}_{int} = -e\bar{\psi}\gamma^\mu\psi A_\mu \end{cases}$$

* When there are interaction, Hamiltonian of each theory cannot be analytically diagonalized.
* If the interaction is weak enough, perturbation may be an effective way
* In real experiments, the details of scatter spectrum isn't be discussed, only several reaction with specific input and output particles has been mentioned!

§1 Interaction Picture and S-matrix

* We will work on interaction Picture. where $\hat{O}_I(t_f,t_i) = \cancel{e^{-i\hat{H}(t_f-t_i)}}\,\hat{T}\exp\left(-i\int_{t_i}^{t_f}\hat{H}_I(t)\,dt\right)$ and $\hat{H}_I$ can be obtain only substitute $\hat{\phi}(x)$ into $\hat{\phi}_I(x)$. $\hat{T}$ → Time order operator.
* Commutative relation and plane-wave expansion's form will not change in different picture.
* Denote the initial and final state as $|i\rangle$, $|f\rangle$, then $|\psi(t)\rangle = \lim_{t_0\to-\infty}\hat{U}(t,t_0)|i\rangle = \lim_{t_0\to+\infty}\hat{U}(t,t_0)|f\rangle$

denote $\langle f|\hat{S}|i\rangle = \lim_{t\to+\infty}\lim_{t_0\to-\infty}\langle f|\hat{U}(t,t_0)|i\rangle$ → In QFT, $|i\rangle = \hat{a}^\dagger_{\vec{p},\sigma}\hat{b}^\dagger_{\vec{p},s}\cdots|0\rangle$ particle state then, S-operator has a Dyson series: $\hat{S}=\hat{T}\exp\left\{-i\int_{-\infty}^{+\infty}dt\,\hat{H}_I(t)\right\}$

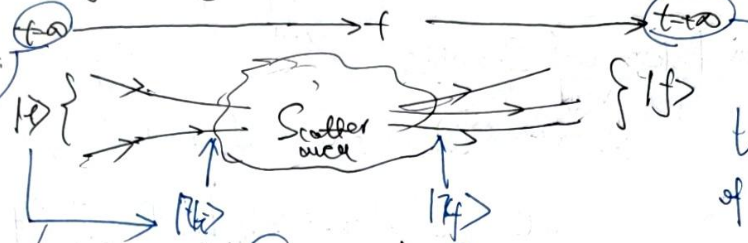
and it can be written as series: $\hat{S}=\sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\int \underbrace{d^4x_1\cdots d^4x_n}\,\hat{T}\{\mathcal{H}_{int}(x_1)\cdots\mathcal{H}_{int}(x_n)\}$.

QFT has been into the theory $\Rightarrow$ (Lorentz invariant)

$\boxed{*1}$ The series has the same order of coupling constant, rely on the times of $\mathcal{H}(x)$ appears.

For perturbation theory, usually take several order of the Dyson series

$\boxed{*2}$ Scatter process:



Without effect of interaction

Without effect of interaction

and there should be $\langle\psi_f|\psi_i\rangle=\langle f|\hat{S}|i\rangle$ $\to$ here $\hat{H}_0|i\rangle=E_i|i\rangle$

$\hat{H}_0|f\rangle=E_f|f\rangle$

$\hat{H}|\psi_i\rangle=E_i|\psi_i\rangle$

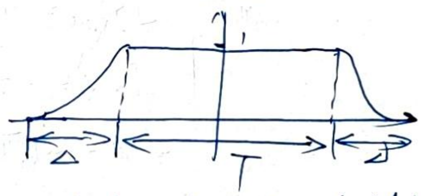
$\hat{H}|\psi_f\rangle=E_f|\psi_f\rangle$

From $|i\rangle$ to $|\psi_i\rangle$, the particle will be dressed!

The dressed process is corresponding to the electron self-energy correction in following section.

$\hat{V}_I\to f(t)\hat{V}_I$, with

A model to describe this process:



and the real situation should be $T\to\infty$, $\Delta\to\infty$, while $\frac{\Delta}{T}\to 0$

$\boxed{*3}$ Any symmetry will all be included in S-operator.

Ex: (Time-space translation): $S_{\beta\alpha}=\langle\psi_\beta|\psi_\alpha\rangle=\langle\psi_\beta|\hat{O}^\dagger(\mathrm{I},a)\,\hat{O}(\mathrm{I},a)|\psi_\alpha\rangle$

Since $\hat{O}(\mathrm{I},a)=e^{-i\hat{\vec{P}}\cdot\vec{a}}$, then: $S_{\beta\alpha}=\langle\psi_\beta|e^{+i\hat{\vec{P}}\cdot\vec{a}}\,e^{-i\hat{\vec{P}}\cdot\vec{a}}|\psi_\alpha\rangle$

$=e^{-i\vec{a}\cdot(\vec{P}_\alpha-\vec{P}_\beta)}\langle\psi_\beta|\psi_\alpha\rangle$

$=e^{-i\vec{a}\cdot(\vec{P}_\alpha-\vec{P}_\beta)}S_{\beta\alpha}$

Total 4-momentum of initial state

Total 4-momentum of final state

then: $S_{\beta\alpha}\supset\delta(\vec{P}_\alpha-\vec{P}_\beta)$

4-momentum is conservative.

we can write: $S_{\beta\alpha}=\underbrace{\delta_{\beta\alpha}}_{\text{0-order}}+(2\pi)^4\delta^{(4)}(\vec{P}_\alpha-\vec{P}_\beta)\,i\mathcal{M}$ $\to$ Main part of T-matrix

§2 Observables in interaction field theory I.

* For particle physics, it's impossible to gain information of <u>final state</u> directly in experiments.
* The computation of field theory should find some observables, which is also detectable in experiments.
* The most important observable: Cross section σ, which is directly ~~cor~~ relative to S-matrix

1. Transition probability

* ~~Recall~~ the structure of S-matrix: $\underline{S} = \underline{I} + i\underline{T}$ and $S_{\beta\alpha} = \delta_{\beta\alpha} + (2\pi)^4 \delta^{(4)}(\vec{P}_\alpha - \vec{P}_\beta)\, i\mathcal{M}$

$\mathcal{M}$ ↓ Invariant amplitude.

* What we have known: a) $|i\rangle$ and $|f\rangle$. (Particles with specific momentum $\vec{p}$ and inner freedom degree σ or λ).

b) The interaction theory $\mathcal{L}_{int}$.

determined → $\boxed{\underset{\sim}{\mathcal{M}}}$ (In perturbation theory, up to a demanding order).

We will discuss how to determine it in following section.

Define transition probability:

$$P_{f\to i} = \frac{|\langle f|i\hat{T}|i\rangle|^2}{\langle i|i\rangle\langle f|f\rangle} = \frac{|T_{fi}|^2}{\langle i|i\rangle\langle f|f\rangle}$$ and since $T_{fi} = (2\pi)^4 \delta^{(4)}(\vec{P}_i - \vec{P}_f)\, i\mathcal{M}$.

we have:
$$P_{f\to i} = \frac{(2\pi)^8 \delta^4(\vec{P}_i - \vec{P}_f)\,|\mathcal{M}|^2}{\langle i|i\rangle\langle f|f\rangle} = \frac{(2\pi)^4 \boxed{(2\pi)^4\delta^{(4)}(0)}\,\delta^{(4)}(\vec{P}_i - \vec{P}_f)\,|\mathcal{M}|^2}{\langle i|i\rangle\langle f|f\rangle}$$

$$= \check{V}\check{T}\cdot\frac{(2\pi)^4\delta^{(4)}(\vec{P}_i - \vec{P}_f)}{\langle i|i\rangle\langle f|f\rangle}|\mathcal{M}|^2$$

$$\delta^{(4)}(0) = \frac{1}{(2\pi)^4}\int d^4\vec{x}\, e^{i\vec{p}\cdot\vec{x}}\Big|_{\vec{p}=0} = \frac{1}{(2\pi)^4}\int d^4\vec{x} = \frac{1}{(2\pi)^4}\check{V}\cdot\check{T}$$

$$\boxed{\frac{P_{f\to i}}{\check{V}\check{T}}} = \frac{(2\pi)^4\delta^{(4)}(\vec{P}_i - \vec{P}_f)\,|\mathcal{M}|^2}{\langle i|i\rangle\langle f|f\rangle}$$

Suppose the whole space take the box-normalization with volume $\check{V}$, time window $\check{T}$.

⇓

The average of volume and time will become a ~~b~~ finite value

⇓

Divergence of $P_{f\to i}$ only come from the infiniteness of Space and Time!

* Now let's discuss ~~the~~ a more specific situation: $N_\alpha \to N_\beta$ process.
  - Initial state: $|i\rangle = \prod_{i=1}^{N_\alpha} \sqrt{2E_{p_i}}\,\hat{a}^\dagger_{p_i}|0\rangle$ (here ignore the inner freedom degree index).

    $E_{p_i} = (|\vec{p}_i|^2 + m^2)^{1/2}$

  - Final state: $|f\rangle = \prod_{j=1}^{N_\beta} \sqrt{2E_{p_j}}\,\hat{a}^\dagger_{p_j}|0\rangle$

and thus $\vec{P}_i = \sum_{i=1}^{N_\alpha} \vec{p}_i$ $\quad \vec{P}_f = \sum_{j=1}^{N_\beta} \vec{p}_j$

$$\frac{P(i\to f)}{V T} = \frac{(2\pi)^4 \delta^{(4)}(\vec{P}_i - \vec{P}_f)|M|^2}{\langle i|i\rangle\langle f|f\rangle}$$

$\circ$ Inner product of $\langle i|i\rangle$ and $\langle f|f\rangle$.

$$\langle i|i\rangle = \langle 0| \prod_{i=1}^{N_\alpha} \sqrt{2E_{p_i}}\, \hat{a}_{p_i} \cdot \prod_{j=1}^{N_\alpha} \sqrt{2E_{p_j}}\, \hat{a}^\dagger_{p_j} |0\rangle$$

If $i=1,2,\cdots,N_\alpha$ are all particles excited from different field, then

$$\langle i|i\rangle = \prod_{i=1}^{N_\alpha} \langle 0|\sqrt{2E_{p_i}}\,\hat{a}_{p_i} \sqrt{2E_{p_i}}\,\hat{a}^\dagger_{p_i}|0\rangle = \prod_{i=1}^{N_\alpha} 2E_{p_i}\cdot[\hat{a}_{p_i}, \hat{a}^\dagger_{p_i}]_\pm = \prod_{i=1}^{N_\alpha} 2E_{p_i}\cdot(2\pi)^3\delta^{(3)}(\vec{0})$$

$$= \prod_{i=1}^{N_\alpha} 2E_{p_i}\, V = V^{N_\alpha}\prod_{i=1}^{N_\alpha} 2E_{p_i}$$

If there are particles excited from same field, then there will be structure.

$$\langle i|i\rangle = \langle 0|\sqrt{2E_{p_1}2E_{p_2}}\,\hat{a}_{p_1}\hat{a}_{p_2}\cdot\sqrt{2E_{p_1}2E_{p_2}}\,\hat{a}^\dagger_{p_2}\hat{a}^\dagger_{p_1}|0\rangle \cdot \underbrace{\prod_{i=3}^{N_\alpha}\langle 0|\sqrt{2E_{p_i}}\,\hat{a}_{p_i}\sqrt{2E_{p_i}}\,\hat{a}^\dagger_{p_i}|0\rangle}_{=V^{N_\alpha-2}\prod_{i=3}^{N_\alpha}2E_{p_i}}$$

Take only $i=1,2$ excited from same field as example

$$= 2E_{p_1}2E_{p_2}\underbrace{\langle 0|\hat{a}_{p_1}\hat{a}_{p_2}\hat{a}^\dagger_{p_2}\hat{a}^\dagger_{p_1}|0\rangle}\cdot V^{N_\alpha-2}\prod_{i=3}^{N_\alpha}2E_{p_i}$$

|| Inner product of double particle state.

$$(2\pi)^6\{\underbrace{\delta(p_2-p_2)\delta(p_1-p_1)}_{V^2} + \underbrace{\delta(p_1-p_2)\delta(p_2-p_1)}_{=0}\}$$

$$= V^{N_\alpha}\prod_{i=1}^{N_\alpha}2E_{p_i}$$

$\Rightarrow$ If No same particles in initial state: $\langle i|i\rangle = V^{N_\alpha}\prod_{i=1}^{N_\alpha}2E_{p_i}$

If No ——— in final state: $\langle f|f\rangle = V^{N_\beta}\prod_{j=1}^{N_\beta}2E_{p_j}$

Thus:
$$\frac{P(i\to f)}{VT} = \frac{(2\pi)^4\delta^{(4)}(\vec{P}_i-\vec{P}_f)|M|^2}{V^{N_\alpha+N_\beta}\prod_{i=1,\,p_i\in|i\rangle}^{N_\alpha}2E_{p_i}\cdot\prod_{j=1,\,p_j\in|f\rangle}^{N_\beta}2E_{p_j}} = V^{-(N_\alpha+N_\beta)}\Big\{\prod_{p_i\in|i\rangle}2E_{p_i}\prod_{p_j\in|f\rangle}2E_{p_j}\Big\}^{-1}$$
$$\times(2\pi)^4\delta\Big(\sum_{i\in|i\rangle}\vec{p}_i-\sum_{j\in|f\rangle}\vec{p}_j\Big)|M|^2$$

thus we have: $R(i\to f) \equiv \dfrac{P(i\to f)}{T} = V^{1-N_\alpha-N_\beta}\Big\{\prod_{p_i\in|i\rangle}\prod_{p_j\in|f\rangle}2E_{p_i}2E_{p_j}\Big\}^{-1}\times(2\pi)^4\delta\Big(\sum_{p_i\in|i\rangle}\vec{p}_i-\sum_{p_j\in|f\rangle}\vec{p}_j\Big)|M|^2$

* Here we have the results of $N_\alpha\to N_\beta$, but with $p_1\cdots p_{N_\alpha}\to\boxed{p_1\cdots p_{N_\beta}}$ → A specific momentum which gives $|f\rangle$

However, in real experiments, we cannot control the momentum in final state.

thus we have to sum over the momentum of final state, which refers to N_β-body integral of phase space.

In volume $V=L^3$ box, $\vec{p}=\frac{2\pi}{L}(k_1\vec{e}_x+k_2\vec{e}_y+k_3\vec{e}_z)$ with $k_1,k_2,k_3\in\mathbb{Z}$, then.

$R(i\to f)$ is actually $dR(i\to f)$. where Page 4 $\vec{p}$ contains small area of momentum

$$\sum_{k_1,k_2,k_3} = \sum_{\vec{k}} \to \frac{V}{(2\pi)^3}\int d\vec{p}$$

thus $R(i\to\Sigma f) \equiv \prod_{j=1}^{N_f} \frac{V}{(2\pi)^3}\int d^3\vec{p}_j\ R(i\to f)$ → or $dR(i\to f)$

$$= V^{1-N_i}\left(\prod_{i=1}^{N_i} 2E_{p_i}\right)^{-1}\cdot \prod_{j=1}^{N_f}\int \frac{d^3\vec{p}_j}{(2\pi)^3 2E_j}(2\pi)^4\delta^{(4)}\left(\sum_{i=1}^{N_i}\vec{p}_i-\sum_{j=1}^{N_f}\vec{p}_j\right)\cdot |\mathcal{M}|^2$$

* It is possible that final states contains same particle! Thus we should modify the formula

$$R(i\to\Sigma f) = \frac{1}{S}V^{1-N_i}\left(\prod_{i=1}^{N_i}2E_{p_i}\right)^{-1}\underbrace{\prod_{j=1}^{N_f}\int\frac{d^3\vec{p}_j}{(2\pi)^3 2E_j}(2\pi)^4\delta^{(4)}\left(\sum_{i=1}^{N_i}\vec{p}_i-\sum_{j=1}^{N_f}\vec{p}_j\right)}_{\int d\Pi_{N_f}}\cdot|\mathcal{M}_{fi}|^2$$

$$= \frac{1}{S}V^{1-N_i}\left(\prod_{i=1}^{N_i}2E_{p_i}\right)^{-1}\cdot\int \boxed{d\Pi_{N_f}}\cdot|\mathcal{M}_{fi}|^2 \qquad d\Pi_{N_f} = \prod_{j=1}^{N_f}\frac{d^3\vec{p}_j}{(2\pi)^3 2E_j}(2\pi)^4\delta^{(4)}(\cdots)$$

N_f-body phase space integrals.

Lorentz-invariant!

2. $N_i=1$ process: Decay width.

* $N_i=1$ process just like: $n\to p+e^-+\bar{\nu}_e$ (β^- decay). There may be many decay branch, each branch will have corresponding decay width, which is partial decay width

* $dR(i\to f) = \frac{1}{S}\cdot\frac{1}{2E_{p_i}}\cdot d\Pi_{N_f}\cdot|\mathcal{M}_{fi}|^2$

$2E_{p_i}$ → Energy of initial particle

$dR(i\to f)$: partial decay width for the process $i\to f$, and there will be $R(i\to f)\equiv\Gamma_f = \frac{1}{S}\cdot\frac{1}{2E_{p_i}}\int|\mathcal{M}_{fi}|^2 d\Pi_{N_f}$.

Not a Lorentz invariant! —lead to→ Clock-slower effect

In general, choose the reference frame to let $\vec{p}_i=0$. then $\Gamma_f = \frac{1}{S\cdot 2m}\int|\mathcal{M}_{fi}|^2 d\Pi_{N_f}$.

* In experiment, we have found: $\frac{dn(t)}{dt} = -n(t)\Gamma \Rightarrow n(t)=n(0)e^{-\Gamma t}$.

define the life of one particle as $\tau = \frac{1}{n(0)}\underbrace{\int_0^\infty \frac{t}{|dn/dt|}dt}$ $= \frac{1}{\Gamma}$

$\Downarrow$

$$\int_0^\infty t\cdot\frac{n(0)}{|dn/dt|}dt \propto \int_0^\infty t e^{-t/\tau}dt \propto \langle t\rangle$$

$\langle t\rangle$. Average life-time of particles.

* Decay condition?

In center of mass system, since energy conservation, there should be:

$$m = \sum_{j=1}^{N_f}E_{p_j} = \sum_{j=1}^{N_f}\sqrt{|\vec{p}_j|^2+m_j^2} \geq \sum_{j=1}^{N_f}m_j$$

m: Mass of particle before decay. $\sum m_j$: Sum over the mass of particles after decay.

thus if a decay branch $i\to f$ can happen, there should be: $m\boxed{>}\sum_{j=1}^{N_f}m_j$

Cannot take equal! else the $d\Pi_f=0$.

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Homework 1 Consider the interaction between two real scalar field ϕ and Φ, with Lagrangian:

$$\mathcal{L} = \mathcal{L}_{KG}(\phi) + \mathcal{L}_{KG}(\Phi) - \mu\Phi\phi\phi$$

The interaction term allows a decay process $\Phi \to \phi\phi$, when $M > 2m$ (M: mass of Φ; m: mass of ϕ). Suppose this condition is met, calculate the life-time of Φ in this branch up to the lowest order of μ.

Solution: According to the discussion former, there will be $\tau = \frac{1}{\Gamma}$, while.

$$\Gamma = \frac{1}{2!}\cdot\frac{1}{2M}\int |\mathcal{M}_{fi}|^2 d\Pi_2.$$

To calculate $\mathcal{M}_{fi}$, let's consider S-matrix up to $O(\mu)$, then

$$iT_{fi} = \langle \vec{p}_1\vec{p}_2 | -i\int d^4x \cdot (-\mu\Phi(x)\phi(x)\phi(x)) \,|\vec{p}=0\rangle$$

$$= i\mu\int d^4x \cdot \langle \vec{p}_1\vec{p}_2|\phi(x)\phi(x)\Phi(x)|\vec{p}=0\rangle$$

$$= 2i\mu\int d^4x \,\langle \vec{p}_1\vec{p}_2|\phi(x)\phi(x)\Phi(x)|\vec{p}=0\rangle$$

$$= 2i\mu\int d^4x \cdot e^{-i(\vec{p}_1+\vec{p}_2)\cdot\vec{x}}\frac{1}{\sqrt{2E_{p_1}\cdot 2E_{p_2}\cdot 2M}}\cdot \underbrace{\langle \vec{p}_1|\hat{a}^\dagger_{p_1}}_{(2\pi)\sqrt{2E_{p_1}}\langle 0|}\underbrace{\langle \vec{p}_2|\hat{a}^\dagger_{p_2}}_{\sqrt{2E_{p_2}}\langle 0|}\cdot\underbrace{\hat{a}_{p=0}|\vec{p}=0\rangle}_{\sqrt{2M}}$$

$$= 2i\mu\cdot\int d^4x\cdot e^{-i(\vec{p}_1+\vec{p}_2)\cdot\vec{x}} = (2\pi)^4\delta^{(4)}(\vec{p}_1+\vec{p}_2-\vec{p})\cdot 2i\mu$$

thus $i\mathcal{M}_{fi} = 2i\mu$, and $|\mathcal{M}_{fi}|^2 = 4\mu^2$, then:

$$\Gamma = \frac{1}{4M}\cdot 4\mu^2\int d\Pi_2 = \frac{\mu^2}{M}\int\frac{d\vec{p}_1\, d\vec{p}_2}{(2\pi)^6\, 2E_{p_1}2E_{p_2}}\cdot(2\pi)^4\delta^{(4)}(\vec{p}_1+\vec{p}_2-\vec{p})$$

$$= \frac{\mu^2}{M}\int\frac{d\vec{p}_1\, d\vec{p}_2}{(2\pi)^2\, 2E_{p_1}2E_{p_2}}\cdot\delta(\vec{p}_1+\vec{p}_2)\,\delta(E_{p_1}+E_{p_2}-M)$$

$$= \frac{\mu^2}{4\pi^2 M}\int\frac{d\vec{p}}{4E_p^2}\cdot\delta(2E_p-M) = \frac{\mu^2}{4\pi^2 M}\cdot\frac{1}{4}\cdot 4\pi\cdot\int_0^\infty\frac{p^2dp}{E_p^2}\delta(2E_p-M)$$

$$= \frac{\mu^2}{4\pi M}\int_0^\infty\frac{p^2}{E_p^2}\delta(2E_p-M)\,dp.$$

Since $E_p^2 = p^2+m^2$, then $E_p dE_p = p\,dp$. we have.

$$\Gamma = \frac{\mu^2}{4\pi M}\int_m^\infty\frac{p}{E_p}\delta(2E_p-M)\,dE_p = \frac{\mu^2}{4\pi M}\int_m^\infty\left(1-\frac{m^2}{E_p^2}\right)^{1/2}\cdot\frac{1}{2}\delta\!\left(E_p-\frac{M}{2}\right)dE_p$$

Rely on $M>2m$ $\rightarrow$ $= \frac{\mu^2}{8\pi M}\left(1-\frac{4m^2}{M^2}\right)^{1/2}$

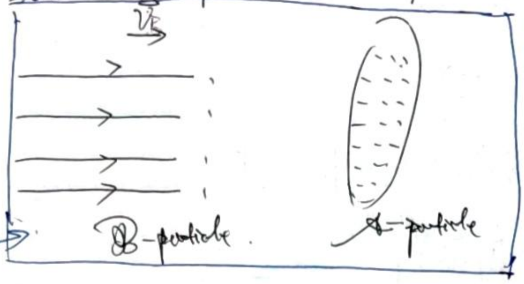
other-wise $E_p > m > \frac{M}{2}$, cannot be integral.

thus we have $\tau = \frac{1}{\Gamma} = \frac{8\pi M}{\mu^2}\left(1-\frac{4m^2}{M^2}\right)^{-1/2}$. $\square$

Part 6

3. $N_i=2$ process — Cross section

* To discuss a $2\to2$ process, we should first discuss the so-called beam-target engine:



Volume V

$\boxed{\text{Take the A-static reference frame}}$, then velocity of B particles are same as $\vec{v}_B$

- Flux density of B: $j_B = \dfrac{n_B A|\vec{v}_B|t}{At} = n_B|\vec{v}_B|$

* Suppose there are N_A particles on the target, then times of scattering $N = N_A j_B \underbrace{\sigma}_{\text{Cross section}} t$.

$N \propto \sigma$: Given j_B and N_A, σ will reflect the strength of scattering.

N per time per volume: $R = \dfrac{N}{Vt} = \dfrac{N_A}{V} = n_A n_B \sigma |\vec{v}_B|$. V should be regarded as the whole space.

* Now suppose only one B and A particle, then $n_A = n_B = \dfrac{1}{V}$, thus $R = \dfrac{1}{V^2}\sigma|\vec{v}_B|$

then scatter-times per second was $\boxed{\mathcal{R}} = RV = \dfrac{1}{V}\sigma|\vec{v}_B|$.

One scatter $\Rightarrow$ one transition, thus $\mathcal{R}$ is transition rate too!

$$\Rightarrow \sigma = \frac{V\mathcal{R}}{|\vec{v}_B|} = \frac{V}{|\vec{v}_B|}\mathcal{R}(i\to f) = \frac{V}{|\vec{v}_B|}\cdot\frac{1}{S}V^{1-N_i}\left(\prod_{i=1}^{N_i}2E_{p_i}\right)^{-1}\int d\Pi_{N_f}\,|\mathcal{M}_{fi}|^2\Bigg|_{N_i=N_f=2}$$

$$=\frac{1}{S}\cdot\frac{1}{2E_{p_1}\cdot 2E_{p_2}\cdot|\vec{v}_B|}\int d\Pi_2\cdot|\mathcal{M}_{fi}|^2$$

and also the differential version: $d\sigma = \dfrac{1}{4E_{p_1}E_{p_2}|\vec{v}_B|}|\mathcal{M}_{fi}|^2 d\Pi_2$ (For specific output momentum).

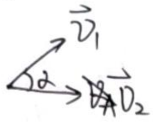
* Now we should modify this formula and generalize it to any reference frame ↳ σ should be Lorentz invariant.

Introduce so-called Møller velocity: $v_{rel} \equiv \dfrac{1}{E_{p_1}E_{p_2}}\sqrt{(\vec{p}_1\cdot\vec{p}_2)^2 - \underbrace{m_1^2 m_2^2}_{\text{Mass of input two particles}}}$

then $d\sigma = \dfrac{1}{4\underbrace{E_{p_1}E_{p_2}v_{rel}}_{=\sqrt{(\vec{p}_1\cdot\vec{p}_2)^2 - m_1^2m_2^2}}}|\mathcal{M}_{fi}|^2 d\Pi_2$ → a Lorentz invariant value.

$$= \frac{1}{4\sqrt{(\vec{p}_1\cdot\vec{p}_2)^2 - m_1^2m_2^2}}|\mathcal{M}_{fi}|^2 d\Pi_2$$

- How to understand moller velocity? Suppose in one reference frame, we have

 and $p_1^a = E_{p_1}(1, \vec{v}_1)$. $p_2^a = E_{p_2}(1, \vec{v}_2)$.

then. $\vec{p}_1 \cdot \vec{p}_2 = E_{p_1}E_{p_2} - \vec{p}_1 \cdot \vec{p}_2 = E_{p_1}E_{p_2}(1 - \vec{v}_1 \cdot \vec{v}_2)$

Since $(\vec{v}_1 \cdot \vec{v}_2)^2 = v_1^2 v_2^2 \cos^2\alpha = v_1^2 v_2^2 (1 - \sin^2\alpha) = v_1^2 v_2^2 - |\vec{v}_1 \times \vec{v}_2|^2$.

thus:

$$(\vec{p}_1 \cdot \vec{p}_2)^2 = E_{p_1}^2 E_{p_2}^2 \left\{ 1 - 2\vec{v}_1 \cdot \vec{v}_2 + (\vec{v}_1 \cdot \vec{v}_2)^2 \right\}$$

$$= E_{p_1}^2 E_{p_2}^2 \left\{ 1 \underbrace{- 2\vec{v}_1 \cdot \vec{v}_2}_{|\vec{v}_1 - \vec{v}_2|^2 - v_1^2 - v_2^2} + v_1^2 v_2^2 - |\vec{v}_1 \times \vec{v}_2|^2 \right\}$$

$$= E_{p_1}^2 E_{p_2}^2 \left\{ |\vec{v}_1 - \vec{v}_2|^2 - |\vec{v}_1 \times \vec{v}_2|^2 + (1 - v_1^2)(1 - v_2^2) \right\}$$

$$= E_{p_1}^2 E_{p_2}^2 \left(|\vec{v}_1 - \vec{v}_2|^2 - |\vec{v}_1 \times \vec{v}_2|^2 \right) + \underbrace{(E_{p_1}^2 - \vec{p}_1^2)}_{m_1^2} \underbrace{(E_{p_2}^2 - \vec{p}_2^2)}_{m_2^2}$$

which means:

$$\sqrt{(\vec{p}_1 \cdot \vec{p}_2)^2 - m_1^2 m_2^2} = E_{p_1}E_{p_2} \sqrt{|\vec{v}_1 - \vec{v}_2|^2 - |\vec{v}_1 \times \vec{v}_2|^2} \Rightarrow v_{rel} = \sqrt{\underbrace{|\vec{v}_1 - \vec{v}_2|^2}_{= |\vec{v}_{rel}|^2} - |\vec{v}_1 \times \vec{v}_2|^2}$$

Reference velocity in Newton mechanism

* Consider a special reference frame — Center of mass reference frame.

- Characteristic: ① $\vec{p}_1 + \vec{p}_2 = 0 \Rightarrow \vec{v}_1 + \vec{v}_2 = 0$, or $\vec{v}_1 \times \vec{v}_2 = 0 \rightarrow v_{rel} = |\vec{v}_1 - \vec{v}_2|$.

② Can define: $E_{cm} = E_1 + E_2 = E_1' + E_2'$

and at this time: $(\vec{p}_1 + \vec{p}_2)^2 = (E_{p_1} + E_{p_2})^2 - \underbrace{(\vec{p}_1 - \vec{p}_2)^2}_{=0} = E_{cm}^2$

$d\sigma = \frac{1}{4E_{p_1}E_{p_2} v_{rel}} |\mathcal{M}_{fi}|^2 d\Pi_2$

∴ Interestingly: $E_{cm} = (\vec{p}_1 + \vec{p}_2)^2$. while $(\vec{p}_1 + \vec{p}_2)^2$ is Lorentz invariant

E_{cm} is Lorentz invariant!

⇓

In any reference frame, we can use $E_{cm} = (\vec{p}_1 + \vec{p}_2)^2$ to describe the cross section.

- Set the scattering procs.  with $\vec{p}_A + \vec{p}_B = \vec{p}_1 + \vec{p}_2 = 0$.

$$d\Pi_2 = \frac{d^3p_1 d^3p_2}{(2\pi)^6 4E_{p_A}E_{p_B}} (2\pi)^4 \underbrace{\delta^{(4)}(\vec{p}_A + \vec{p}_B - \vec{p}_1 - \vec{p}_2)}_{\shortparallel} = \frac{d^3\vec{p}_1}{(2\pi)^2 4E_{p_1}E_{p_2}} \delta\left(E_{cm} - \sqrt{p_1^2 + m_1^2} - \sqrt{p_1^2 + m_2^2}\right)$$

$\delta(E_A + E_B - E_{p_1} - E_{p_2}) \delta^{(3)}(\vec{p}_A + \vec{p}_B - \vec{p}_1 - \vec{p}_2)$

$= \delta(E_{cm} - E_{p_1} - E_{p_2}) \underbrace{\delta^{(3)}(\vec{p}_A + \vec{p}_B - \vec{p}_1 - \vec{p}_2)}_{\delta(\vec{p}_1 - \vec{p}_2)}$

$\shortparallel$

$\frac{d^3\vec{p}}{16\pi^2 E_{p_1}^2} \delta\left(E_{cm} - \sqrt{p_1^2 + m_1^2} - \sqrt{p_1^2 + m_2^2}\right)$

then $d\Pi_2 = \frac{d\Omega \cdot p^2 dp}{16\pi^2 E_{p_1} E_{p_2}} \delta\left(E_{cm} - \sqrt{p^2+m_1^2} - \sqrt{p^2+m_2^2}\right)$

$$\boxed{\delta(f(x)) = \sum_i \frac{\delta(x-x_i)}{|f'(x_i)|} \quad \text{where } f(x_i)=0}$$

Suppose k will satisfy $E_{cm} - \sqrt{k^2+m_1^2} - \sqrt{k^2+m_2^2} = 0$, and $k>0$, then we will have:

$$d\Pi_2 = \frac{d\Omega \cdot k^2}{16\pi^2 E_k^2} \left| \frac{d}{dp}\left(E_{cm} - \sqrt{p^2+m_1^2} - \sqrt{p^2+m_2^2}\right) \right|^{-1}_{p=k}$$

$$= \frac{k^2 d\Omega}{16\pi^2 E_k^2} \cdot \left| \frac{k}{-\sqrt{k^2+m_1^2}} + \frac{k}{-\sqrt{k^2+m_2^2}} \right|^{-1} = \frac{k^2 d\Omega}{16\pi^2 E_k^2} \left| \frac{k}{E_{k_1}} + \frac{k}{E_{k_2}} \right|^{-1}$$

$$= \frac{k^2 d\Omega}{16\pi^2 E_k^2} \cdot \frac{E_{k_1} E_{k_2}}{k \cdot (E_{k_1}+E_{k_2})} = \frac{k\, d\Omega}{16\pi^2 E_{k_1} E_{k_2}} \cdot \frac{E_{k_1}E_{k_2}}{E_{k_1}+E_{k_2}} = \frac{k\, d\Omega}{16\pi^2 (E_{k_1}+E_{k_2})}$$

$$= \frac{k\, d\Omega}{16\pi^2 E_{cm}}$$

$E_{k_1}+E_{k_2} = E_{cm}$ < since $\vec{p}_1+\vec{p}_2=0$, while $p_1=p_2=k$

$$\therefore d\sigma = \frac{1}{4E_{p_A}E_{p_B}\underbrace{|\vec{v}_1-\vec{v}_2|}} \cdot |\mathcal{M}_{fi}|^2 \cdot \frac{k\, d\Omega}{16\pi^2 E_{cm}} = \frac{1}{64\pi^2} \frac{1}{E_{p_A}E_{p_B}|\vec{v}_A-\vec{v}_B|} \cdot \frac{k}{E_{cm}} |\mathcal{M}_{fi}|^2 d\Omega.$$

V_{rel} in CM-reference frame

Notice that $\vec{v}_A - \vec{v}_B = \frac{\vec{p}_A}{\sqrt{p_A^2+m_A^2}} - \frac{\vec{p}_B}{\sqrt{p_B^2+m_B^2}} = \frac{\vec{p}_A}{E_{p_A}} - \frac{\vec{p}_B}{E_{p_B}}$

□ If the particles input has the same mass, which means $m_A=m_B$, then

$$|\vec{v}_A - \vec{v}_B| = \frac{|\vec{p}_A - \vec{p}_B|}{E_{p_A}} = \frac{2|\vec{p}_A|}{E_{p_A}} \qquad E_{p_A} = \sqrt{p_A^2+m_A^2} = \sqrt{p_B^2+m_B^2} = E_{p_B}$$

$$\therefore \frac{1}{E_{p_A}E_{p_B}|\vec{v}_A-\vec{v}_B|} = \frac{E_{p_A}}{E_{p_A}E_{p_B}\cdot 2|p_A|} = \frac{E_{cm}/2}{\frac{E_{cm}^2}{4}\cdot 2|p_A|} = \frac{1}{E_{cm}|p_A|}$$

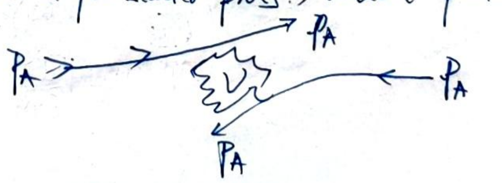
$$\therefore d\sigma\Big|_{m_A=m_B} = \frac{1}{64\pi^2} \frac{1}{E_{cm}|p_A|} \frac{k}{E_{cm}} |\mathcal{M}_{fi}|^2 d\Omega = \frac{1}{64\pi^2 E_{cm}^2} \frac{|p_1|}{|p_A|} |\mathcal{M}_{fi}|^2 d\Omega$$

□ If $m_A=m_B=m_1=m_2$ (Particles in the process have all the same mass).

$$E_{cm} \overset{!}{=} 2\sqrt{k^2+m^2} = 2\sqrt{p_A^2+m^2} \Longrightarrow k = p_A.$$

$$\therefore d\sigma\Big|_{\boxed{m_A=m_B=m_1=m_2}} = \frac{1}{64\pi^2 E_{cm}^2} |\mathcal{M}_{fi}|^2 d\Omega.$$

Usually describe a pair scatter process, without particle change.



§3 Wick Theorem

* We have discussed the critical observables in interaction theory, which is transition rate and cross section.
* If we can get the information of $\langle f|\hat{S}|i\rangle$, we can get the invariant amplitude $i\mathcal{M}$.

and finally get the cross section $d\sigma = \frac{1}{4E_{pA}E_{pB}V_{\text{relative}}}|\mathcal{M}_{fi}|^2 d\Pi_2$.

* The only mission is how to calculate $i\mathcal{M}$, or how to compute $\langle f|\hat{S}|i\rangle$ up to a specific perturbation level. Since

$$\langle f|\hat{S}|i\rangle = \hat{T}\exp i\left\{-\int_{-\infty}^{+\infty} d^4\vec{x}\, \langle f|\hat{\mathcal{H}}_{int}(\vec{x})|i\rangle\right.$$

$$= \delta_{fi} - i\int_{-\infty}^{+\infty} d^4\vec{x}\, \langle f|\hat{\mathcal{H}}_{int}(\vec{x})|i\rangle + \frac{(-i)^2}{2!}\int_{-\infty}^{+\infty} d^4\vec{x}\, d^4\vec{y}\, \langle f|\hat{T}\,\hat{\mathcal{H}}_{int}(\vec{x})\hat{\mathcal{H}}_{int}(\vec{y})|i\rangle$$

while $\mathcal{H}_{int}(\vec{x})$ always take the structure of $\hat{\Phi}_1(\vec{x})\hat{\Phi}_2(\vec{x})\cdots\hat{\Phi}_n(\vec{x})$, thus we need to deal with

$$\langle f|\hat{T}\,\hat{\Phi}_1(\vec{x})\cdots\hat{\Phi}_n(\vec{x})\hat{\Phi}_1(\vec{y})\cdots\hat{\Phi}_m(\vec{y})\cdots|i\rangle$$

* Since field operators will be expanded as particle operators, thus we can firstly discuss particle operator series!

1. Normal order and decomposition of field operator as positive/negative part.

* Define the so-called normal order of particle operators, e.g.

$$\mathcal{N}(\hat{a}_p\hat{a}_k^\dagger\hat{a}_q\hat{a}_l^\dagger) = \underbrace{a_k^\dagger a_l^\dagger}\, \underbrace{a_p a_q}\cdot\epsilon$$

ϵ is a sign-factor, which will appear for fermion operators.

Produce in front of the annihilation

At this Order, we can easily found: $\langle 0|\mathcal{N}(\cdots)|0\rangle = 0$!

produce Annihilation $\rightarrow$ One operator cannot annihilate the vacuum!

* Decomposition of field operators: Any field operators must have positive and negative energy part.

Guaranteed by casuality!

e.g. KG field: $\hat{\phi}(x) = \hat{\phi}^{(+)}(x) + \hat{\phi}^{(-)}(x)$

$$\begin{cases} \hat{\phi}^{(+)}(x) = \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\hat{a}_p e^{-i\vec{p}\cdot\vec{x}} \\ \hat{\phi}^{(-)}(x) = \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\boxed{\hat{b}_p^\dagger} e^{i\vec{p}\cdot\vec{x}} \end{cases}$$

Positive part only contains annihilation operator!

Dirac field: $\hat{\psi}(x) = \hat{\psi}^{(+)}(x) + \hat{\psi}^{(-)}(x)$

$$\begin{cases} \hat{\psi}^{(+)}(x) = \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_\sigma \hat{a}_{p,\sigma}u(\vec{p},\sigma) e^{-i\vec{p}\cdot\vec{x}} \\ \hat{\psi}^{(-)}(x) = \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_\sigma \boxed{\hat{b}_{p,\sigma}^\dagger}v(\vec{p},\sigma) e^{i\vec{p}\cdot\vec{x}} \end{cases}$$

Negative part only contains production operator!

Thus Normal operator can also be used in field operators. such as.

$$\mathcal{N}\{\Phi_a(x)\Phi_b(y)\} = \mathcal{N}\left\{\left(\Phi_a^{(+)}(x)+\Phi_a^{(-)}(x)\right)\left(\Phi_b^{(+)}(y)+\Phi_b^{(-)}(y)\right)\right\}$$

$$= \mathcal{N}\left(\Phi_a^{(+)}(x)\Phi_b^{(+)}(y)\right)+\mathcal{N}\left(\Phi_a^{(+)}(x)\Phi_b^{(-)}(y)\right)+\mathcal{N}\left(\Phi_a^{(-)}(x)\Phi_b^{(+)}(y)\right)+\mathcal{N}\left(\Phi_a^{(-)}(x)\Phi_b^{(-)}(y)\right)$$

$$= \Phi_a^{(+)}(x)\Phi_b^{(+)}(y) + \underbrace{\boxed{\epsilon_{ab}}}_{\text{potential sign factor}}\Phi_b^{(-)}(y)\Phi_a^{(+)}(x) + \Phi_a^{(-)}(x)\Phi_b^{(+)}(y) + \Phi_a^{(-)}(x)\Phi_b^{(-)}(y)$$

if Φ_b, Φ_a are all fermionic operator!

one can find that: $\mathcal{N}\{\Phi_a(x)\Phi_b(y)\} = \underline{\epsilon_{ab}}\,\mathcal{N}\{\Phi_b(y)\Phi_a(x)\}$

↳ Commutative relation won't change under normalized operation

* What about $\hat{T}\{\Phi_a(x)\Phi_b(y)\}$?

$$\hat{T}\{\hat{\Phi}_a(x)\hat{\Phi}_b(y)\} = \theta(x^0-y^0)\hat{\Phi}_a(x)\hat{\Phi}_b(y) + \theta(y^0-x^0)\epsilon_{ab}\hat{\Phi}_b(y)\hat{\Phi}_a(x)$$

① For $x^0-y^0>0$. Notice that :

$$\hat{\Phi}_a(x)\hat{\Phi}_b(y) = \mathcal{N}\left(\hat{\Phi}_a(x)\hat{\Phi}_b(y)\right) - \epsilon_{ab}\Phi_b^{(-)}(y)\Phi_a^{(+)}(x) + \Phi_a^{(+)}(x)\Phi_b^{(-)}(y)$$

$$= \mathcal{N}\left(\hat{\Phi}_a(x)\hat{\Phi}_b(y)\right) + \left[\hat{\Phi}_a^{(+)}(x), \hat{\Phi}_b^{(-)}(y)\right]_{\epsilon_{ab}}$$

→ if $\epsilon_{ab}=+1$. take "−"
$\epsilon_{ab}=-1$ take "+"

② For $y^0-x^0>0$. there will be.

$$\epsilon_{ab}\hat{\Phi}_b(y)\hat{\Phi}_a(x) = \epsilon_{ab}\mathcal{N}\left(\hat{\Phi}_b(y)\hat{\Phi}_a(x)\right) + \epsilon_{ab}\left[\hat{\Phi}_b^{(-)}(y), \hat{\Phi}_a^{(+)}(x)\right]_{\epsilon_{ab}} = \epsilon_{ab}\mathcal{N}\left(\hat{\Phi}_a(x)\hat{\Phi}_b(y)\right) + \left[\hat{\Phi}_b^{(-)}(y), \hat{\Phi}_a^{(+)}(x)\right]_{\epsilon_{ab}}$$

Since when $x^0>y^0$, $\left[\hat{\Phi}_a^{(+)}(x), \hat{\Phi}_b^{(-)}(y)\right]_{\epsilon_{ab}} = \langle 0|\hat{\Phi}_a^{(+)}(x)\hat{\Phi}_b^{(-)}(y)|0\rangle =$

when $y^0<x^0$ $\epsilon_{ab}\left[\hat{\Phi}_b^{(-)}(y), \hat{\Phi}_a^{(+)}(x)\right]_{\epsilon_{ab}} = \underset{\epsilon_{ab}}{\langle}0|\hat{\Phi}_b^{(-)}(y)\hat{\Phi}_a^{(+)}(x)|0\rangle$

thus. Finally: $\hat{T}\{\hat{\Phi}_a(x)\hat{\Phi}_b(y)\} = \mathcal{N}\left(\hat{\Phi}_a(x)\hat{\Phi}_b(y)\right) + \langle 0|\hat{T}\hat{\Phi}_a(x)\hat{\Phi}_b(y)|0\rangle$

* here we introduce the contraction between two operators: $\overline{\Phi_a(x)\Phi_b}(y) = \langle 0|\hat{T}\hat{\Phi}_a(x)\hat{\Phi}_b(y)|0\rangle$

and thus:
$$\overline{\Phi_a(x)\Phi_b}(y) = \begin{cases} \left[\hat{\Phi}_a^{(+)}(x), \hat{\Phi}_b^{(-)}(y)\right]_{\epsilon_{ab}} & x^0 \geq y^0. \\ \epsilon_{ab}\left[\hat{\Phi}_b^{(-)}(y), \hat{\Phi}_a^{(+)}(x)\right]_{\epsilon_{ab}} & x^0<y^0. \end{cases}$$

interestingly. this is the definition of Feynman propagator! $\overline{\Phi_a(x)\Phi_b}(y) = \begin{cases} D_F(x-y) & (\Phi_a, \Phi_b \text{ are same type of field}) \\ 0 & \text{(Not same type)} \end{cases}$

We can understand that no contraction can be assert between different field

* The most generalized conclusion — Wick theorem:

$$\hat{T}\{\Phi_{a_1}(x_1)\Phi_{a_2}(x_2)\cdots\Phi_{a_n}(x_n)\} = N\{\Phi_{a_1}(x_1)\cdots\Phi_{a_n}(x_n) + \underbrace{(\text{All possible contraction})}\}\ !$$

Add all kinds of contraction line.

◦ Why it is useful? : Consider the expectation in vaccum, with possible contraction

$$\langle 0|N(\hat{\Phi}_1\hat{\Phi}_2\hat{\Phi}_3\cdots\hat{\Phi}_n)|0\rangle$$

The only non-zero possible, is in one type of contraction, where all operators has been connected into contraction line!

* However, in the calculation of $\langle i|\hat{S}|f\rangle$, we should calculate $\langle i|N(\hat{\Phi}_1\hat{\Phi}_2\cdots\hat{\Phi}_n)|f\rangle$?

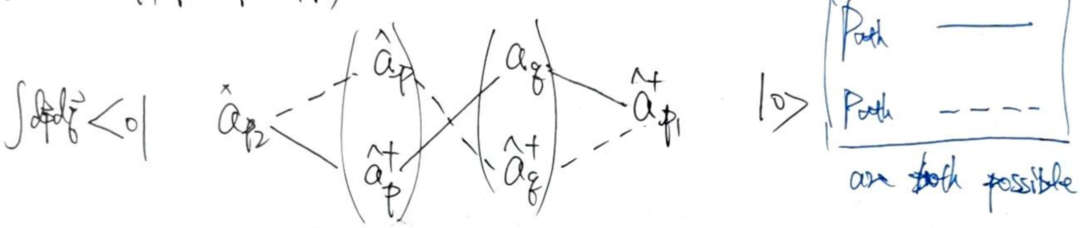
how can it become easier to calculate?

$$\hat{\phi}(x_1)|\vec{p}_1\rangle = \int\frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\left(a_p e^{-i\vec{p}\cdot\vec{x}_1} + \hat{a}_p^+ e^{i\vec{p}\cdot\vec{x}_1}\right)\cdot\sqrt{2E_{p_1}}\,\hat{a}_{p_1}^+|0\rangle$$

↳ Suppose it is a real KG-field

$$\langle p_2|\hat{\phi}(x_2) = \int\frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sqrt{2E_p}\langle 0|\hat{a}_{p_2}\left(a_p e^{-i\vec{p}\cdot\vec{x}_2} + \hat{a}_p^+ e^{i\vec{p}\cdot\vec{x}_1}\right)$$

how can $\langle p_2|\hat{\phi}(x_2)\hat{\phi}(x_1)|\vec{p}_1\rangle \neq 0$?



but what about $\langle p_2|\hat{T}\hat{\phi}(x_2)\hat{\phi}(x_1)|p_1\rangle \neq 0$?

$$\text{Since } \langle p_2|\hat{T}\hat{\phi}(x_2)\hat{\phi}(x_1)|p_1\rangle = \underline{\langle p_2|N(\hat{\phi}(x_2)\hat{\phi}(x_1))|p_1\rangle} + D_F(p_2-p_1)$$

⇓

Only this path satisfy normal order!

thus: we can introduce another contraction.

$$\hat{\phi}(x)|p_1\rangle = \hat{\phi}^{(+)}(x)|p_1\rangle, \quad \langle p_2|\hat{\phi}(x) = \langle p_2|\hat{\phi}^{(-)}(x) \quad \langle p_1|p_2\rangle = \langle p_1|p_2\rangle = 2E_p(2\pi)^3\delta^{(3)}(\vec{p}_1-\vec{p}_2)$$

And thus we have found: $\langle p_2|\hat{T}\hat{\phi}(x_2)\hat{\phi}(x_1)|p_1\rangle = \langle p_2|\hat{\phi}(x_2)\hat{\phi}(x_1)|p_1\rangle + \langle p_2|\hat{\phi}(x_2)\hat{\phi}(x_1)|p_1\rangle$

$\therefore \langle i|\hat{T}\Phi_{a_1}(x_1)\cdots\Phi_{a_n}(x_n)|f\rangle = N\{\text{All possible contraction within operators and external state}\}$

§4 Feynman Diagram. Take ϕ^4 theory as example.

Let's first summary what we have learned:

(1) The main goal of interaction field theory is to determine the observables like cross section.

$$d\sigma = \frac{1}{4E_A E_B V_{rel}} |\mathcal{M}_{if}|^2 d\Pi_2 \qquad d\Pi_2 = \int \frac{d\vec{p}_1 d\vec{p}_2}{(2\pi)^6 4E_{p_1}E_{p_2}} (2\pi)^4 \delta^{(4)}(\vec{p}_A + \vec{p}_B - \vec{p}_1 - \vec{p}_2)$$

(2) The $|\mathcal{M}_{if}|^2$ in $d\sigma$ comes from S-matrix, where $S_{\alpha\beta} = \delta_{\alpha\beta} + (2\pi)^4 \delta^{(4)}(\vec{p}_\alpha - \vec{p}_\beta)\, i\mathcal{M}_{\alpha\beta}$

(3) In interaction picture, $S_{\alpha\beta}$ can be expanded in series, take interaction coupling constant as perturbation order.

$$S_{\alpha\beta} = S^{(0)}_{\alpha\beta} + S^{(1)}_{\alpha\beta} + S^{(2)}_{\alpha\beta} + \cdots$$

$$S^{(0)}_{\alpha\beta} \to \delta_{\alpha\beta}, \quad S^{(1)}_{\alpha\beta} \to -i\int d^4\vec{x}\, \langle\alpha|\hat{\mathcal{H}}_{int}(\vec{x})|\beta\rangle, \quad S^{(2)}_{\alpha\beta} \to \frac{(-i)^2}{2!}\int d^4\vec{x}\, d^4\vec{y}\, \langle\alpha|\hat{\mathcal{T}}\mathcal{H}_{int}(\vec{x})\mathcal{H}_{int}(\vec{y})|\beta\rangle$$

(4) According to Wick theory, each level of $S^{(i)}_{\alpha\beta}$ is the sum over of all complete contraction. Thus the only mission is to compute all possible contraction, which contributes to $i\mathcal{M}$

* Consider ϕ^4 theory, where $\mathcal{L} = \mathcal{L}_{KG}[\phi] - \frac{\lambda}{4!}\phi^4$. thus $\mathcal{H}_{int} = \frac{\lambda}{4!}\phi^4$

In this theory, we can only consider KG particle to KG particle. More specific, we will discuss $\phi\phi \to \phi\phi$ process.

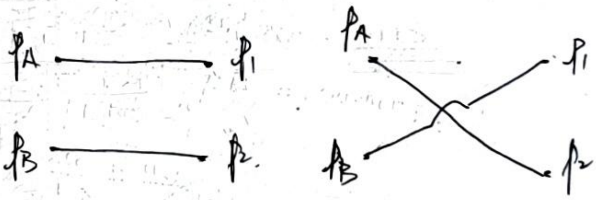
* Zero-order: This is the order that will not contribute to $i\mathcal{M}$. We can see that:

$$S_{fi} = \langle p_1 p_2 | p_A p_B\rangle = \langle p_1 p_2 | p_A p_B\rangle + \langle p_1 p_2 | p_A p_B\rangle$$

$$= 4E_{p_A}E_{p_B}(2\pi)^6 \delta^{(3)}(\vec{p}_1 - \vec{p}_A)\delta^{(3)}(\vec{p}_2 - \vec{p}_B)$$
$$+ 4E_{p_A}E_{p_B}(2\pi)^6 \delta^{(3)}(\vec{p}_1 - \vec{p}_B)\delta^{(3)}(\vec{p}_2 - \vec{p}_A)$$

We can use two diagram to record these two type of contraction:

Those two diagrams are unconnected.

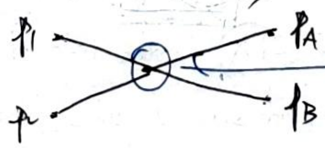


* First order:

$$S^{(1)}_{fi} = \langle p_1 p_2 | S^{(1)} | p_A p_B\rangle = -i\frac{\lambda}{4!}\int d^4\vec{x}\, \langle p_1 p_2 | \hat{\mathcal{T}}\phi(x)\phi(x)\phi(x)\phi(x) | p_A p_B\rangle$$

There are several types of contraction:

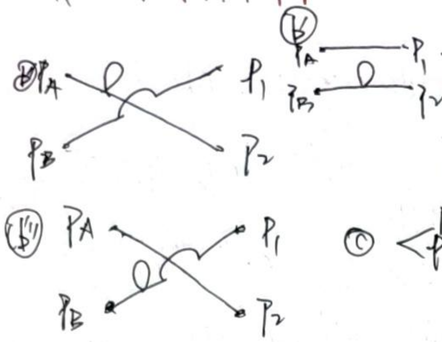
ⓐ $\langle p_1 p_2 | \phi\phi\phi\phi | p_A p_B\rangle$ which can be represented in diagram.

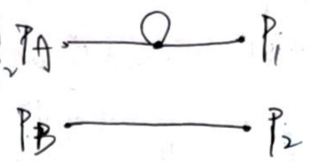


Use a vertex to correspond an interaction center that with position $\vec{x}$

Actually this diagram contains # of contraction approach is 4!

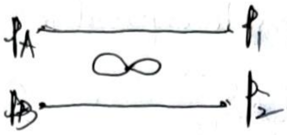
$S_{fi}^{(1)} = -\frac{i\lambda}{4!}\langle p_1 p_2|\phi\phi\phi\phi|p_A p_B\rangle$ ⓑ $\langle p_1 p_2|\phi\phi\phi\phi|p_A p_B\rangle$, which can be denoted in diagram:





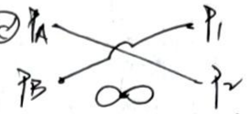
- This diagram contains $2\times4\times2\times3 = 12$ contractions
- It is also concorrected diagram.

ⓒ $\langle p_1 p_2|\phi\phi\phi\phi|p_A p_B\rangle$, which can be denoted in:

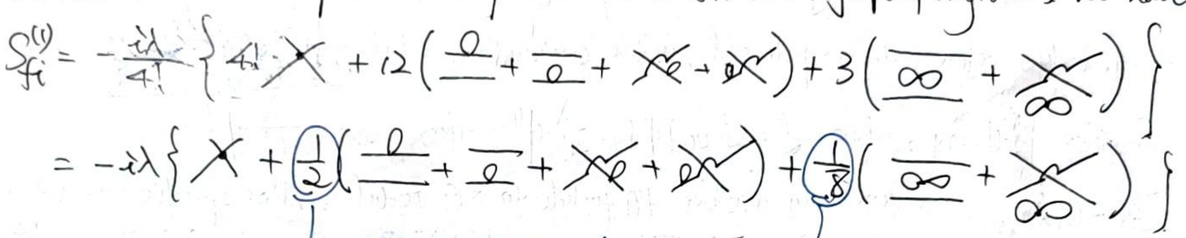


- This diagram contains $\frac{1}{2}\binom{4}{2} = \cancel{6}\ 3$

Similarly ⓒ'



Thus we have considered all possibilities of contraction. In the language of diagram, we have



These are symmetry factor for each diagrams, which can be gain just from the symmetry of each diagram.

Let's find what are those diagram's value:

$$-i\lambda \times = -i\lambda\int\langle p_1 p_2|\phi\phi\phi\phi|p_A p_B\rangle d^4\vec{x} = -i\lambda\int\langle p_1 p_2|\phi^{(-)}\phi^{(-)}\phi^{(+)}\phi^{(+)}|p_A p_B\rangle d^4\vec{x}.$$

In ϕ^4 theory: $\phi^{(-)}|p_A\rangle = \int\frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\hat{a}_p\, e^{-i\vec{p}\cdot\vec{x}}\sqrt{2E_{p_A}}\,\hat{a}^\dagger_{p_A}|0\rangle$

$$= \int\frac{d^3\vec{p}}{(2\pi)^3}\sqrt{\frac{E_{p_A}}{E_p}}\, e^{-i\vec{p}\cdot\vec{x}}\,[a_p, \hat{a}^\dagger_{p_A}]|0\rangle = e^{-i\vec{p}_A\cdot\vec{x}}|0\rangle$$

$[a_p, \hat{a}^\dagger_{p_A}] \to (2\pi)^3\delta(\vec{p}-\vec{p}_A)$

similarly $\langle p_1|\hat{\phi}^{(+)} = \langle 0|e^{i\vec{p}_1\cdot\vec{x}}$.

thus there will be: $-i\lambda \times = -i\lambda\int d^4\vec{x}\, e^{-i(\vec{p}_A+\vec{p}_B-\vec{p}_1-\vec{p}_2)\cdot\vec{x}}$

$$= (-i\lambda)\cdot(2\pi)^4\delta^{(4)}(\vec{p}_A+\vec{p}_B-\vec{p}_1-\vec{p}_2).$$

In fact we have found that $-i\lambda$ is the role of $i\mathcal{M}$ in $\langle f|\hat{S}|i\rangle$!

- Usually, we can introduce several rules for the diagram:
  ① Vertex $\times = -i\lambda$.
  ② Around the vertex, 4-momentum should conserved.

then: $\times = -i\lambda = i\mathcal{M}$!

□ What about other diagrams? Take an example:

$$= \frac{-i\lambda}{2}\int \langle p_1 p_2 | \phi\phi\phi\phi | p_A p_B\rangle d^4\vec{x} = \frac{-i\lambda}{2}\int \langle p_1|\hat{\phi}^{(-)}(x)\cdot D_F(0)\cdot\hat{\phi}^{(+)}(x)|p_A\rangle \cdot \langle p_2|p_B\rangle\, d^4\vec{x}$$

① Diagram absorb the interaction $(-i\lambda)$

② Also absorb the symmetry factor

$$= \frac{-i\lambda}{2}\int d^4\vec{x}\cdot e^{i\vec{p}_1\cdot\vec{x}}\cdot\left(\int d^4\vec{p}\,\frac{i}{\vec{p}^2-m^2+i\varepsilon}\right)\cdot e^{-i\vec{p}_A\cdot\vec{x}}\cdot d^4\vec{x} \cdot (2\pi)^4\cdot 2E_{\vec{p}_B}\,\delta^{(4)}(\vec{p}_2-\vec{p}_B)$$

It cannot be written as the form of $\boxed{i\mathcal{M}}$

⇓

Unconnected diagram won't support $i\mathcal{M}$

$$\Leftarrow \left\{ = \frac{-i\lambda}{2}\cdot(2\pi)^4\, 2E_{\vec{p}_B}\,\delta^{(4)}(\vec{p}_2-\vec{p}_B)\cdot(2\pi)^4\delta^{(4)}(\vec{p}_1-\vec{p}_A)\cdot\int d^4\vec{p}\,\frac{i}{\vec{p}^2-m^2+i\varepsilon}\right.$$

※ Our conclusion: The contribution to $i\mathcal{M}$ is the completely connected diagram.

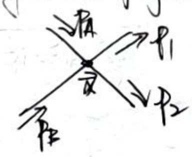
These diagram is called Feynman diagram!

And: Up to $O(\lambda)$ order, we have found $S_{fi}^{(1)} =$

$$= -i\mathcal{M}\cdot(2\pi)^4\delta^{(4)}(\vec{p}_A+\vec{p}_B-\vec{p}_1-\vec{p}_2)$$

We can only give the Feynman diagram in momentum space, that is

$i\mathcal{M}^{(1)} =$

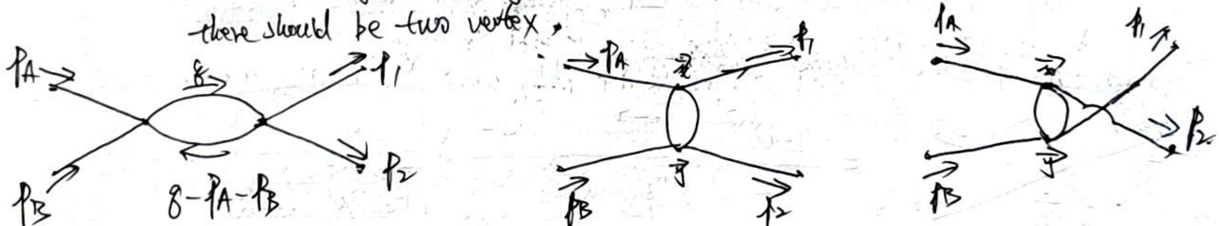


$= -i\lambda$ and on the vertex $\vec{x}$, 4-momentum conservation should be conserved.

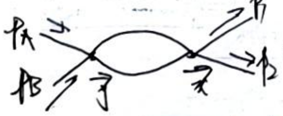
Interestingly: Only the vertex-rule reflect the form of interaction.

The rule of external line (which is trivial for real KG) is independent of theory

* Second order: Let us analyse the diagram in $O(\lambda^2)$, which means in these diagram, there should be two vertex,



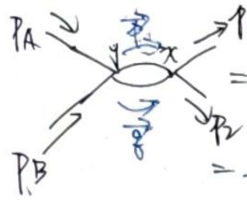
There are three basic type of $O(\lambda^2)$ connected diagram. According to Feynman rule of ϕ^4 theory, and they all containing a "circle". How to deal with the un-determined $\vec{q}$?



$$= (-i\lambda)^2\cdot\frac{1}{2}\int \langle p_1 p_2|\phi_x\phi_x\phi_x\phi_x\phi_y\phi_y\phi_y\phi_y|p_A p_B\rangle d^4\vec{x}\, d^4\vec{y}$$

① We have considered symmetry factor $1/2$.

② The factor $\frac{1}{2!}$ in Dyson Series has been canceled, since we have $\vec{x}\leftrightarrow\vec{y}$ in this diagram



$$= \frac{(-i\lambda)^2}{2}\int d^4x\, d^4y \left\langle p_1 p_2 \middle| \phi_x\phi_x\phi_x\phi_x\phi_y\phi_y\phi_y\phi_y \middle| p_A p_B \right\rangle$$

$$= \frac{(-i\lambda)^2}{2}\int d^4x\, d^4y \cdot e^{-i(\vec{p}_A+\vec{p}_B)\cdot y}\, e^{i(\vec{p}_1+\vec{p}_2)\cdot x} \cdot D(x-y)\, D(x-y)$$

$$= \frac{(-i\lambda)^2}{2}\int d^4x \cdot d^4y \cdot \underbrace{e^{-i(\vec{p}_A+\vec{p}_B)\cdot y}}\, e^{i(\vec{p}_1+\vec{p}_2)\cdot x} \int \frac{d^4\vec{p}}{(2\pi)^4}\frac{d^4\vec{q}}{(2\pi)^4}\frac{i}{\vec{p}^2-m^2+i\varepsilon}\cdot\frac{i}{\vec{q}^2-m^2+i\varepsilon}\cdot \underbrace{e^{-i\vec{p}\cdot(x-y)}\cdot e^{-i\vec{q}\cdot(x-y)}}$$

$$\underline{e^{-i(\vec{p}_A+\vec{p}_B-\vec{p}-\vec{q})y}\cdot e^{i(\vec{p}_1+\vec{p}_2-\vec{p}-\vec{q})x}}$$

$\Downarrow$ integrated from $\int d^4x \cdot d^4y$

$$= \frac{(-i\lambda)^2}{2}\int \frac{d^4\vec{p}}{(2\pi)^4}\cdot\frac{d^4\vec{q}}{(2\pi)^4}\cdot \boxed{(2\pi)^4\delta^{(4)}(\vec{p}_A+\vec{p}_B-\vec{p}-\vec{q})\cdot(2\pi)^4\delta^{(4)}(\vec{p}_1+\vec{p}_2-\vec{p}-\vec{q})}\cdot\frac{i}{\vec{p}^2-m^2+i\varepsilon}\cdot\frac{i}{\vec{q}^2-m^2+i\varepsilon}$$

$$\underbrace{(2\pi)^4\delta^{(4)}(\vec{p}_A+\vec{p}_B-\vec{p}_1-\vec{p}_2)}\cdot(2\pi)^4\delta^{(4)}(\vec{p}_A+\vec{p}_B-\vec{p}-\vec{q})$$

$\Downarrow$

Not the part of $i\mathcal{M}$

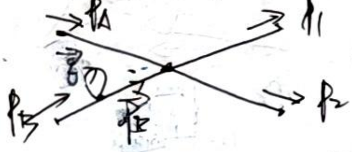
$$= (2\pi)^4\delta^{(4)}(\vec{p}_A+\vec{p}_B-\vec{p}_1-\vec{p}_2)\cdot\underbrace{\frac{(-i\lambda)^2}{2}\int\frac{d^4\vec{q}}{(2\pi)^4}\cdot\frac{i}{\vec{q}^2-m^2+i\varepsilon}\cdot\frac{i}{(\vec{q}-\vec{p}_A-\vec{p}_B)^2+i\varepsilon}}_{i\mathcal{M}}$$

thus there are another two Feynman rule:

(1) Inner line $\xrightarrow[\vec{q}]{} = D_F(\vec{q})$, since inner line is the contraction between field operators

(2) All undetermined momentum should be integrated as $\int\frac{d^4\vec{q}}{(2\pi)^4}$.

II There are other possible diagram, such as



$$= \frac{(-i\lambda)^2}{2}\cdot\int\frac{d\vec{q}}{(2\pi)^4}\cdot\frac{i}{\vec{q}^2-m^2+i\varepsilon}\cdot\boxed{\frac{i}{\vec{p}_B^2-m^2+i\varepsilon}}$$

This type of Feynman diagram with loop on tails will always give out a divergence! $\Longleftarrow$ $\begin{cases}\text{Since } \vec{p}_B \text{ is on shell, there should be } \vec{p}_B^2 = m^2 \text{ !} \\ \text{thus } \boxed{} \text{ is a divergent structure!}\end{cases}$

$\Downarrow$

In following chapter, these loops gives the effect of <u>self-energy correction</u>! which will be dealed with under renormalization, and the final effect is:

All tails-loop diagram's sum-over = Amputated!

* Now we can give a summary for ϕ^4-theory: ~~iM =~~
  - $i\mathcal{M}$ = All fully connected amputated Feynman diagram.
    and each Feynman Diagram represents the one final results for contraction.
  - We have several Feynman rule:
    (1) External line is 1, since $\hat{\phi}(x)|\vec{p}\rangle = e^{-i\vec{p}\cdot\vec{x}}|0\rangle$
    (2) Inner line is Feynman propagator $\xrightarrow[\vec{p}]{} = \dfrac{i}{\vec{p}^2 - m^2 + i\varepsilon}$] Independent of interaction form, only rely on the kinds of field.
    (3) Vertex: $\times = (-i\lambda)$. → Rely on ϕ^4 interaction
    (4) At each vertex, the 4-momentum is conserved. This is the results of $\int d^4\vec{x}\, e^{-i(\cdots)\vec{p}}$
    (5) All undetermined momentum should be integrated.
    (6) Each diagram should divide their symmetry factor

§5. Feynman Rule for Complex field: Scalar QED

In Scalar QED, $\mathcal{L} = D_\mu\phi D^\mu\phi^* - m^2\phi\phi^* - \frac{1}{4}F^{\mu\nu}F_{\mu\nu}$, where U(1)-gauge symmetry has emerged.

Here $\underline{D_\mu = \partial_\mu + ieA_\mu}$, then the Lagrangian will become: $\mathcal{L} = \mathcal{L}_{KG}[\phi] + \mathcal{L}_{Max}[A^\mu]$
↳ Require of U(1) gauge symmetry
$$-ieA_\mu\underbrace{(\phi^*\partial^\mu\phi - (\partial^\mu\phi^*)\phi)} + e^2|\phi|^2 A_\mu A^\mu.$$

$$\hat{S} = \exp i\left(+\int d^4\vec{x}\,\mathcal{L}_{int}\right) \longleftarrow$$ ~~but~~ we can be inspired by path integral quantization

< In scalar QED, there will be derivation coupling!

and here $\mathcal{L}_{int} = -ieA_\mu(\phi^*\partial^\mu\phi - (\partial^\mu\phi^*)\phi) + e^2\phi\phi^* A_\mu A^\mu$.

** External line: Notice that → Scalar particle excited from ϕ-field (π介子)

- $\hat{\phi}(x)|\pi(\vec{p})\rangle = \int \frac{d^3\vec{q}}{(2\pi)^3}\hat{a}_q e^{-i\vec{q}\cdot\vec{x}}\sqrt{\frac{E_p}{E_q}}\hat{a}^\dagger_p|0\rangle = \boxed{e^{-i\vec{p}\cdot\vec{x}}}|0\rangle$, thus we have $\xrightarrow{\vec{p}}$ - - - = 1

  → Same phase factor.

  and also $\langle\pi(\vec{p})|\hat{\phi}^\dagger(x) = e^{-i\vec{p}\cdot\vec{x}}\langle 0|$, thus $\xrightarrow{\vec{p}}$ - - - = 1.

- For the conjugation, $\hat{\phi}^\dagger(x)|\bar{\pi}(\vec{p})\rangle = \int \frac{d^3\vec{q}}{(2\pi)^3}\frac{1}{\sqrt{2E_q}}\hat{b}_q e^{-i\vec{q}\cdot\vec{x}}\cdot\sqrt{2E_p}\hat{b}^\dagger_p|0\rangle = \boxed{e^{-i\vec{p}\cdot\vec{x}}}|0\rangle$

  thus for the positive particle, it's external line has the same rule.

  and we use so-called particle flow arrow to distinguish the particle and anti-particle.

- For photons, there will be $\hat{A}_\mu(x)|\gamma(\vec{p},\lambda)\rangle = \int\frac{d^3\vec{q}}{(2\pi)^3}\frac{1}{\sqrt{2E_q}}\sum_{\lambda'=0}^{3} a_{\vec{q},\lambda'}\varepsilon_\mu(\vec{q},\lambda')e^{-i\vec{q}\cdot\vec{x}}\cdot\sqrt{2E_p}\,\hat{a}^\dagger_{\vec{p},\lambda}|0\rangle$

  A nontrivial external line. $\longleftarrow$ $= \boxed{\varepsilon_\mu(\vec{p},\lambda)}e^{-i\vec{p}\cdot\vec{x}}|0\rangle$

We can sum the Feynman rule for external line:

$S = e^{+i\int d^4\vec{x}\, \mathcal{L}_{int}}$

$\mathcal{L}_{int} = -ieA_\mu(\phi^* \overleftrightarrow{\partial} \phi) + e^2 A_\mu A^\mu \phi^* \phi$

Same direction is positive particle; Anti-direction is anti-particle

$\vec{p}$ - - - → - - - $= \vec{p}$ - - ← - - $\vec{p} = 1$

$\vec{p},\lambda$ ~~~ $= \varepsilon_\mu(\vec{p},\lambda)$ ~~~ $\vec{p},\lambda = \varepsilon^*_\nu(\vec{p},\lambda)$

μ → this Lorentz index should be contracted by interaction vertex

** Vertex-rule:

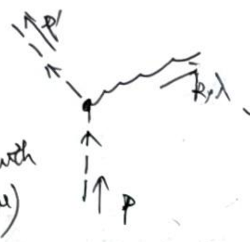
- Interaction of $e^2 A_\nu A^\mu \phi \phi^*$ is easy to impose Feynman rule:

$= (2) i e^2 g^{\mu\nu}$ → Symmetry factor for photon line

this vertex will give second order $O(e^2)$

- To impose the Feynman rule of $\underbrace{-ieA_\mu(\phi^* \overleftrightarrow{\partial} \phi)}_{\text{Vertex of}}$, let consider the $O(e)$ full diagram.

e.g.1 (Process with particle)



$\sim \langle \pi(p')\gamma(k,\lambda) | i\int d^4\vec{x}\, (-ieA^\mu(\phi^* \overleftrightarrow{\partial_\mu} \phi)) | \pi(p) \rangle$

$= e\int d^4\vec{x}\, \langle e(p')\gamma(k,\lambda) | A^\mu \phi^* \overleftrightarrow{\partial_\mu} \phi | e(p) \rangle$

$= e\int d^4\vec{x}\, \varepsilon^{*\mu}(k,\lambda) e^{i\vec{k}\cdot\vec{x}} \left(e^{i\vec{p}'\cdot\vec{x}} \overleftrightarrow{\partial_\mu} e^{-i\vec{p}\cdot\vec{x}}\right)$

$= (2\pi)^4 \delta^{(4)}(\vec{k}+\vec{p}'-\vec{p})\, \varepsilon^\mu(k,\lambda) \boxed{-ie(p_\mu + p'_\mu)}$

Come from the role of vertex

thus we can impose

$= -ie(p_\mu + p'_\mu)$ → will be contracted will photon's external line

e.g.2 (Process with anti-particle).

$\sim \langle \pi^+(p_1)\pi^-(p_2) | i\int d^4\vec{x}\, (-ieA^\mu(\phi^* \overleftrightarrow{\partial_\mu} \phi)) | \gamma(k,\lambda) \rangle$

$= e\int d^4\vec{x}\, \varepsilon^\mu(k,\lambda) e^{-i\vec{k}\cdot\vec{x}} \left(e^{-i\vec{p}_1\cdot\vec{x}} \overleftrightarrow{\partial_\mu} e^{+i\vec{p}_2\cdot\vec{x}}\right)$

$= (2\pi)^4 \delta^{(4)}(\vec{k}-\vec{p}_1-\vec{p}_2)\, \underline{e\cdot i(p_{2\mu} - p_{1\mu})}\, \varepsilon^\mu(k,\lambda)$

$= -ie(p_{1\mu} - p_{2\mu})$

e.g.3

$t\uparrow$

$\sim i\int d^4\vec{x} \langle \pi^+(p')\gamma(k,\lambda) | -ieA^\mu(\phi^* \overleftrightarrow{\partial_\mu} \phi) | \pi^+(p) \rangle$

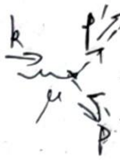
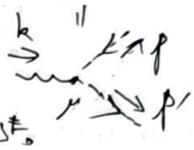
$= e\int d^4\vec{x}\, \varepsilon^\mu(k,\lambda) e^{i\vec{k}\cdot\vec{x}} \left(e^{-i\vec{p}\cdot\vec{x}} \overleftrightarrow{\partial_\mu} e^{i\vec{p}'\cdot\vec{x}}\right)$

$= (2\pi)^4 \delta^{(4)}(\cdots)\, \varepsilon^\mu(k,\lambda) \cdot e\, i(p'_\mu + p_\mu)$

$t\uparrow$

$= ie(p_\mu + p'_\mu)$

From these, we can get the Feynman rule of vertex

 $= -ie(p_\mu + p'_\mu)$  $= +ie(p_\mu + p'_\mu)$  $= -ie(p_\mu - p'_\mu)$ $=$ 

Here are something worthy to be mentioned:

① In all diagrams of scalar QED, the arrow of particle flow is continuous.
this is the results of charge conservation, coming from U(1) gauge symmetry.

② In scalar QED theory, for a process with photon in or out, amplitude $\mathcal{M}$ must take the form.
$\mathcal{M} \equiv \mathcal{M}^\mu \varepsilon_\mu(\vec{k},\lambda)$, which is the requirement of photon external line.
And there would be so-called Ward identity : $\boxed{\mathcal{M}^\mu k_\mu \overset{!}{=} 0}$

For example, process: $i\mathcal{M}\{\gamma(\vec{k},\lambda) \to \pi^+(p)\pi^-(p')\} =$ 

$$= \varepsilon^\mu(\vec{k},\lambda)\cdot(-ie)(p_\mu - p'_\mu) = -ie(p'_\mu - p_\mu)\cdot\varepsilon^\mu(\vec{k},\lambda)$$

thus we have $\mathcal{M}_\mu = -ie(p'_\mu - p_\mu)$, while $\vec{k} = \vec{p} + \vec{p}'$ [$\delta^{(4)}(\vec{k}-\vec{p}-\vec{p}')$], thus $\vec{p}, \vec{p}'$ is on shell

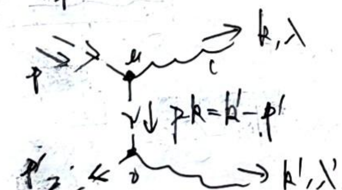
$$\mathcal{M}_\mu k^\mu = -ie(p'_\mu - p_\mu)(p'^\mu + p^\mu) = -ie(\vec{p}^2 - \vec{p}'^2) = -ie(m-m) = 0$$

③ The photon also has a helicity index, while in the computation of cross section, this index will not appear. In real experiments, we can set the particles in different helicity half and half, while the particles outside should be fully considered, no matter the proportion of two helicity. Thus when we count $|\mathcal{M}|^2$ into $d\sigma$, we should take the average of inner freedom degree, and sum over the output's inner freedom degree.

<u>Homework2</u>: Consider the lowest order of process $\pi^+\pi^- \to 2\gamma$. Write the amplitude and check Ward identity
And try to derive the cross section $d\sigma$.

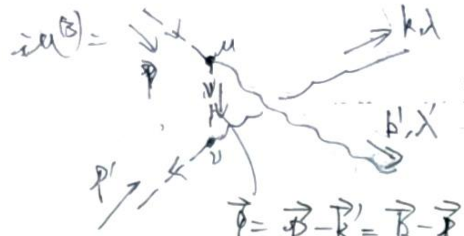
There are several type of Feynman diagram for this process

$i\mathcal{M}^{(1)} =$  $= 2ie^2 g^{\mu\nu} \varepsilon^*_\mu(k,\lambda)\,\varepsilon^*_\nu(k',\lambda')$

$i\mathcal{M}^{(2)} =$  $= (-ie)(p_\mu + p_\mu - k_\mu)\cdot\varepsilon^\mu(\vec{k},\lambda)\cdot\dfrac{i}{(\vec{k}-\vec{p})^2 - m^2 + i\varepsilon}\cdot$

$\cdot(+ie)(p'_\mu + p'_\mu - k'_\mu)\,\varepsilon^\nu(\vec{k}',\lambda')$

$$= e^2\,(2p_\mu - k_\mu)\,\varepsilon^\mu(\vec{k},\lambda)\,(2p'_\nu - k'_\nu)\,\varepsilon^\nu(\vec{k}',\lambda')\cdot\frac{i}{(\vec{p}-\vec{k})^2 - m^2 + i\varepsilon}$$

$i\mathcal{M}^{(3)} =$ 

$$= (-ie)(2p_\mu - k'_\mu)\varepsilon^\mu(k',\lambda')\frac{i}{(\vec p-\vec k')^2 - m^2 + i\epsilon}\cdot(-ie)(k_\nu - 2p'_\nu)\varepsilon^\nu(k,\lambda)$$

$$\vec q = \vec p - \vec k' = \vec k - \vec p' \qquad = e^2(2p_\mu - k'_\mu)\varepsilon^\mu(k',\lambda')\frac{i}{(\vec p-\vec k')^2 - m^2 + i\epsilon}(2p'_\nu - k_\nu)\varepsilon^\nu(k,\lambda)$$

Thus we have:

$$i\mathcal{M} = 2ie^2\underbrace{\varepsilon^\mu(k,\lambda)}_{g_{\mu\nu}}\varepsilon^\nu(k',\lambda') + ie^2\frac{(2p_\mu - k_\mu)(2p'_\nu - k'_\nu)\varepsilon^\mu(k,\lambda)\varepsilon^\nu(k',\lambda')}{\underbrace{(\vec p-\vec k)^2 - m^2 + i\epsilon}_{\underbrace{\vec p^2}_{m^2} + \underbrace{\vec k^2}_{0} - 2\vec p\cdot\vec k - m^2 = -2\vec p\cdot\vec k}} + ie^2\frac{(2p_\mu - k'_\mu)(2p'_\nu - k_\nu)\varepsilon^\mu(k',\lambda')\varepsilon^\nu(k,\lambda)}{\underbrace{(\vec p-\vec k')^2 - m^2 + i\epsilon}_{-2\vec p\cdot\vec k'}}$$

and thus $i\mathcal{M}^\mu = \left(2ie^2 g^{\mu\nu} - ie^2\frac{(2p^\mu - k^\mu)(2p'^\nu - k'^\nu)}{2\vec p\cdot\vec k} - ie^2\frac{(2p'^\mu - k'^\mu)(2p^\nu - k'^\nu)}{2\vec p\cdot\vec k'}\right)\varepsilon_\nu(k',\lambda')$

with $\mathcal{M} = \mathcal{M}^\mu\varepsilon_\mu(k,\lambda)$, thus.

$$\mathcal{M}^\mu k_\mu = e^2\left\{2g^{\mu\nu} - \frac{1}{2\vec p\cdot\vec k}(2p^\mu - k^\mu)(2p'^\nu - k'^\nu) - \frac{1}{2\vec p\cdot\vec k'}(2p'^\mu - k'^\mu)(2p^\nu - k'^\nu)\right\}\varepsilon_\nu(k',\lambda')k_\mu$$

Notice that $\frac{1}{2\vec p\cdot\vec k}(2p^\mu - k^\mu)(2p'^\nu - k'^\nu)\varepsilon_\nu(k',\lambda')k_\mu$ — For physical photon $\vec k\cdot\vec\varepsilon(k,\lambda) = 0$

$$= \frac{1}{2\vec p\cdot\vec k}(2\vec p\cdot\vec k - \underbrace{\vec k^2}_{0})\cdot 2p'^\nu\varepsilon_\nu(k',\lambda') = 2\vec p'\cdot\vec\varepsilon(k',\lambda')$$

, since $\vec k^2 = 0$ is the on-shell condition

And: $\frac{1}{2\vec p\cdot\vec k'}(2p'^\mu - k'^\mu)(2p^\nu - k'^\nu)\varepsilon_\nu(k',\lambda')k_\mu$

$$= \underbrace{\frac{\vec p'\cdot\vec k}{\vec p\cdot\vec k'}}_{1}\cdot 2p^\nu\varepsilon_\nu(k',\lambda') = 2\vec p\cdot\vec\varepsilon(k',\lambda')$$

1, since we have $\vec p + \vec p' = \vec k + \vec k' \Leftrightarrow \vec p - \vec k' = \vec k - \vec p'$

$$\Updownarrow$$

$$\vec p\cdot\vec k' = \vec k\cdot\vec p'$$

And: $2g^{\mu\nu}\varepsilon_\nu(k',\lambda')k_\mu = 2\vec k\cdot\varepsilon_\nu(k',\lambda')$

thus we have: $\mathcal{M}^\mu k_\mu = e^2\left\{2\vec k - 2\vec p' - 2\vec p\right\}\cdot\vec\varepsilon(k',\lambda') = 2e^2\left\{\underbrace{\vec k - \vec p' - \vec p + \vec k'}_{0}\right\}\cdot\vec\varepsilon(k',\lambda')$ $= 0$

∴ We have checked the Ward identity is satisfied for $\mathcal{O}(e^2)$-order.

Now we introduce $\sum_{\lambda,\lambda'} |\mathcal{M}|^2$. Firstly we want to point out $\sum_{\lambda,\lambda'} \varepsilon_\mu^*(k,\lambda)\varepsilon_\nu(k,\lambda')\mathcal{M}^\mu(k)\mathcal{M}^\nu(k)$

$= -g_{\mu\nu}\mathcal{M}^\mu(k)\mathcal{M}^\nu(k)$

From Ward identity, we have $k_\mu \mathcal{M}^\mu = 0$, thus:

$$\sum_{\lambda,\lambda'=1,2} |\mathcal{M}|^2 = \sum_{\lambda,\lambda'=1,2} \varepsilon_\mu^*(k,\lambda)\varepsilon_\nu(k,\lambda)\mathcal{M}^\mu(k)\mathcal{M}^\nu(k)$$

$$= \Big(-g_{\mu\nu} \underbrace{- \frac{k_\mu k_\nu}{(\vec{k}\cdot\vec{n})^2} + \frac{k_\mu n_\nu + k_\nu n_\mu}{\vec{k}\cdot\vec{n}}}_{0,\ \text{from Ward identity}}\Big)\mathcal{M}^\mu(k)\mathcal{M}^\nu(k) = -g_{\mu\nu}\mathcal{M}^\mu(k)\mathcal{M}^\nu(k)$$

Polarization sum over

Since we have $-i\mathcal{M} = ie^2\Big(2g^{\mu\nu} - \frac{(2p^\mu - k^\mu)(2p'^\nu - k'^\nu)}{2\vec{p}\cdot\vec{k}} - \frac{(2p'^\mu - k^\mu)(2p^\nu - k'^\nu)}{2\vec{p}\cdot\vec{k}'}\Big)\varepsilon_\mu(k,\lambda)\varepsilon_\nu(k',\lambda)$

thus there will be:

$$\sum_{\lambda,\lambda'} |\mathcal{M}|^2 = \sum_{\lambda,\lambda'} e^4 \{\cdots\}^{\mu\nu}\{\cdots\}^{\rho\sigma}\varepsilon_\mu(k,\lambda)\varepsilon_\nu(k',\lambda)\varepsilon_\rho^*(k,\lambda)\varepsilon_\sigma^*(k',\lambda)$$

$$= e^4\{\cdots\}^{\mu\nu}\{\cdots\}^{\rho\sigma} g_{\mu\rho}g_{\nu\sigma} = e^4\{\cdots\}^{\mu\nu}\{\cdots\}_{\mu\nu}$$

$$\{\cdots\}^{\mu\nu}\varepsilon_\mu(k,\lambda)\varepsilon_\nu(k',\lambda) \underset{k^\mu\varepsilon_\mu(k,\lambda)=0}{=} \Big\{2g^{\mu\nu} - \frac{2p^\mu p'^\nu}{\vec{p}\cdot\vec{k}} - \frac{2p'^\mu p^\nu}{\vec{p}\cdot\vec{k}'}\Big\}\varepsilon_\mu(k,\lambda)\varepsilon_\nu(k',\lambda)$$

$$\therefore \{\cdots\}^{\mu\nu}\{\cdots\}_{\mu\nu} = 4g^{\mu\nu}g_{\mu\nu} + \frac{4}{(\vec{p}\cdot\vec{k})^2}\vec{p}^2\cdot\vec{p}'^2 + \frac{4\vec{p}'^2\vec{p}^2}{(\vec{p}\cdot\vec{k}')^2}$$

$$- \frac{4\vec{p}\cdot\vec{p}'}{(\vec{p}\cdot\vec{k})} - \frac{4\vec{p}'\cdot\vec{p}}{(\vec{p}\cdot\vec{k}')} + \frac{4(\vec{p}\cdot\vec{p}')^2}{(\vec{p}\cdot\vec{k})(\vec{p}\cdot\vec{k}')}$$

$$= 4\Big(4 + m^4\Big\{\frac{1}{(\vec{p}\cdot\vec{k})^2} + \frac{1}{(\vec{p}\cdot\vec{k}')^2}\Big\} - (\vec{p}\cdot\vec{p}')\cdot\Big\{\frac{1}{\vec{p}\cdot\vec{k}} - \frac{1}{\vec{p}\cdot\vec{k}'}\Big\} + \frac{(\vec{p}\cdot\vec{p}')^2}{(\vec{p}\cdot\vec{k})(\vec{p}\cdot\vec{k}')}\Big)$$

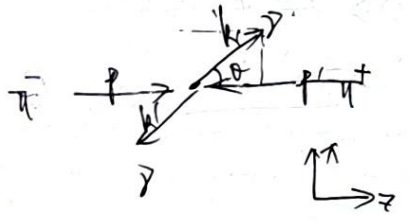
In CM-frame. let's suppose:

$$p^a = (E\ 0\ 0\ \beta E)$$
$$p'^a = (E\ 0\ 0\ -\beta E)$$
$$k = (E,\ E\sin\theta,\ 0,\ E\cos\theta)$$
$$k' = (E,\ E\sin\theta,\ 0,\ E\cos\theta)$$

$$\begin{cases} \vec{p}\cdot\vec{k} = E^2(1-\beta\cos\theta) \\ \vec{p}\cdot\vec{k}' = E^2(1+\beta\cos\theta) \\ \vec{p}\cdot\vec{p}' = E^2(1+\beta^2) \end{cases}$$

, thus there will be:

$$\{\cdots\}^{\mu\nu}\{\cdots\}_{\mu\nu} = 4\Big\{4 + \frac{m^4}{E^4}\Big(\frac{1}{(1-\beta\cos\theta)^2} + \frac{1}{(1+\beta\cos\theta)^2}\Big)$$
$$- (1+\beta^2)\cdot\Big(\frac{1}{1-\beta\cos\theta} - \frac{1}{1+\beta\cos\theta}\Big)$$
$$+ \frac{(1+\beta^2)^2}{(1-\beta\cos\theta)(1+\beta\cos\theta)}\Big\}$$



$$= 4\left\{4 + \frac{m^4}{E^4}\cdot\frac{\gamma + 2\beta^2\cos^2\theta}{(1-\beta\cos\theta)^2(1+\beta\cos\theta)^2} - (1+\beta^2)\cdot\frac{2\beta\cos\theta + 1+\beta^2}{(1-\beta\cos\theta)(1+\beta\cos\theta)}\right\}$$

$$= 8\left\{2 + \frac{(1-\beta^2)^2(1+\beta^2\cos^2\theta)}{(1-\beta^2\cos^2\theta)^2} - \frac{1-\beta^4}{(1-\beta^2\cos^2\theta)}\right\} = \{\cdots\}^{\mu\nu}\{\cdots\}_{\mu\nu}$$

thus we have $\sum_{\lambda,\lambda'}|\mathcal{M}|^2 = 8e^4\left\{2 - \frac{1-\beta^4}{(1-\beta^2\cos^2\theta)} + \frac{(1-\beta^2)^2(1+\beta^2\cos^2\theta)}{(1-\beta^2\cos^2\theta)^2}\right\}$.

Next, about the so-called $d\Pi_2$, we have:

$$d\Pi_2 = \frac{d\vec{k}_1 d\vec{k}_2}{(2\pi)^6 4E^2}(2\pi)^4\delta^{(4)}(\vec{p}_1+\vec{p}_2-\vec{k}_1-\vec{k}_2) =$$

$$= \frac{d\vec{k}_1 d\vec{k}_2}{(2\pi)^2 4E^2}\delta(E_{cm}-|\vec{k}_1|-|\vec{k}_2|)\delta^{(3)}(\vec{k}_1+\vec{k}_2) = \frac{d\vec{k}}{(2\pi)^2 4E^2}\delta(E_{cm}-2E) = \frac{k^2 d\Omega}{(2\pi)^2\cdot 4E^2}\delta(E_{cm}-2E)$$

$$\therefore \left.\frac{d\sigma}{d\Omega}\right|_{cm} = \frac{1}{4E^2|\vec{v}_1-\vec{v}_2|}\cdot\frac{E^2}{4\pi^2\cdot 4E^2}\sum_{\lambda,\lambda'}|\mathcal{M}|^2 \qquad |\vec{v}_1-\vec{v}_2| = \frac{2|\vec{p}_1|}{E} = 2\beta.$$

$$\therefore \left.\frac{d\sigma}{d\Omega}\right|_{cm} = \frac{1}{32\pi^2\beta\cdot 4E^2}|\mathcal{M}|^2 = \frac{e^4}{4\cdot 2\pi^2\beta E_{cm}^2}\left(2 - \frac{1-\beta^4}{(1-\beta^2\cos^2\theta)} + \frac{(1-\beta^2)^2(1+\beta^2\cos^2\theta)}{(1-\beta^2\cos^2\theta)^2}\right)$$

□

§6 Feynman rule for fermion: Yukawa theory

In this section, we will talk about the Feynman rule with fermion field. Let's consider the so-called Yukawa theory, where $\mathcal{L}_{int} = -\lambda\phi(x)\bar{\psi}(x)\psi(x)$, and λ takes the role of coupling constant.

** External line: Since there are electronical field, thus we should consider external line for electron.

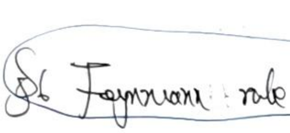
since $\psi(x)|e^-(\vec{p},s)\rangle = \int\frac{d^3\vec{q}}{(2\pi)^3}\cdot\frac{1}{\sqrt{2E_q}}\sum_r \hat{a}_{\vec{q},r}\vec{u}(\vec{q},r)e^{-i\vec{q}\cdot\vec{x}}\sqrt{2E_p}\cdot\hat{a}^\dagger_{\vec{p},s}|0\rangle = \vec{u}(\vec{p},s)\cdot e^{-i\vec{p}\cdot\vec{x}}|0\rangle$

thus we can introduce electronical external line: $\vec{p},s \rightarrow\!\!\!\!\rightarrow\; = \vec{u}(\vec{p},s)$

and similarly, there should be. $\rightarrow\!\!\!\!\rightarrow\; \vec{p},s = \bar{u}(\vec{p},s)$ Fermion is anti order of $\bar{v}, \vec{v}$

and for anti-fermion, there should be $\boxed{\vec{p},s \rightleftarrows\; = \bar{v}(\vec{p},s) \quad \rightleftarrows \vec{p},s = \vec{v}(\vec{p},s)}$

** Vertex rule. Let's consider a simple interaction:



$\sim \langle e^-(p',r)\pi(q)|-i\int d^4\vec{x}\cdot\lambda\phi(x)\bar{\psi}(x)\psi(x)|e^-(p,s)\rangle$

$= -i\lambda\int d^4\vec{x}\cdot\langle e^-(p',r)\pi(q)|\phi(x)\bar{\psi}(x)\psi(x)|e^-(p,s)\rangle$

$= -i\lambda\int d^4\vec{x}\cdot\bar{u}(p',r)u(p,s)\cdot e^{-i(\cdots)} = 2\pi^{(4)}\delta^{(4)}(\cdots)\,\bar{u}(p',r)u(p,s)\,\boxed{-i\lambda}$

Since $i\mathcal{M}$ is c-number, we should write the amplitude against the direction of fermion flow

Contributed from vertex

thus the contribution of vertex should be:  $= -i\lambda$. One can check this vertex rule is independent of fermion or anti-fermion, and the chronomic order for line!

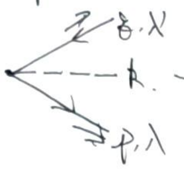
Sometimes Yukawa theory has a more general form $\mathcal{L}_{int} = -\lambda\phi(x)\bar{\psi}(x)\Gamma\psi(x)$, at this time $= -i\lambda\Gamma$

** Rule of fermion sign:

- For fermion, different contraction approach may gives out an extra sign due to Fermion statistics. (or may not)

- If we consider a low-order transaction, that only one diagram is considered in the amplitude $\mathcal{M}$, the fermion sign is no matter. However if there are several diagrams, the relative sign will be important.

- For example, let's consider the procs below:



$$\sim \langle e^-(p,\lambda)\,\pi(k)\,e^+(q,\lambda')\,|\,-i\int d^4\vec{x}\cdot\varkappa\,\phi(x)\,\bar{\psi}(x)\,\psi(x)\,|0\rangle$$

$$= -i\varkappa\int d^4\vec{x}\,\langle e^-(p,\lambda)\,\pi(k)\,e^+(q,\lambda')\,|\,\phi(x)\,\bar{\psi}(x)\,\psi(x)\,|0\rangle$$

- Here $\phi(x)$ will not give out sign, the contraction line is figured below the formula.

Fermion contraction line is cross over, to extract with the bra-state, the field operators should do some essential exchange

$$= -i\varkappa\int d^4\vec{x}\,\langle e^-(p,\lambda)\,\pi(k)\,e^+(q,\lambda')\,|\,(-1)\,\phi(x)\,\psi_a(x)\,\bar{\psi}_a(x)\,|0\rangle$$

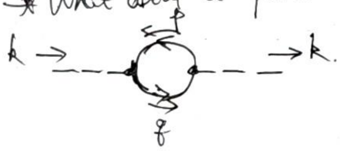
$$= -i\varkappa\cdot(2\pi)^4\delta^{(4)}(\cdots)\cdot v_a(q,\lambda')\,\bar{u}_a(p,\lambda) = (2\pi)^4\delta^{(4)}(\cdots)\boxed{(-1)}\,i\varkappa\,\bar{u}(p,\lambda)\,v(q,\lambda')$$

Come from the fermion sign.

* In fact, if we record the out state as $\langle e^+(q,\lambda')\,\pi(k)\,e^-(p,\lambda)|$, the sign will vanish. For a single diagram, the sign is no matter, so two convention of out state is equivalent.

* If there are more diagrams, we set a uniform convention of input and output state, and only move field operator

* What about a fermion loop? Let consider the procs:



$$\sim \langle\pi(k)\,|\,(-i)^2\int d^4\vec{x}\,d^4\vec{y}\cdot\varkappa^2\,\phi(x)\,\bar{\psi}(x)\,\psi(x)\,\phi(y)\,\bar{\psi}(y)\,\psi(y)\,|\,\pi(k)\rangle$$

$$= (-i\varkappa)^2\int d^4\vec{x}\,d^4\vec{y}\cdot\langle\pi(k)\,|\,\phi(x)\,\bar{\psi}(x)\,\psi(x)\,\bar{\psi}(y)\,\psi(y)\,\phi(y)\,|\,\pi(k)\rangle$$

Notice that we only have the rule of $\psi\bar{\psi}$, which is a matrix of Feynman propagator S_F.

thus $\bar{\psi}_a(x)\,\psi_a(x)\,\bar{\psi}_b(y)\,\psi_b(y) = \psi_a(x)\,\bar{\psi}_b(y)\,\bar{\psi}_a(x)\,\psi_b(y) = -\psi_a(x)\,\bar{\psi}_b(y)\cdot\psi_b(y)\,\bar{\psi}_a(x)$

$$= -S_{F,ab}(x-y)\,S_{F,ba}(y-x) = -\mathrm{tr}(S_F\cdot S_F)$$

In momentum space, there will be:

$$\bar{\psi}(x)\,\psi(x)\,\bar{\psi}(y)\,\psi(y) = -\mathrm{Tr}\left\{\frac{\not{p}-m}{\vec{p}^2-m^2+i\epsilon}\cdot\frac{\not{q}-m}{\vec{q}^2-m^2+i\epsilon}\right\}$$

thus: ① Fermion loop will gives out another (-1) sign

② Propagator in fermion loop should be traced over!

By now, we have got the Feynmann rule for fermion field:

(1) External line: $p,s \longrightarrow\!\!\!\!\!\!\!\rightarrow\ = u(p,s)$, $\longrightarrow\ p,s = \bar u(p,s)$

$p,s \longleftarrow\ = \bar v(p,s)$ $\qquad \longleftarrow\ p,s = v(p,s)$

(2) There may be an extra "-1" in the diagrams.

(3) In Fermion loop, the propagator should be traced over, and always an another "-1"

Homework 3. In this section, we will discuss the socalled Rutherford Scattering. Here the interaction Hamiltonian is $\hat H_I = \int d^3\vec x \cdot e\bar\psi(x)\gamma^\mu\psi(x)A_\mu(x)$. $\psi(x)$ is the usual Dirac field, but $A_\mu(x)$ will be treated as an classical field function.

(a) Show that T-matrix element for electron scattering off a localized classical potential is, to lowest order, $\langle \vec p'|i\hat T|\vec p\rangle = -ie\bar u(p')\gamma^\mu u(p)\tilde A_\mu(\vec p'-\vec p)$. where $\tilde A_\mu$ is momentum-space

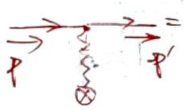
At the lowest order, we have: $\langle \vec p'|i\hat T|\vec p\rangle = \langle \vec p'|-ie\int d^4\vec x\, \hat T\,\bar\psi(\vec x)\gamma^\mu\psi(\vec x)A_\mu(\vec x)|\vec p\rangle$

$$= -ie\int d^4\vec x\cdot A_\mu(\vec x)\langle \vec p'|\hat T\,\bar\psi(x)\gamma^\mu\psi(x)|\vec p\rangle = -ie\int d^4\vec x\cdot A_\mu(\vec x)\cdot\langle \vec p'|\bar\psi(x)\psi(x)|\vec p\rangle$$

$$= -ie\int d^4\vec x\cdot A_\mu(\vec x)\,\bar u(p')\gamma^\mu u(p)\,e^{-i(\vec p-\vec p')\cdot\vec x}$$

$$= -ie\bar u(\vec p')\gamma^\mu u(\vec p)\tilde A_\mu(\vec p'-\vec p)$$

(b) Now suppose $A_\mu(\vec x)$ is time-independent, thus $\tilde A_\mu(\vec p'-\vec p) \supseteq \delta(E_{p'}-E_p)$, thus it's natural to define $\langle \vec p'|i\hat T|\vec p\rangle = iM\cdot 2\pi\delta(E_{p'}-E_p)$, and gives out the Feynman rule for the vertex



$= -ie\gamma^\mu\tilde A_\mu(\vec q)$. At this time, $\tilde A_\mu$ is 3d-Fourier transformation, and $\vec q = \vec p'-\vec p$. Now, show that the cross section for scattering off a time-independent localized potential is $d\sigma = \frac{1}{v_i}\cdot\frac{1}{2E_i}\cdot\frac{d^3\vec p_f}{(2\pi)^3}\frac{1}{2E_f}|M(p_i\to p_f)|^2\cdot(2\pi)\delta(E_f-E_i)$ where v_i is the particle's initial velocity. Integrate over $|p_f|$ to find $\frac{d\sigma}{d\Omega}$.

First we consider the transition rate: $P_{i\to f} = \frac{|\langle f|i\hat T|i\rangle|^2}{\langle i|i\rangle\langle f|f\rangle}$

Notice that $|\langle f|i\hat T|i\rangle|^2 = |(2\pi)\delta(E_f-E_i)\cdot iM|^2 = (2\pi)^2\delta(E_f-E_i)\delta(0)|M|^2$ Time window

suppose we take box-renormalization, we have $\delta(0) = \frac{1}{2\pi}\int dt\, e^{-iEt}\Big|_{E=0} = \frac{\hat T}{2\pi}$

$\therefore |\langle f|i\hat T|i\rangle|^2 = 2\pi\hat T\delta(E_f-E_i)|M|^2$, and we have $\langle i|i\rangle = 2E_i\cdot(2\pi)^3\delta^{(3)}(\vec 0) = 2E_i\boxed{\hat V}$ Volume of box

$\langle f|f\rangle = 2E_f\hat V$.

thus there will be $\dot P_{i\to f} = \frac{P_{i\to f}}{\hat T} = \frac{2\pi\delta(E_f-E_i)|M|^2}{2E_i\cdot 2E_f\cdot\hat V^2}$

~~Next~~. we have $R(i\to\sum f) = \frac{V}{(2\pi)^3}\int d^3p_f \cdot R_{i\to f} = \frac{V}{(2\pi)^3}\int d^3p_f \cdot \frac{2\pi\delta(E_f-E_i)|\mathcal{M}|^2}{2E_i\cdot 2E_f\cdot V^2}$

$$= \frac{1}{V}\cdot\frac{1}{2E_i}\int\frac{d^3p_f}{(2\pi)^3 2E_f} 2\pi\delta(E_f-E_i)|\mathcal{M}|^2$$

thus $\sigma = \frac{R\cdot V}{V_i} = \frac{1}{V_i}\cdot\frac{1}{2E_i}\int\frac{d^3p_f}{(2\pi)^3 2E_f}2\pi\delta(E_f-E_i)|\mathcal{M}|^2$

which means. $d\sigma = \frac{1}{2E_i\cdot V_i}\frac{d^3p_f}{(2\pi)^3 2E_f}2\pi\delta(E_f-E_i)|\mathcal{M}|^2$

If $\mathcal{M}$ only rely on $|p_f|$, then $d^3p_f = p_f^2 dp_f\cdot d\Omega$. and $\frac{d\sigma}{d\Omega} = \int\frac{1}{2E_i\cdot V_i}\cdot\frac{p_f^2 dp_f}{(2\pi)^3\cdot 2E_f}\cdot 2\pi\delta(E_f-E_i)|\mathcal{M}|^2$

notice that $\delta(E_f-E_i) = \frac{E_i\delta(p_f-p_i)}{p_i}$, thus.

$E_f = \sqrt{p_f^2+m^2}$

$$\frac{d\sigma}{d\Omega} = \frac{1}{2E_i V_i}\int\frac{dp_f\, p_f^2}{(2\pi)^3 2E_f}\cdot 2\pi\cdot\frac{E_i}{p_i}\delta(p_f-p_i)\cdot|\mathcal{M}|^2$$

$$= \frac{1}{2E_i V_i}\cdot\frac{p_i^2}{(2\pi)^3\cdot 2E_i}\cdot 2\pi\cdot\frac{E_i}{p_i}|\mathcal{M}(p_i,\theta)|^2$$

$\hookrightarrow p_i/E_i$

$$= \frac{1}{16\pi^2}|\mathcal{M}(p_i,\theta)|^2$$

here amplitude only rely on the mode of input momentum.

(c) Consider the scattering in Coulomb potential, where $A^0 = \frac{Ze}{4\pi r}$. Work in NR limit, there will be $\frac{d\sigma}{d\Omega} = \frac{\alpha^2 Z^2}{4m^2 v^4 \sin^4\frac{\theta}{2}}$, which is so-called Rutherford formula. Firstly derive the formula in relative case, and ~~then~~ take ~~relati~~ NR-limit.

For Coulomb potential. only $A^0 = \frac{Ze}{4\pi r}$, while we can set $A^i = 0$. then.

$$\tilde{A}_0(q) = \int d^3r\, e^{-i\vec{q}\cdot\vec{r}}\frac{Ze}{4\pi r} = \frac{Ze}{4\pi}\int d\Omega\cdot r e^{-i\vec{q}\cdot\vec{r}}dr = \frac{Ze}{2}\int_0^\infty r dr\int_{-1}^1 e^{-iqr\cos\theta}d\cos\theta$$

$$= \frac{Ze}{2}\int_0^\infty r dr\cdot\left(e^{-iqr}-e^{iqr}\right) = \frac{Ze}{q^2}.$$

thus we have $i\mathcal{M} = $  $= \bar{u}(\vec{p}',s')\cdot(-ie\gamma^\mu\tilde{A}_\mu(\vec{p}'-\vec{p}))\cdot u(\vec{p},s)$

$$= -ie\cdot\tilde{A}_\mu(\vec{p}'-\vec{p})\,\bar{u}(\vec{p}',s')\gamma^\mu u(\vec{p},s)$$

$\therefore |\mathcal{M}(p,s\to p',s')|^2 = +e^2\tilde{A}_\mu(\vec{p}'-\vec{p})\tilde{A}_\nu(\vec{p}'-\vec{p})\,\bar{u}(\vec{p}',s')\gamma^\mu u(\vec{p},s)\left(\bar{u}(\vec{p}',s')\gamma^\nu u(\vec{p},s)\right)^*$

and $\sum_{spin}|\mathcal{M}(p,s\to p',s')|^2 = +\frac{e^2}{2}\tilde{A}_\mu(\vec{p}'-\vec{p})\tilde{A}_\nu(\vec{p}'-\vec{p})\sum_{s,s'}\bar{u}(p',s')\gamma^\mu u(p,s)\,\bar{u}(p,s)\gamma^\nu u(p',s')$

$$= +\frac{e^2}{2}\tilde{A}_\mu(\cdots)\tilde{A}_\nu(\cdots)\,\mathrm{Tr}\left\{\gamma^\mu\left(\sum_s u(p,s)\bar{u}(p,s)\right)\cdot\gamma^\nu\cdot\sum_{s'}u(p',s')\bar{u}(p',s')\right\}$$

$$= +\frac{e^2}{2}\tilde{A}_\mu(\cdots)\tilde{A}_\nu(\cdots)\,\mathrm{Tr}\left\{\gamma^\mu(\not{p}+m)\gamma^\nu(\not{p}'+m)\right\}$$

Thus we have: $\frac{1}{2}\sum_{spin}|M|^2 = \frac{+e^2}{2}\tilde{A}_\mu(p'-p)\tilde{A}_\nu(p'-p)\,\mathrm{Tr}\{\gamma^\mu(\not p+m)\gamma^\nu(\not p'+m)\}$.

$$\mathrm{Tr}\{\gamma^\mu(\not p+m)\gamma^\nu(\not p'+m)\} = \mathrm{Tr}\{\gamma^\mu \not p \gamma^\nu \not p'\} + m^2 \underbrace{\mathrm{Tr}\{\gamma^\mu\gamma^\nu\}}_{4g^{\mu\nu}}$$

$$\mathrm{Tr}\{\gamma^\mu \not p \gamma^\nu \not p'\} = \mathrm{Tr}\{\gamma^\mu\gamma^\rho\gamma^\nu\gamma^\sigma\}\, p_\rho p'_\sigma$$

$$= 4(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\nu}g^{\rho\sigma} + g^{\mu\sigma}g^{\rho\nu})\, p_\rho p'_\sigma$$

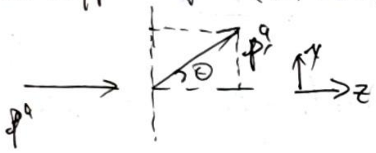
$$= 4(p^\mu p'^\nu - g^{\mu\nu}\vec{p}\cdot\vec{p}' + p'^\mu p^\nu)$$

$$\therefore \frac{1}{2}\sum_{spin}|M|^2 = \frac{+e^2}{2}\tilde{A}_\mu(\cdots)\tilde{A}_\nu(\cdots)\cdot\{4p^\mu p'^\nu - 4g^{\mu\nu}\vec{p}\cdot\vec{p}' + 4p'^\mu p^\nu + 4m^2 g^{\mu\nu}\}$$

$$-4g^{\mu\nu}(\vec{p}\cdot\vec{p}' - m^2)$$

$$= +2e^2\left\{\vec{\tilde{A}}^2(m^2 - \vec{p}\cdot\vec{p}') + 2(\vec{p}\cdot\vec{\tilde{A}})(\vec{p}'\cdot\vec{\tilde{A}})\right\}$$

Now suppose $p^a = (E, 0, 0, vE)$, $p'^a = (E, vE\sin\theta, 0, vE\cos\theta)$



thus $\tilde{A}_0(p'-p) = \dfrac{Ze}{|\vec{p}'-\vec{p}|^2} = \dfrac{Ze}{(vE)^2((1-\cos\theta)^2+\sin^2\theta)}$

$$= \frac{Ze}{(vE)^2(2-2\cos\theta)} = \frac{Ze}{4v^2E^2\sin^2\theta/2}$$

and $\vec{\tilde{A}}^2 = A_0^2 = \dfrac{Z^2e^2}{16v^4E^4\sin^4(\theta/2)}$. $\vec{p}\cdot\vec{p}' = E^2 - v^2E^2\cos\theta = E^2(1-v^2\cos\theta)$

$$\vec{p}\cdot\tilde{A} = \frac{Ze}{4v^2E\sin^2\frac{\theta}{2}} = \vec{p}'\cdot\tilde{A}$$

$$\therefore \frac{1}{2}\sum_{spin}|M|^2 = 2e^2\left\{A_0^2(m^2 - E^2(1-v\cos\theta)) + 2E^2A_0^2\right\}$$

$$= 2e^2A_0^2\{m^2 - E^2(1-v\cos\theta) + 2E^2\} = +2e^2A_0^2(m^2 + E^2(1+v\cos\theta))$$

8×16

$$\text{thus: } \frac{d\sigma}{d\Omega} = \frac{1}{16\pi^2}\cdot\frac{1}{2}\sum_{spin}|M|^2 = +\frac{e^2A_0^2}{8\pi^2}(m^2 + E^2(1+v\cos\theta))$$

$$= +\frac{Z^2e^4}{16\cdot 8\pi^2v^4E^4\sin^4(\theta/2)}\underbrace{(m^2 + E^2(1+v\cos\theta))}_{\parallel}$$

$$= +\frac{Z^2e^4\cdot 2E^2(1-v^2\sin^2\frac{\theta}{2})}{16\times 8\pi^2v^4E^4\sin^4\theta/2}$$

$$(E^2 - p^2 + E^2 + E^2v^2\cos\theta)$$
$$= E^2(2 - v^2 + v^2\cos\theta)$$
$$= E^2(2 - 2v^2\sin^2\tfrac{\theta}{2}) = 2E^2(1 - v^2\sin^2\tfrac{\theta}{2})$$

$$= \frac{Z^2e^4(1-v^2\sin^2\frac{\theta}{2})}{64\pi^2v^4E^2\sin^4\frac{\theta}{2}}$$

$$\alpha = \frac{e^2}{4\pi}$$

$$= \frac{Z^2\alpha^2(1-v^2\sin^2\frac{\theta}{2})}{4v^4E^2\sin^4\frac{\theta}{2}}$$

In NR-limit, $v\to 0$, $E\to m$, then

$$\left.\frac{d\sigma}{d\Omega}\right|_{NR} = \frac{Z^2\alpha^2}{4m^2v^4\sin^4\frac{\theta}{2}}$$

§7 Discussion of Correlation Function

* In former section, we are all discussed the computation of S-matrix. In this section, we will consider the computation of Correlation function, which can also be translated in Feynman diagram.
* The main difference of two-point correlation function between free and interaction theory will be the ground state. This difference will give out a so-called Kallen spectrum function.
* Since Correlation Function and S-matrix can all be represented in Feynman Diagram, there will be some relation connecting them. That is so-called LSZ-formula.

1. Feynman Diagram for correlation function.

In interaction field theory, the worthy attentioned correlation function is $\langle\Omega|\hat{T}\{\hat{\phi}_H(x)\hat{\phi}_H(y)\}|\Omega\rangle$

Interaction ground state　　Heisenberg picture.

Since $e^{-i\hat{H}T}\underbrace{|0\rangle}_{\text{Vacuum state.}} = \sum_n e^{-i\hat{H}T}|n\rangle\langle n|0\rangle$

$$= e^{-iE_0T}|\Omega\rangle\langle\Omega|0\rangle + \sum_{n\neq 0} e^{-iE_nT}|n\rangle\langle n|0\rangle$$

Let's take $T\to\infty(1-i\epsilon)$ limits, then we will have $e^{-i\hat{H}T}|0\rangle = e^{-iE_0T}|\Omega\rangle\langle\Omega|0\rangle$, $T\to\infty(1-i\epsilon)$

$$e^{-iE_n\infty(1-i\epsilon)} = e^{-iE_n\infty}\underbrace{e^{-E_n\infty}}_{\to 0}$$

$$|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \frac{e^{-i\hat{H}T}|0\rangle}{e^{-iE_0T}\langle\Omega|0\rangle}$$

Since T is large enough, it is okay to set:

$$|\Omega(t_0)\rangle = \lim_{T\to\infty(1-i\epsilon)} \left(e^{-iE_0(T+t_0)}\langle\Omega|0\rangle\right)^{-1} e^{-i\hat{H}(T+t_0)}|0\rangle$$

$$e^{-i\hat{H}(T+t_0)} = \boxed{e^{-i\hat{H}(t_0-(-T))}\cdot e^{-iH_0(-T-t_0)}} = \hat{O}_I(t_0,-T)$$

$$\hookrightarrow e^{-iH_0(-T-t_0)}|0\rangle = |0\rangle$$

$$= \lim_{T\to\infty(1-i\epsilon)} \left(e^{-iE_0(T+t_0)}\langle\Omega|0\rangle\right)^{-1} \hat{O}_I(t_0,-T)|0\rangle$$

$$|\Omega(t_0)\rangle \propto \hat{O}_I(t_0,-T)|0\rangle$$

$$= \lim_{T\to\infty} \left(e^{-iE_0(t_0-(-T))}\langle\Omega|0\rangle\right)^{-1}\hat{O}_I(t_0,-T)|0\rangle$$

$$|0\rangle_{-\infty} \xrightarrow{\hat{O}_I} |\Omega(t_0)\rangle_{t_0}$$

A usual time evolutionary process.

Similarly, since $\langle 0|e^{-i\hat{H}T} = \langle 0|\Omega\rangle\langle\Omega|e^{-iE_0T} + \cdots$

there will be $\langle\Omega| = \left(e^{-iE_0T}\langle 0|\Omega\rangle\right)^{-1}\langle 0|e^{-i\hat{H}T}$

$$= \left(e^{-iE_0(T-t_0)}\langle 0|\Omega\rangle\right)^{-1}\langle 0|\underbrace{e^{-iH(T-t_0)}}_{e^{-iH_0(T-t_0)}}$$

$$= \lim_{T\to\infty(1-i\epsilon)} \left(e^{-iE_0(T-t_0)}\langle 0|\Omega\rangle\right)^{-1}\langle 0|\hat{O}_I(T,t_0)$$

since we have $|\Omega(t_0)\rangle = \lim_{T\to\infty(1-i\epsilon)} \left(e^{-iE_0(T+t_0)} \langle\Omega|0\rangle\right)^{-1} \hat{U}(t_0,-T)|0\rangle$

$$\langle\Omega(t_0)| = \lim_{T\to\infty(1-i\epsilon)} \left(e^{-iE_0(T-t_0)} \langle 0|\Omega\rangle\right)^{-1} \langle 0|\hat{U}(T,t_0)$$

thus we have:

$$\langle\Omega|\hat{T}\{\hat\phi_H(x)\hat\phi_H(y)\}|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|\hat{U}(T,t_0)\hat{T}\{\hat\phi_H(x)\hat\phi_H(y)\}\hat{U}(t_0,-T)|0\rangle}{e^{-2iE_0T}|\langle 0|\Omega\rangle|^2}$$

$\hat\phi_H(y) \to \hat{U}^\dagger(y^0,t_0)\hat\phi_I(y)\hat{U}(y^0,t_0)$

$\hat\phi_H(x) = \hat{U}^\dagger(x^0,t_0)\hat\phi_I(x)\hat{U}(x^0,t_0)$

$$= \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|\hat{T}\{\hat{U}(T,x^0)\hat\phi(\vec{x})\hat{U}(x^0,y^0)\hat\phi(y)\hat{U}(y^0,-T)\}|0\rangle}{e^{-2iE_0T}|\langle 0|\Omega\rangle|^2}$$

$$= \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|\hat{T}\{\phi(\vec{x})\phi(\vec{y})\hat{U}(T,-T)\}|0\rangle}{e^{-2iE_0T}|\langle 0|\Omega\rangle|^2}$$

Since $\langle\Omega|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \left(e^{-2iE_0T}\langle 0|\Omega\rangle\right)^{-1}\cdot\langle 0|\hat{U}(-T,T)|0\rangle \overset{!}{=} 1$

$$= \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|\hat{T}\{\phi(\vec{x})\phi(\vec{y})\hat{U}(T,-T)\}|0\rangle}{\langle 0|\hat{U}(T,-T)|0\rangle} \qquad e^{-2iE_0T}|\langle 0|\Omega\rangle|^2 \overset{!}{=} \langle 0|\hat{U}(-T,T)|0\rangle$$

where is $\hat{U}(T,-T)$? We should have: $\hat{U}(T,-T) = \exp i\left(-\int_{-T}^{T} dt\cdot \hat{H}_I(t)\right)$

thus. $$\langle\Omega|\hat{T}\{\phi_H(\vec{x})\phi_H(\vec{y})\}|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|\hat{T}\,\hat\phi(\vec{x})\hat\phi(\vec{y})\,e^{-i\int_{-T}^{T}dt\,\hat{H}_I(t)}|0\rangle}{\langle 0|e^{-i\int_{-T}^{T}\hat{H}_I(t)\,dt}|0\rangle}$$

This is so-called Gell-Mann-Low formula. $\leftarrow$ $$= \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|\hat{T}\{\phi(\vec{x})\phi(\vec{y})\,S\}|0\rangle}{\langle 0|\hat{S}|0\rangle}$$

(1) Numerator: Since $\hat{S} = 1 - \underbrace{i\int d^4\vec{x}\cdot\hat{T}(\mathcal{H}_{int}(\vec{x}))}_{S^{(1)}} + \underbrace{\frac{(-i)^2}{2!}\int d^4\vec{x}\,d^4\vec{y}\cdot\hat{T}(\mathcal{H}_{int}(\vec{x})\mathcal{H}_{int}(\vec{y}))}_{S^{(2)}}\}$

thus in a specific order, there will be:

$$\langle 0|\hat{T}\,\phi(\vec{x})\phi(\vec{y})\,S^{(n)}|0\rangle = \frac{(-i)^n}{n!}\int d^4\vec{x}_1\cdots d^4\vec{x}_n\cdot\langle 0|\hat{T}\{\phi(x)\phi(y)\cdot\phi\phi\cdots\phi\}|0\rangle$$

$$\to = \frac{(-i)^n}{n!}\int d^4\vec{x}_1\cdots d^4\vec{x}_n\cdot(\text{Sum over of all contraction}).$$

Results of Wick theorem.

e.g. For ϕ^4 theory. $\mathcal{H}_{int} = \frac{i\lambda}{4!}\phi^4(\vec{x})$, the correlation function in the first order of λ.

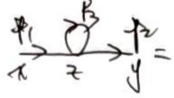
$$\langle 0|\hat{T}\,\phi(x)\phi(y)\cdot\left(\frac{-i\lambda}{4!}\right)\int d^4z\cdot\phi^4(z)|0\rangle = \frac{-i\lambda}{4!}\int d^4z\cdot\langle 0|\hat{T}\,\phi_x\phi_y\phi_z\phi_z\phi_z\phi_z|0\rangle$$

$$= \frac{-i\lambda}{4!}\int d^4z\left\{3\langle 0|\phi_x\phi_y\phi_z\phi_z\phi_z\phi_z|0\rangle + 12\langle 0|\phi_x\phi_y\phi_z\phi_z\phi_z\phi_z|0\rangle\right\}$$

$= -i\lambda \int d^4z \cdot \left\{ \frac{1}{8}\langle 0|\phi_x\phi_y\phi_z\phi_z\phi_z\phi_z|0\rangle + \frac{1}{2}\langle 0|\phi_x\phi_y\phi_z\phi_z\phi_z\phi_z|0\rangle \right.$

In the same rule, we can use Feynman diagram to represent different contraction.

Symmetry factor

Since  $= -i\lambda\int d^4z \cdot D(x-z)D(z-y)D(z-z)$

$$= -i\lambda\int d^4z \cdot \frac{d^4\vec{p}_1\, d^4\vec{p}_2\, d^4\vec{p}_3}{(2\pi)^4(2\pi)^4(2\pi)^4}\frac{i}{\vec{p}_1^{\,2}-m^2+i\epsilon}e^{-i\vec{p}_1\cdot(x-z)}\frac{i}{\vec{p}_3^{\,2}-m^2+i\epsilon}e^{-i\vec{p}_3\cdot(z-y)}\frac{i}{\vec{p}_2^{\,2}-m^2+i\epsilon}$$

$e^{-i\vec{p}_1\cdot\vec{x}}\cdot e^{i\vec{p}_1\cdot\vec{z}}$ $\quad e^{-i\vec{p}_3\cdot\vec{z}}\, e^{i\vec{p}_3\cdot\vec{y}}$ $\quad \int d^4\vec{z} \rightarrow (2\pi)^4\delta(\vec{p}_1-\vec{p}_3)$

$$= -i\lambda\int\frac{d^4\vec{p}_1\, d^4\vec{p}_2}{(2\pi)^4(2\pi)^4}\cdot D(\vec{p}_1)D(\vec{p}_1)D(\vec{p}_2)\cdot e^{-i\vec{p}_1\cdot(\vec{x}-\vec{y})}$$

$$= -i\lambda\int\frac{d^4\vec{p}}{(2\pi)^4}e^{-i\vec{p}\cdot(\vec{x}-\vec{y})}\boxed{\left(\int\frac{d^4\vec{k}}{(2\pi)^4}D(\vec{k})\right)D(\vec{p})D(\vec{p})}$$

$\| \;\int d^4(\vec{x}-\vec{y})\cdot e^{-i\vec{p}\cdot(\vec{x}-\vec{y})} \times\langle 0|\phi(x)\phi(y)\hat{S}^{(1)}|0\rangle$

or $\langle 0|\phi(x)\phi(y)\hat{S}^{(1)}|0\rangle$ in momentum space.

thus we can get the Feynman rule for correlation function:

① There are so-called external point (instead of external line)

② Inner line will support Feynman propagator

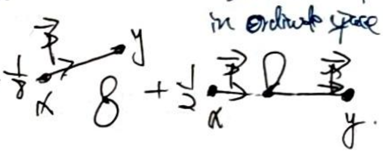
③ Vertex will gives out $(-i\lambda)$ in ϕ^4 theory

④ At each vertex, 4-momentum is conserved.

⑤ Undetermined momentum should be integrated.

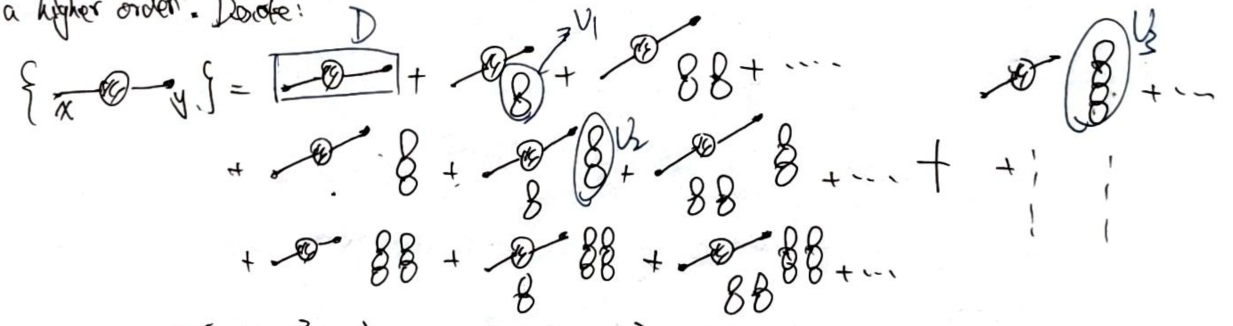
⑥ Divide the symmetry factor.

⑦ $\xrightarrow{\;\vec{p}\;}\!\bullet_x = e^{i\vec{p}\cdot\vec{x}}$. $\quad {}_x\!\bullet\!\xrightarrow{\;\vec{p}\;} = e^{-i\vec{p}\cdot\vec{x}}$. This rule will be used to find in ordinate space.

Then we will have: $\int d^4\vec{x}\, e^{-i\vec{p}\cdot(\vec{x}-\vec{y})}\langle 0|\phi(x)\phi(y)\hat{S}^{(1)}|0\rangle = \frac{1}{8}$ 

(2) Bubbles and denominator.

For each diagram, we can induce the bubbles diagram, which contributes to $\langle 0|\phi(x)\phi(y)\hat{S}|0\rangle$ in a higher order. Denote:



$$= D(1+V_1+V_1^2+\cdots) + DV_3(1+V_1+V_1^2+\cdots)$$
$$+DV_2(1+V_1+V_2^2+\cdots) + DV_3V_2(1+V_1+\cdots)\ \cdots \quad = De^{V_1}e^{V_2}e^{V_3}\cdots$$
$$+DV_2^2(1+V_1+V_2^2+\cdots) + DV_3V_2^2(\cdots)$$
$$+\cdots \qquad\qquad \cancel{+}+\cdots$$

However, notice that $\langle 0|\hat{S}|0\rangle$ = Sum over of all bubbles $= e^{V_1+V_2+\cdots}$

$$\therefore \frac{\langle 0|\phi(x_1)\phi(x_2)\hat{S}|0\rangle}{\langle 0|\hat{S}|0\rangle} = \sum_D \frac{D e^{V_1}\cdots e^{V_2}}{e^{V_1}\cdots e^{V_2}} = \sum \boxed{D}$$

Sum over of non-bubble diagrams!

* We can think that, the only effect of bubbles is to let $|0\rangle \to |\Omega\rangle$

2. Spectrum representation of two-point Correlation Function

* For free field theory: $\langle 0|\hat{T}\phi(x)\phi(y)|0\rangle = \int \frac{d^4\vec{p}}{(2\pi)^4}\cdot\frac{i}{\vec{p}^2-m^2+i\varepsilon}e^{-i\vec{p}\cdot(\vec{x}-\vec{y})} = D_F(x-y)$

What about interaction theory?

* Introduce the complete relation: $\mathbb{I} = |\Omega\rangle\langle\Omega| + \sum_{\boxed{\lambda}}\int\frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{2E_{\vec{p}}(\lambda)}|\lambda_{\vec{p}}\rangle\langle\lambda_{\vec{p}}|$

Total momentum

Particle combination

then. $\langle\Omega|\hat{T}\phi_H(x)\phi_H(y)|\Omega\rangle = \hat{T}\underbrace{\langle\Omega|\phi_H(x)|\Omega\rangle\langle\Omega|\phi_H(y)|\Omega\rangle}_{\text{Usually zero}} + \sum_\lambda\int\frac{d^3\vec{p}}{(2\pi)^3}\cdot\frac{1}{2E_{\vec{p}}(\lambda)}\hat{T}\underline{\langle\Omega|\phi_H(x)|\lambda_{\vec{p}}\rangle}\langle\lambda_{\vec{p}}|\phi_H(y)|\Omega\rangle$

$$= \sum_\lambda\int\frac{d^3\vec{p}}{(2\pi)^3}\cdot\frac{1}{2E_{\vec{p}}(\lambda)}|\langle\Omega|\phi_H(0)|\lambda_0\rangle|^2\cdot \underset{\hat{T}}{e}^{-i\vec{p}\cdot(\vec{x}-\vec{y})}\Big|_{p^0=E_{\vec{p}}}$$

$$\langle\Omega|\underbrace{e^{-i\vec{p}\cdot\vec{x}}}_{=1}\phi(0)e^{-i\vec{p}\cdot\vec{x}}|\lambda_{\vec{p}}\rangle = \langle\Omega|\phi(0)|\lambda_{\vec{p}}\rangle e^{-i\vec{p}\cdot\vec{x}}\Big|_{p^0=E_{\vec{p}}}$$

$$= \langle\Omega|\Lambda^{-1}\Lambda\phi(0)\Lambda^{-1}\Lambda|\lambda_{\vec{p}}\rangle$$

A boost to let $\vec{p}$ to $\vec{0}$

$$= \langle\Omega|\phi(0)|\lambda_0\rangle e^{-i\vec{p}\cdot\vec{x}}\Big|_{p^0=E_{\vec{p}}}$$

$$= \sum_\lambda\int\frac{d^4\vec{p}}{(2\pi)^4}\cdot|\langle\Omega|\phi_H(0)|\lambda_0\rangle|^2\frac{e^{-i\vec{p}\cdot(\vec{x}-\vec{y})}}{\vec{p}^2-m_\lambda^2+i\varepsilon}$$

Total mass for particles "λ"

$$= \sum_\lambda|\langle\Omega|\phi_H(0)|\lambda_0\rangle|^2\cdot D_F(\vec{x}-\vec{y};m_\lambda)$$

We can introduce a continuous function $\rho(m^2) = \sum_\lambda(2\pi)\delta(m^2-m_\lambda^2)\cdot|\langle\Omega|\phi(0)|\lambda_0\rangle|^2$.

then: $\langle\Omega|\hat{T}\hat{\phi}(x)\hat{\phi}(y)|\Omega\rangle = \int_0^\infty\frac{dm^2}{2\pi}\cdot\rho(m^2)\cdot D_F(\vec{x}-\vec{y};m^2)$

• Immediately, for free theory. $\underset{\downarrow}{\rho(m^2)} = \delta(m^2-m^2)$. then $\langle 0|\hat{T}\phi_H(x)\phi_H(y)|0\rangle = D_F(\vec{x}-\vec{y},m)$

The probability to found the components in $\phi(x)|0\rangle$ when invariant mass is M.

• The specific form of $\rho(m^2)$ is rely on the interaction form. Generally, it takes the distribution.

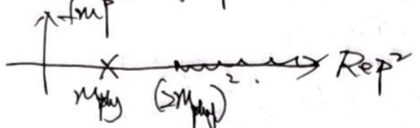
$m_{phy} \neq m$ (Lagrangian).

m_{phy} is the mass measured in experiments!

* In momentum space:

$$\int d^4x \, e^{i\vec{p}\cdot\vec{x}} \langle \Omega | T \phi_H(x) \phi_H(y) | \Omega \rangle = \int_0^{\infty} \frac{d\mu^2}{2\pi} \rho(\mu^2) \cdot \frac{i}{\vec{p}^2 - \mu^2 + i\varepsilon} \equiv G^{(2)}(\vec{p})$$

According to the distribution of $\rho(\mu^2)$, thus $G^{(2)}(\vec{p})$ may have the analytical structure.



thus generally, we can denote $\rho(\mu^2) = Z\delta(\mu^2 - m_{phy}^2) + \sigma(\mu^2)$, then

$$G^{(2)}(\vec{p}) = \frac{iZ}{\vec{p}^2 - m_{phy}^2 + i\varepsilon} + \int_{(2m_{phy})^2}^{\infty} \frac{d\mu^2}{2\pi} \frac{i\sigma(\mu^2)}{\vec{p}^2 - \mu^2 + i\varepsilon} \quad (*)$$

∘ What is factor Z? Since $\rho(\mu^2) = \sum_\lambda (2\pi)\delta(\mu^2 - m_\lambda^2) \left| \langle \Omega | \phi_H(0) | \lambda_0 \rangle \right|^2$.

if we take $\mu \to m_{phy}$, then $\rho(m_{phy}^2) = 2\pi \left| \langle \Omega | \phi_H(0) | \vec{p} = 0, m_{phy} \rangle \right|^2$ ~~$\delta(\mu^2)$~~

thus it can be argued that $\sqrt{Z} = \langle \Omega | \phi(0) | \vec{p} = 0, m_{phy} \rangle = \langle \Omega | \phi_H(0) | \vec{p}, m_{phy} \rangle$

* Factor Z is so-called field-strength renormalization factor.

∘ It can be proved that $Z \leq 1$

* Now let's turn back to perturbation theory. Introduce so called one-particle irreducible diagram (1PI), then define:

$-i\Sigma(\vec{p}) = $ —(1PI)— $= \sum$ of all irreducible diagrams.

Thus $G^{(2)}(\vec{p}) = \xrightarrow{\vec{p}}\bullet + \bullet\xrightarrow{\vec{p}}(1PI)\xrightarrow{\vec{p}}\bullet + \bullet\xrightarrow{\vec{p}}(1PI)\xrightarrow{\vec{p}}(1PI)\xrightarrow{\vec{p}}\bullet + \cdots$

$$= D_F(\vec{p}; m) + D_F(\vec{p}; m)\left(-i\Sigma(\vec{p}^2)\right) D_F(\vec{p}; m) + \cdots$$

$$= \frac{i}{\vec{p}^2 - m^2 - \Sigma(\vec{p}^2)}$$

Notice that according to $(*)$, there should be a pole of $\vec{p}^2 = m_{phy}^2$ when $\vec{p}^2 = m_{phy}^2$

which means $\left. \vec{p}^2 - m^2 - \Sigma(\vec{p}^2) \right|_{\vec{p}^2 = m_{phy}^2} \overset{!}{=} 0.$

Thus we can get the physical mass is $m_{phy}^2 = m^2 + \Sigma(m_{phy}^2)$

∘ Near the pole ($\vec{p}^2 \to m_{phy}^2$), we have $\Sigma(\vec{p}^2) = \Sigma(m_{phy}^2) + \left. \frac{d\Sigma(\vec{p}^2)}{d\vec{p}^2} \right|_{\vec{p}^2 = m_{phy}^2} (\vec{p}^2 - m_{phy}^2)$

$$\therefore \vec{p}^2 - m^2 - \Sigma(\vec{p}^2) \approx \vec{p}^2 - \underbrace{m^2 - \Sigma(m_{phy}^2)}_{m_{phy}^2} + \left. \frac{d\Sigma(\vec{p}^2)}{d\vec{p}^2} \right|_{\vec{p}^2 = m_{phy}^2} (\vec{p}^2 - m_{phy}^2)$$

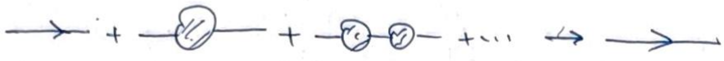
$$= (\vec{p}^2 - m_{phy}^2) \cdot \left(1 + \left. \frac{d\Sigma(\vec{p}^2)}{d\vec{p}^2} \right|_{\vec{p}^2 = m_{phy}^2} \right)$$

And we have: $G^{(2)}(\vec{p})\Big|_{\vec{p}^2\to m_{phy}^2} = \frac{iZ}{\vec{p}^2 - m_{phy}^2 + i\epsilon} \stackrel{!}{=} \frac{i}{\vec{p}^2 - m^2 + \Sigma(\vec{p}^2)}$

while $\vec{p}^2 - m^2 + \Sigma(\vec{p}^2)\Big|_{\vec{p}^2\to m_{phy}^2} = (\vec{p}^2 - m_{phy}^2)\left(1 + \frac{d\Sigma(\vec{p}^2)}{d\vec{p}^2}\Big|_{\vec{p}^2 = m_{phy}^2}\right)$

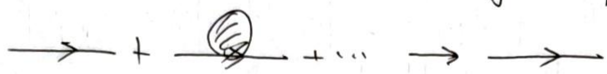
thus we can get the factor: $Z = \left(1 + \frac{d\Sigma(\vec{p}^2)}{d\vec{p}^2}\Big|_{\vec{p}^2 = m_{phy}^2}\right)^{-1}$

This is mean we can let



for substitute. $m \to m_{phy}$, and introduce $Z = \left(1 + \frac{d\Sigma(\vec{p}^2)}{d\vec{p}^2}\Big|_{\vec{p}^2 = m_{phy}^2}\right)^{-1}$

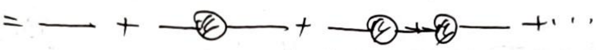
** We can see that, what we have done can be changed into Feynman diagram in S-matrix. where



can be regarded as the amputation process! Thus the factor Z and the value of $\boxed{\Sigma}$ will become the essential correction for amputation

Can be computed perturbationly.

** For Dirac theory, there are a same process to deal with  and here,

$-i\Sigma(\not{p}) =$ 

there will be $m_{phy} = m + \Sigma(m_{phy})$, $Z^{-1} = 1 - \frac{d\Sigma(\not{p})}{d\not{p}}\Big|_{\not{p} = m_{phy}}$

3. LSZ-formula

The LSZ formula gives out how to get S-matrix from $(n+m)$ point Correlation function.

* Consider a $m \to n$ process, there will be: (Take ϕ^4-theory as example)

$$\prod_{i=1}^{n}\int d^4x_i\, e^{i\vec{p}_i\cdot\vec{x}_i}\cdot\prod_{j=1}^{m}\int d^4y_j\, e^{-i\vec{k}_j\cdot\vec{x}_j}\,\langle\Omega|T\hat{\phi}_H(x_1)\cdots\hat{\phi}_H(x_n)\hat{\phi}_H(y_1)\cdots\hat{\phi}_H(y_m)|\Omega\rangle\Big|_{\substack{\vec{p}_i^2 = m_i \\ \vec{k}_j^2 = m_j}}$$

$$= \prod_{i=1}^{n}\frac{i\sqrt{Z}}{\vec{p}_i^2 - m_{i,phy}^2 + i\epsilon}\prod_{j=1}^{m}\frac{i\sqrt{Z}}{\vec{k}_j^2 - m_{j,phy}^2 + i\delta}\cdot\langle\vec{p}_1\cdots\vec{p}_n|i\hat{T}|\vec{k}_1\cdots\vec{k}_m\rangle$$

* From LSZ, we can find the so-called crossing symmetry... that is

$\langle\phi(p)\cdots|i\hat{T}|\alpha\rangle = \langle\cdots|i\hat{T}|\phi^*(-p)\alpha\rangle$ ← For any field theory.

e.g. in QED, there will be

QED and Renormalization

* In this section we will focus on the QED theory, which is come from the U(1) gauge symmetry, that $\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F^{\mu\nu}F_{\mu\nu}$; and $D_\mu = \partial_\mu + ieA_\mu$, and thus.

$$\mathcal{L}_{QED} = \bar{\psi}(i\not{\partial} - m)\psi - \frac{1}{4}F^{\mu\nu}F_{\mu\nu} - e\bar{\psi}\not{A}\psi, \text{ and } \mathcal{L}_{int} = -e\bar{\psi}\gamma^\mu\psi A_\mu.$$

* In some case, the process may refer to two kinds of spin-1/2 particles, for instance, $e^+e^- \to \mu^+\mu^-$, which become the most basic process and we will discuss. In here, we can suppose our $\mathcal{L}_{int}$ as.

$$\mathcal{L}_{int} = -e\bar{\psi}_e\gamma^\mu\psi A_\mu - e\bar{\psi}_{(\mu)}\gamma^\mu\psi_{(\mu)}A_\mu.$$

* Except the basic tree diagram, we have found that the radiation diagram will give out the so-called UV divergence. We will discuss three classical types of loop-diagram in QED, and analyse how to deal with the infinitiveness problem.

§1 $e^+e^- \to \mu^+\mu^-$

will be contracted with photon line.

The basic vertex rule in QED theory is  $= -ie\gamma^\mu$, thus we can give out the lowest order of this diagram:

$$= i\mathcal{M} = \bar{v}(p',s')(-ie\gamma^\mu)u(p,s) \cdot \frac{-ig^{\mu\nu}}{q^2+i\varepsilon} \cdot \bar{u}(k,r)\underbrace{\gamma^\nu}_{(-ie)} \cdot v(k',r')$$

$$= ie^2 \cdot \frac{g^{\mu\nu}}{q^2+i\varepsilon}\,\bar{v}(p',s')\gamma^\mu u(p,s)\bar{u}(k,r)\gamma^\nu v(k',r')$$

$$\therefore |\mathcal{M}|^2 = e^2 \cdot \frac{g^{\mu\nu}}{q^2+i\varepsilon} \cdot \frac{g^{\rho\sigma}}{q^2+i\varepsilon}\left(\bar{v}(p',s')\gamma^\mu u(p,s)\right)\left(\bar{u}(k,r)\gamma^\nu v(k',r')\right) \cdot \left(\bar{u}(p,s)\gamma^\rho v(p',s')\right)\left(\bar{v}(k',r')\gamma^\sigma u(k,r)\right)$$

$$= e^2\frac{g^{\mu\nu}}{q^2+i\varepsilon} \cdot \frac{g^{\rho\sigma}}{q^2+i\varepsilon}\mathrm{Tr}\left\{\gamma^\mu u(p,s)\bar{u}(p,s)\gamma^\rho v(p',s')\bar{v}(p',s')\right\}$$

$$\cdot \mathrm{Tr}\left\{\gamma^\nu v(k',r')\bar{v}(k',r')\gamma^\sigma u(k,r)\bar{u}(k,r)\right\}$$

Now do the average for initial state, and sum over for final state, there will be:

$$\frac{1}{4}\sum_{s,s'}\sum_{r,r'}|\langle\mathcal{M}\rangle|^2 = e^2\frac{g^{\mu\nu}}{q^2+i\varepsilon}\frac{g^{\rho\sigma}}{q^2+i\varepsilon}\mathrm{Tr}\left\{\gamma^\mu(\not{p}+m_e)\gamma^\rho(\not{p}'-m_e)\right\} \cdot \mathrm{Tr}\left\{\gamma^\nu(\not{k}'-m_\mu)\gamma^\sigma(\not{k}+m_\mu)\right\}$$

Notice that: $\mathrm{Tr}\left\{\gamma^\mu(\not{p}+m_e)\gamma^\rho(\not{p}'-m_e)\right\} = \mathrm{Tr}\left\{\gamma^\mu\not{p}\gamma^\rho\not{p}'\right\} - m_e^2\,\mathrm{Tr}\left\{\gamma^\mu\gamma^\rho\right\}$

$$= 4\left(p^\mu p'^\rho - g^{\mu\rho}\vec{p}\cdot\vec{p}' + p'^\mu p^\rho\right) - 4m_e^2 g^{\mu\rho}$$

$$= 4\left\{p^\mu p'^\rho + p'^\mu p^\rho - (\vec{p}\cdot\vec{p}' + m_e^2)g^{\mu\rho}\right\}$$

Similarly: $\mathrm{Tr}\{\gamma^\nu(\not k'-m_\mu)\gamma^\sigma(\not k+m_\mu)\} = \mathrm{Tr}\{\gamma^\nu \not k' \gamma^\sigma \not k\} - m_\mu^2\,\mathrm{Tr}\{\gamma^\nu\gamma^\sigma\}$

$$= 4\left(k'^\nu k^\sigma - g^{\nu\sigma}\vec{k'}\cdot\vec{k} + k^\nu k'^\sigma\right) - 4m_\mu^2 g^{\nu\sigma}$$

$$= 4\left(k'^\nu k^\sigma + k^\nu k'^\sigma - (\vec{k'}\cdot\vec{k} + m_\mu^2)g^{\nu\sigma}\right)$$

$$\therefore \frac{1}{4}\sum|M|^2 = 4e^4\frac{g_{\mu\nu}g_{\rho\sigma}}{(q^2+i\varepsilon)(q^2+i\varepsilon)}\left(p^\mu p'^\rho + p'^\mu p^\rho - (\vec{p}\cdot\vec{p'}+m_e^2)g^{\mu\rho}\right)\left(k'^\nu k^\sigma + k^\nu k'^\sigma - (\vec{k'}\cdot\vec{k}+m_\mu^2)g^{\nu\sigma}\right).$$

$$= \frac{4e^4}{q^4}\left(p^\mu p'^\rho + p'^\mu p^\rho - (\vec{p}\cdot\vec{p'}+m_e^2)g^{\mu\rho}\right)\left(k'_\mu k_\rho + k_\mu k'_\rho - (\vec{k'}\cdot\vec{k}+m_\mu^2)g_{\mu\rho}\right)$$

$$= \frac{4e^4}{q^4}\cdot\Big\{2(\vec{p}\cdot\vec{k'})(\vec{p'}\cdot\vec{k}) + 4(\vec{p}\cdot\vec{p'}+m_e^2)(\vec{k'}\cdot\vec{k}+m_\mu^2)$$
$$+ 2(\vec{p}\cdot\vec{k})(\vec{p'}\cdot\vec{k'}) - 2(\vec{k'}\cdot\vec{k}+m_\mu^2)(\vec{p}\cdot\vec{p'}) - 2(\vec{p}\cdot\vec{p'}+m_e^2)(\vec{k}\cdot\vec{k'})\Big\}$$

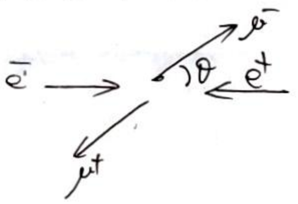
$$= \frac{8e^4}{q^4}\Big\{(\vec{p}\cdot\vec{k'})(\vec{p'}\cdot\vec{k}) + (\vec{p}\cdot\vec{k})(\vec{p'}\cdot\vec{k'}) + m_e^2(\vec{k}\cdot\vec{k'}) + m_\mu^2(\vec{p}\cdot\vec{p'}) + 2m_e^2m_\mu^2\Big\}.$$

For simplicity, since $m_e \ll m_\mu$, we can set $m_e=0$, then

$$\frac{1}{4}\sum_{\text{spin}}|M|^2 = \frac{8e^2}{q^4}\Big\{(\vec{p}\cdot\vec{k'})(\vec{p'}\cdot\vec{k}) + (\vec{p}\cdot\vec{k})(\vec{p'}\cdot\vec{k'}) + m_\mu^2(\vec{p}\cdot\vec{p'})\Big\}.$$

And now, let's consider in the CM-frame. then. and $E^2(1-\beta^2)=m_\mu^2$

Only satisfied in $m_e=0$ limits



and: $p^a=(E,0,0,E)=E(1,0,0,1)$

$p'^a=(E,0,0,-E)=E(1,0,0,-1)$

$k^a=(E,\beta E\sin\theta,0,\beta E\cos\theta)=E(1,\beta\sin\theta,0,\beta\cos\theta)$

$k'=E(1,-\beta\sin\theta,0,\beta\cos\theta)$

$\therefore\ \vec{p}\cdot\vec{k'}=E^2(1-\beta\cos\theta)$, $\vec{p'}\cdot\vec{k}=E^2(1+\beta\cos\theta)$, $\vec{p}\cdot\vec{p'}=2E^2$

$\vec{p}\cdot\vec{k}=E^2(1-\beta\cos\theta)$, $\vec{p'}\cdot\vec{k'}=E^2(1+\beta\cos\theta)$, $q=\vec{p}+\vec{p'}=2E$, $q^4=16E^4$

$$\therefore \frac{1}{4}\sum_{\text{spin}}|M|^2 = \frac{8e^4}{16E^4}\cdot E^4\left\{2(1-\beta\cos\theta)(1+\beta\cos\theta) + \frac{2m_\mu^2}{E^2}\right\}$$

$$= e^4\left\{(1-\beta^2\cos^2\theta)+\frac{m_\mu^2}{E^2}\right\} = e^4\left\{(1-\beta^2\cos^2\theta)+(1-\beta^2)\right\}$$

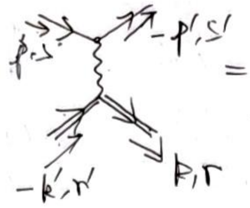
According to the $\left.\frac{d\sigma}{d\Omega}\right|_{CM} = \frac{1}{64\pi^2}\cdot\frac{1}{E_A E_B|v_A-v_B|}\cdot\frac{k}{E_{CM}}|M_{fi}|^2$

there will be: $\left.\frac{d\sigma}{d\Omega}\right|_{CM} = \frac{1}{64\pi^2}\cdot\frac{1}{\frac{E_{CM}^2}{4}\cdot 2}\cdot\frac{\beta E_{CM}/2}{E_{CM}}\,e^4(2-\beta^2-\beta^2\cos^2\theta)$

$$= \frac{e^4}{64\pi^2E_{CM}^2}\cdot(2-\beta^2-\beta^2\cos^2\theta) = \frac{\alpha^2}{4E_{CM}^2}\,\beta(2-\beta^2-\beta^2\cos^2\theta)$$

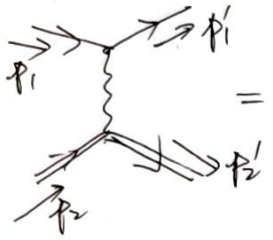
Homework 1. In last chapter, we have discussed Rutherford scattering, and we have got the results that $\frac{d\sigma}{d\Omega}=\frac{Z^2\alpha^2(1-v^2\sin^2\frac{\theta}{2})}{4v^4E^2\sin^4\frac{\theta}{2}}$, this is called Mott-formula. Here is another view. Consider the QED process with $e^-\mu^-\to e^-\mu^-$, and let $m_\mu\to\infty$, then in μ rest frame, we can also get this formula.

According to crossing symmetry, there will be:


=


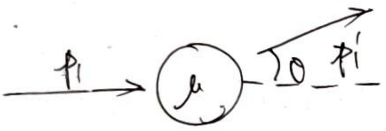
$$=\frac{8e^4}{q^4}\Big\{(p\cdot k')(p'\cdot k)+(p\cdot k)(p'\cdot k')+2m_e^2m_\mu^2+m_e^2(k\cdot k')+m_\mu^2(p\cdot p')\Big\}.$$

then consider a $e^-\mu^-\to e^-\mu^-$: $p\to p_1$, $p'\to -p_1'$, $k'\to -p_2$, $k\to p_2'$

$\frac{1}{4}\sum_{spin}$


$$=\frac{1}{4}\sum_{spin}|M|^2=\frac{8e^4}{q^4}\Big\{(p_1\cdot p_2)(p_1'\cdot p_2')+(p_1\cdot p_2')(p_1'\cdot p_2)+2m_e^2m_\mu^2-m_e^2(p_2\cdot p_2')-m_\mu^2(p_1\cdot p_1')\Big\}$$

In the μ-frame.



$$p_1^a=(E,0,0,\beta E)$$
$$p_1'^a=(E,\beta E\sin\theta,0,\beta E\cos\theta)$$
$$p_2^a=p_2'^a=(m_\mu,0,0,0)$$

and $m_\mu \gg E$.

then $p_1\cdot p_2=p_1'\cdot p_2'=p_1\cdot p_2'=p_1'\cdot p_2=Em_\mu$, $p_1\cdot p_1'=E^2(1-\beta^2\cos\theta)$. $p_2\cdot p_2'=m_\mu^2$.

then: $$\frac{1}{4}\sum_{spin}|M|^2=\frac{8e^4}{q^4}\left(2E^2m_\mu^2+2m_e^2m_\mu^2-m_e^2m_\mu^2-m_\mu^2E^2(1-\beta^2\cos\theta)\right)$$

$$=\frac{8e^4}{q^4}m_\mu^2E^2\left(2+\frac{m_e^2}{E^2}-1+\beta^2\cos\theta\right)$$

$\beta^2=\frac{p^2}{E^2}=1-\frac{m_e^2}{E^2}$, $\frac{m_e^2}{E^2}=1-\beta^2$

$$=\frac{8e^4}{q^4}m_\mu^2E^2\Big(\underbrace{2+1-\beta^2-1+\beta^2\cos\theta}_{=2-\beta^2(1-\cos\theta)=2-2\beta^2\sin^2\frac{\theta}{2}}\Big)$$

$$=\frac{16e^4}{q^4}m_\mu^2E^2\left(1-\beta^2\sin^2\frac{\theta}{2}\right)$$

since $q=p_1-p_1'=(0,-\beta E\sin\theta,0,\beta E(1-\cos\theta))$, then:

$$q^2=-\beta^2E^2(2-2\cos\theta)=-4\beta^2E^2\sin^2\theta/2.\quad q^4=16\beta^4E^4\sin^4\frac{\theta}{2}.$$

$$\therefore \frac{1}{4}\sum_{spin}|M|^2=\frac{e^4m_\mu^2}{\beta^4E^2\sin^4\theta/2}\left(1-\beta^2\sin^2\theta/2\right)$$

$$d\Pi_2 = \frac{d^3\vec{p}'\, d^3\vec{k}'}{(2\pi)^6\, 2E^{(e)}_{p'}\, 2E^{(\mu)}_{k'}} (2\pi)^4\, \delta\left(E^{(e)} + m_\mu - E^{(e)}_{p'} - E^{(\mu)}_{k'}\right) \delta^{(3)}\left(\vec{p} - \vec{p}' - \vec{k}'\right)$$

$$= \frac{d^3\vec{p}'}{(2\pi)^2\, 2E^{(e)}_{p'}\, 2E^{(\mu)}_{\vec{p}-\vec{p}'}} \cdot \delta\left(E^{(e)}_{p'} + E^{(\mu)}_{k'} - E^{(e)} - m_\mu\right) = \frac{p'^2\, dp'\, d\Omega}{(2\pi)^2\, 2E^{(e)}_{p'}\, 2E^{(\mu)}_{(p-p')}} \cdot \left(\frac{p'}{E^{(e)}_{p'}} + \frac{p'-p}{E^{(\mu)}_{(p-p')}}\right)^{-1} \delta(p'-?)$$

Since $m_\mu \to \infty$, thus $E_{p'} \simeq E$, $E_{p-p'} = m_\mu$, and thus $d\Pi_2 = \frac{p'^2\, d\Omega}{16\pi^2 E m_\mu} \cdot \frac{E}{p'} = \frac{p'\, d\Omega}{16\pi^2 m_\mu}$

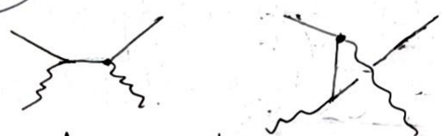
then: $d\sigma = \frac{1}{4Em_\mu \cdot \beta} \cdot d\Pi_2 |\mathcal{M}|^2 = \frac{1}{4Em_\mu \beta} \cdot \frac{pE\, d\Omega}{16\pi^2 m_\mu} \cdot \frac{e^4 m_\mu^2}{\beta^4 E^2 \sin^4\frac{\theta}{2}} \left(1 - \beta^2 \sin^2\frac{\theta}{2}\right)$

$\therefore$ $$\frac{d\sigma}{d\Omega} = \frac{e^4\left(1-\beta^2\sin^2\frac{\theta}{2}\right)}{64\pi^2 \beta^4 E^2 \sin^4\frac{\theta}{2}} = \frac{\alpha^2\left(1-\beta^2\sin^2\frac{\theta}{2}\right)}{4\beta^4 E^2 \sin^4\frac{\theta}{2}}$$

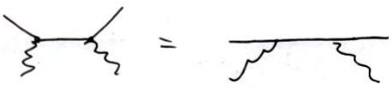
thus we have got the Mott Formula! $\square$

§2 Compton scattering: $e^-\gamma \to e^-\gamma$

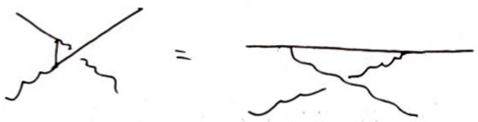
Here are involving two Feynman:



the difference between these two diagram can be seen:

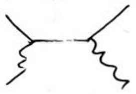


, which is absorb then emit

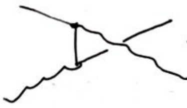


, which is first emit then absorb

Now consider the Fermion sign:



$\sim \langle e^-\gamma | \bar{\psi}\gamma A \psi\, \bar{\psi}\gamma A \psi | e^-\gamma \rangle$

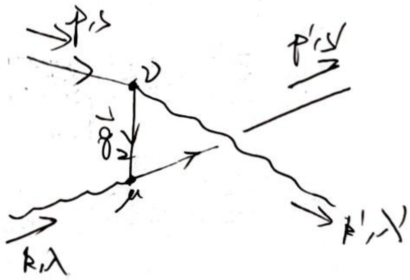


$\sim \langle e^-\gamma | \bar{\psi}\gamma A \psi\, \bar{\psi}\gamma A \psi | e^-\gamma \rangle$

they are the same fermion sign.

thus $i\mathcal{M} =$

$+$



$$= \bar{u}(p',s')(-ie\gamma^\mu) \epsilon^*_\mu(k',\lambda') \cdot \frac{i(\not{q}_1 + m)}{q_1^2 - m^2 + i\epsilon} \cdot (-ie\gamma^\nu) \epsilon_\nu(k,\lambda)\, u(p,s)$$

$$+ \bar{u}(p',s')(-ie\gamma^\mu) \epsilon_\mu(k,\lambda) \frac{i(\not{q}_2 + m)}{q_2^2 - m^2 + i\epsilon} (-ie\gamma^\nu) \epsilon^*_\nu(k',\lambda')\, u(p,s)$$

$$= -ie^2\, \bar{u}(p',s')\gamma^\mu \frac{(\not{q}_1+m)}{q_1^2-m^2+i\varepsilon}\gamma^\nu u(p,s)\cdot \varepsilon^*_\mu(k',\lambda')\varepsilon_\nu(k,\lambda)$$

$$-ie^2\, \bar{u}(p',s')\gamma^\mu \frac{(\not{q}_2+m)}{\vec{q}_2^{\,2}-m^2+i\varepsilon}\gamma^\nu u(p,s)\, \varepsilon_\mu(k,\lambda)\,\varepsilon^*_\nu(k',\lambda')$$

which means: $\mathcal{M} = -\frac{e^2}{\vec{q}_1^{\,2}-m^2+i\varepsilon}\left(\bar{u}(p',s')\gamma^\mu(\not{q}_1+m)\gamma^\nu u(p,s)\right)\varepsilon^*_\mu(k',\lambda')\varepsilon_\nu(k,\lambda)$

$$-\frac{e^2}{\vec{q}_2^{\,2}-m^2+i\varepsilon}\left(\bar{u}(p',s')\gamma^\mu(\not{q}_2+m)\gamma^\nu u(p,s)\right)\varepsilon_\mu(k,\lambda)\varepsilon^*_\nu(k',\lambda')$$

and notice that $\vec{q}_1=\vec{p}+\vec{k}$, then $\vec{q}_1^{\,2}-m^2 = \vec{p}^{\,2}+\vec{k}^{\,2}+2\vec{p}\cdot\vec{k}-m^2 = 2\vec{p}\cdot\vec{k}$

$\vec{q}_2=\vec{p}-\vec{k}'$, then $\vec{q}_2^{\,2}-m^2 = \vec{p}^{\,2}+\vec{k}'^{\,2}-2\vec{p}\cdot\vec{k}'-m^2 = -2\vec{p}\cdot\vec{k}'$

and also. $(\not{q}_1+m)\gamma^\nu u(p,s) = (\not{p}+m)\gamma^\nu u(p,s)+\not{k}\gamma^\nu u(p,s)$

$= 2p^\nu u(p,s) - \gamma^\nu\underbrace{(\not{p}-m)u(p,s)}_{\text{Dirac equation}} + \cdots = 2p^\nu u(p,s) + \not{k}\gamma^\nu u(p,s)$

similar. $(\not{q}_2+m)\gamma^\nu u(p,s) = (\not{p}+m)\gamma^\nu u(p,s) - \not{k}'\gamma^\nu u(p,s)$

$= 2p^\nu u(p,s) - \not{k}'\gamma^\nu u(p,s)$

$$\therefore \mathcal{M} = -\frac{e^2}{2\vec{p}\cdot\vec{k}}\left\{\bar{u}(p',s')\gamma^\mu\left[2p^\nu u(p,s)+\not{k}\gamma^\nu u(p,s)\right]\right\}\varepsilon^*_\mu(k',\lambda')\varepsilon_\nu(k,\lambda)$$

$$-\frac{e^2}{-2\vec{p}\cdot\vec{k}}\left\{\bar{u}(p',s')\gamma^\mu\left[2p^\nu u(p,s)-\not{k}'\gamma^\nu u(p,s)\right]\right\}\varepsilon_\mu(k',\lambda')\varepsilon^*_\nu(k,\lambda)$$

$$= -e^2\varepsilon^*_\mu(k',\lambda')\varepsilon_\nu(k,\lambda)\cdot\bar{u}(p',s')\Big\{\underbrace{\frac{\gamma^\mu\not{k}\gamma^\nu+2\gamma^\mu p^\nu}{2\vec{p}\cdot\vec{k}}+\frac{-\gamma^\nu\not{k}'\gamma^\mu+2\gamma^\nu p^\mu}{-2\vec{p}\cdot\vec{k}}}_{\Gamma^{\mu\nu}}\Big\}u(p,s)$$

$$\therefore \frac{1}{4}\sum_{\text{spin}}|\mathcal{M}|^2 = \frac{e^4}{4}g_{\mu\rho}g_{\nu\sigma}\cdot\left(\bar{u}(p',s')\Gamma^{\mu\nu}u(p,s)\right)\cdot\left(\bar{u}(p,s)\Gamma^{\sigma\rho}u(p',s')\right)$$

$(pab)^\dagger = pba$. since $(\gamma^\nu\gamma^\mu)^\dagger \to (\gamma^\mu)^\dagger(\gamma^\nu)^\dagger \to \gamma^0 \gamma^\mu\gamma^\nu$

$$= \frac{e^4}{4}g_{\mu\rho}g_{\nu\sigma}\sum_{s,s'}\mathrm{Tr}\left\{\Gamma^{\mu\nu}u(p,s)\bar{u}(p,s)\Gamma^{\sigma\rho}u(p',s')\bar{u}(p',s')\right\}$$

$$= \frac{e^4}{4}g_{\mu\rho}g_{\nu\sigma}\boxed{\mathrm{Tr}\left\{\Gamma^{\mu\nu}(\not{p}+m)\Gamma^{\sigma\rho}(\not{p}'+m)\right\}} = M^{\mu\nu\sigma\rho}$$

□ The biggest mission is how to compute.

$$\underline{\mathrm{Tr}\left\{\Gamma^{\mu\nu}(\not{p}+m)\Gamma^{\sigma\rho}(\not{p}'+m)\right\}}$$

$$\mathrm{Tr}\left\{\Gamma^{\mu\nu}(\not p+m)\Gamma^{\rho\sigma}(\not p'+m)\right\}\cdot g_{\mu\rho}g_{\nu\sigma}$$

$$=\mathrm{Tr}\left\{\left(\frac{\gamma^\mu \not k\gamma^\nu+2\gamma^\mu p^\nu}{2\vec p\cdot\vec k}+\frac{\gamma^\nu \not k'\gamma^\mu-2\gamma^\nu p^\mu}{2\vec p\cdot \vec k'}\right)(\not p+m)\left(\frac{\gamma_\nu \not k\gamma_\mu+2\gamma_\nu p_\mu}{2\vec p\cdot\vec k}+\frac{\gamma_\mu \not k'\gamma_\nu-2\gamma_\mu p_\nu}{2\vec p\cdot\vec k'}\right)(\not p'+m)\right\}$$

$$=\frac{\mathrm{I}}{(2\vec p\cdot\vec k)^2}+\frac{\mathrm{II}}{(2\vec p\cdot\vec k')^2}+\frac{\mathrm{III}}{(2\vec p\cdot\vec k)(2\vec p\cdot\vec k')}+\frac{\mathrm{IV}}{(2\vec p\cdot\vec k')(2\vec p\cdot\vec k)}$$

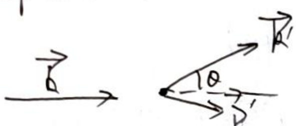
where $\mathrm{I}=\mathrm{Tr}\left\{(\gamma^\mu\not k\gamma^\nu+2\gamma^\mu p^\nu)(\not p+m)(\gamma_\nu\not k\gamma_\mu+2\gamma_\nu p_\mu)(\not p'+m)\right\}$

$\boxed{\text{Tedious work !!!}}$

Finally, we can get the result easily:

$$\frac14\sum_{\text{spin}}|M|^2=2e^4\left\{\frac{\vec p\cdot\vec k'}{\vec p\cdot\vec k}+\frac{\vec p\cdot\vec k}{\vec p\cdot\vec k'}+2m^2\left(\frac{1}{\vec p\cdot\vec k}-\frac{1}{\vec p\cdot\vec k'}\right)+m^4\left(\frac{1}{\vec p\cdot\vec k}-\frac{1}{\vec p\cdot\vec k'}\right)^2\right\}$$

II Compute the cross section: Work in the electron-static frame, and suppose



$$\vec k=(\omega,0,0,\omega)\qquad \vec k'=(\omega',\omega'\sin\theta,0,\omega'\cos\theta)$$
$$\vec p=(m,0,0,0)\qquad p'=(E',\vec p')$$

$$d\Pi_2=\frac{d^3\vec k'\,d^3\vec p'}{(2\pi)^2 4\omega' E'}\cdot\delta(\omega+m-\omega'-E')\,\delta^{(3)}(\vec k+\vec p-\vec k'-\vec p')$$

since $\vec p'$ is on shell, thus $m^2=\vec p'^2=(\vec k+\vec p-\vec k')^2=m^2+2\vec k\cdot\vec p-2\vec k'\cdot\vec p-2\vec k\cdot\vec k'$

$$=m^2+2m\omega-2m\omega'-2\omega\omega'+2\omega\omega'\cos\theta$$
$$=m^2+2m(\omega-\omega')-2\omega\omega'(1-\cos\theta)\stackrel{!}{=}m^2$$

thus $\Rightarrow m(\omega-\omega')=\omega\omega'(1-\cos\theta)\Rightarrow\boxed{\frac{1}{\omega'}-\frac{1}{\omega}=\frac{1-\cos\theta}{m}}$ Compton formula

$$\omega'=\frac{\omega}{1+\frac{\omega}{m}(1-\cos\theta)}$$

then $E'=m+\omega-\omega'=m+\omega-\dfrac{\omega}{1+\frac{\omega}{m}(1-\cos\theta)}$

$$d\Pi_2=\frac{d^3\vec k'}{(2\pi)^2 4\omega' E'}\cdot\delta\left(\omega+m-\omega'-\sqrt{m^2+(\vec k+\vec p-\vec k')^2}\right)$$
$$=\frac{d^3\vec k'}{(2\pi)^2 4\omega' E'}\delta\left(\omega+m-\omega'-\sqrt{m^2+\underbrace{(\omega'\sin\theta)^2+(\omega-\omega'\cos\theta)^2}_{\omega'^2+\omega^2-2\omega\omega'\cos\theta}}\right)$$
$$=\frac{d^3\vec k'}{(2\pi)^2 4\omega' E'}\delta\left(\omega+m-\omega'-\sqrt{m^2+\omega^2+\omega'^2-2\omega\omega'\cos\theta}\right)$$

$d^3k' = k'^2 dk' d\Omega = \omega'^2 d\omega' d\Omega$. thus we have:

$$d\Pi_2 = \frac{\omega'^2 d\omega' d\Omega}{(2\pi)^2 4\omega' E'} \delta(\cdots) = \frac{\omega' d\omega' d\Omega}{(2\pi)^2 \cdot 4E'} \delta\left(\omega + m - \omega' - \sqrt{m^2 + \omega^2 + \omega'^2 - 2\omega\omega'\cos\theta}\right)$$

$$= \frac{\omega' d\Omega}{(2\pi)^2 \cdot 4E'} \cdot \left(1 + \frac{\omega' - \omega\cos\theta}{E'}\right)^{-1} = \frac{\omega' d\cos\theta \cdot d\phi}{(2\pi)^2 \cdot 4E'}\left(1 + \frac{\omega' - \omega\cos\theta}{E'}\right)^{-1}$$

thus we have $d\Pi_2 = \dfrac{\omega' d\cos\theta}{8\pi E'}\left(1 + \dfrac{\omega' - \omega\cos\theta}{E'}\right)^{-1} = \dfrac{\omega' d\cos\theta}{8\pi}\left(E' + \omega' - \omega\cos\theta\right)^{-1}$

$$= \frac{d\cos\theta}{8\pi} \cdot \left(\frac{E'}{\omega'} + 1 - \frac{\omega}{\omega'}\cos\theta\right)^{-1} = \frac{d\cos\theta}{8\pi}\left(\frac{m}{\omega'} + \frac{\omega}{\omega'}(1 - \cos\theta)\right)^{-1}$$

$$\frac{E'}{\omega'} = \frac{m + \omega - \omega'}{\omega'} = \frac{m}{\omega'} + \frac{\omega}{\omega'} - 1$$

$$= \frac{d\cos\theta}{8\pi} \cdot \boxed{\frac{\omega'}{m + \omega(1 - \cos\theta)}} \qquad \omega(1-\cos\theta) = \frac{m(\omega - \omega')}{\omega\omega'}$$

$$= \frac{d\cos\theta}{8\pi} \cdot \frac{(\omega')^2}{m\omega} \qquad \frac{\omega'}{m + \frac{m(\omega - \omega')}{\omega'}} = \frac{(\omega')^2}{\cancel{m\omega}\, m\omega}$$

$$\therefore \frac{d\sigma}{d\cos\theta} = \frac{1}{4\omega m} \cdot \frac{d\cos\theta}{8\pi} \cdot \frac{(\omega')^2}{m\omega} \cdot \left(\frac{1}{4}\sum_{\text{spin}} |M|^2\right)$$

$p \cdot k' = m\omega'$, $p \cdot k = m\omega$

~~and for~~

$$2e^4 \left\{ \frac{\omega'}{\omega} + \frac{\omega}{\omega'} + 2m^2\left(\frac{1}{m\omega} - \frac{1}{m\omega'}\right) + m^4\left(\frac{1}{m\omega} - \frac{1}{m\omega'}\right)^2 \right\}$$

$$\frac{1}{m\omega} - \frac{1}{m\omega'} = -\frac{1}{m^2}(1-\cos\theta) \qquad \left(\frac{1}{m\omega} - \frac{1}{m\omega'}\right)^2 = \frac{1}{m^4}(1-\cos\theta)^2$$

$$= \frac{(\omega')^2}{4m^2\omega^2} \cdot \frac{d\cos\theta}{8\pi} \cdot 2e^4 \left\{ \frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta \right\}$$

$$= \frac{2 \times 16\pi^2 \alpha^2 \, d\cos\theta}{4m^2\omega^2 \times 8\pi}(\cdots) \qquad = 2e^4\left\{\frac{\omega'}{\omega} + \frac{\omega}{\omega'} \underbrace{- 2(1-\cos\theta) + (1-\cos\theta)^2}\right\}$$

$$= \frac{\pi\alpha^2}{m^2}\left(\frac{\omega'}{\omega}\right)^2 \left\{ \frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta \right\} \qquad 2e^4\left\{\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta\right\}$$

$$-2 + \cancel{2\cos\theta} + 1 + \cos^2\theta - \cancel{2\cos\theta} = \cos^2\theta - 1 = -\sin^2\theta$$

Thus we have got: $\dfrac{d\sigma}{d\cos\theta} = \dfrac{\pi\alpha^2}{m^2}\left(\dfrac{\omega'}{\omega}\right)^2 \left\{ \dfrac{\omega'}{\omega} + \dfrac{\omega}{\omega'} - \sin^2\theta \right\}$

Klein–Nishina formula

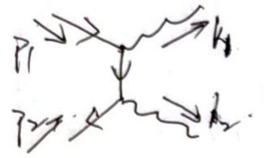
Now take $\omega \to 0$ (Soft photon limits), then $\dfrac{\omega'}{\omega} \to \dfrac{1}{1 + \frac{0}{m}(1-\cos\theta)} = 1$. thus.

$$\left.\frac{d\sigma}{d\cos\theta}\right|_{\omega\to 0} = \frac{\pi\alpha^2}{m^2}(1 + \cos^2\theta)$$

and $\sigma_{\text{total}} = \dfrac{8\pi\alpha^2}{3m^2}$ ← Thomson cross section.

Homework2. Consider the pross of $e^+e^- \to 2\gamma$. Check the Word identity and derive the cross section.

The Feynman Diagram in the lowest order are:



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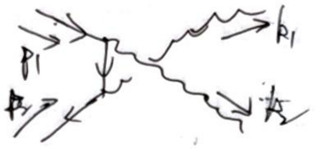
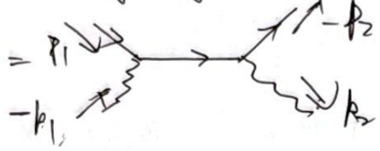


$= i\mathcal{M}(e^+e^- \to 2\gamma)$

According to the cross symmetry, we have:

$i\mathcal{M}(e^+e^- \to 2\gamma) =$


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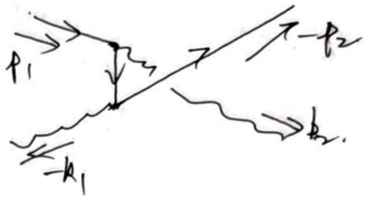


$= i\mathcal{M}(e^-\gamma \to e^-\gamma)$

thus we can get $i\mathcal{M}(e^+e^- \to 2\gamma)$ from the results of $e^-\gamma \to e^-\gamma$.

$$\frac{1}{4}\sum_{spin}|\mathcal{M}|^2 = 2e^4\left\{\frac{\vec{p}_1\cdot\vec{k}_2}{-\vec{p}_1\cdot\vec{k}_1} + \frac{-\vec{p}_1\cdot\vec{k}_1}{\vec{p}_1\cdot\vec{k}_2} + 2m^2\left(\frac{1}{-\vec{p}_1\cdot\vec{k}_1} - \frac{1}{\vec{p}_1\cdot\vec{k}_2}\right) + m^4\left(\frac{1}{-\vec{p}_1\cdot\vec{k}_1} - \frac{1}{\vec{p}_1\cdot\vec{k}_2}\right)^2\right\}$$

The overall minus sign is the results of Fermion statistic, and should be canceled

$$= 2e^4\left\{\frac{\vec{p}_1\cdot\vec{k}_2}{\vec{p}_1\cdot\vec{k}_1} + \frac{\vec{p}_1\cdot\vec{k}_1}{\vec{p}_1\cdot\vec{k}_2} + 2m^2\left(\frac{1}{\vec{p}_1\cdot\vec{k}_1} + \frac{1}{\vec{p}_1\cdot\vec{k}_2}\right) + m^4\left(\frac{1}{\vec{p}_1\cdot\vec{k}_1} + \frac{1}{\vec{p}_1\cdot\vec{k}_2}\right)^2\right\}$$

In center of mass frame: with $E = (p^2+m^2)^{1/2}$

with $p_1 = (E, 0, 0, p)$ $\quad k_1 = (E, E\sin\theta, 0, E\cos\theta)$

$p_2 = (E, 0, 0, -p)$ $\quad k_2 = (E, -E\sin\theta, 0, -E\cos\theta)$

and there will be. $\vec{p}_1\cdot\vec{k}_2 = E^2 + pE\cos\theta$.

$\vec{p}_1\cdot\vec{k}_1 = E^2 - pE\cos\theta$

$$\text{then } \frac{1}{4}\sum_{spin}|\mathcal{M}|^2 = 2e^4\left\{\frac{E+p\cos\theta}{E-p\cos\theta} + \frac{E-p\cos\theta}{E+p\cos\theta} + \frac{2m^2}{E}\left(\frac{1}{E-p\cos\theta} + \frac{1}{E+p\cos\theta}\right) + \frac{m^4}{E^2}\left(\frac{1}{E-p\cos\theta} + \frac{1}{E+p\cos\theta}\right)^2\right\}$$

According to the conclusion of $d\sigma$ in center of mass frame:

$$d\sigma = \frac{1}{64\pi^2}\cdot\frac{1}{E^2\cdot 2p/E}\cdot\frac{E}{E_{cm}}\,d\Omega\cdot\left(\frac{1}{4}\sum_{spin}|\mathcal{M}|^2\right)$$

$$= \frac{1}{64\pi^2 E_{cm}}\cdot\frac{1}{2p}\,d\Omega\cdot 2e^4\{\cdots\} = \frac{e^4}{64\pi^2}\cdot\frac{d\Omega}{pE_{cm}}\{\cdots\} = \frac{\alpha^2}{4pE_{cm}}\{\cdots\}\,d\Omega.$$

$$\therefore \frac{d\sigma}{d\cos\theta} = \frac{\pi\alpha^2}{2pE_{cm}}\left\{\frac{2(E^2+p^2\cos^2\theta)}{E^2-p^2\cos^2\theta} + \frac{2m^2}{E}\cdot\frac{2E}{E^2-p^2\cos^2\theta} + \frac{m^4}{E^2}\cdot\frac{4E^2}{(E^2-p^2\cos^2\theta)^2}\right\}.$$

$$= \frac{\pi\alpha^2}{2pE}\left\{\frac{E^2+p^2\cos^2\theta}{m^2+p^2\sin^2\theta} + \frac{2m^2}{m^2+p^2\sin^2\theta} - \frac{2m^4}{(m^2+p^2\sin^2\theta)^2}\right\}$$

$$\mathcal{M} = -e^2 \epsilon^*_\mu(k',\lambda')\epsilon_\nu(k,\lambda)\,\bar u(p',s')\left\{\frac{\gamma^\mu \slashed{k}\gamma^\nu + 2\gamma^\mu p^\nu}{2\vec p\cdot\vec k} + \frac{\gamma^\nu\slashed{k}'\gamma^\mu - 2\gamma^\nu p^\mu}{2\vec p\cdot\vec k'}\right\}u(p,s)$$

$$= -e^2\epsilon^*_\mu(k_2,\lambda')\epsilon^*_\nu(k_1,\lambda)\,\underbrace{\overline{u}(-p_2,s')}_{\bar v(p_2,s')}\left\{\frac{-\gamma^\mu\slashed{k}_1\gamma^\nu+2\gamma^\mu p_1^\nu}{-2\vec p_1\cdot\vec k_1} + \frac{\gamma^\nu\slashed{k}_2\gamma^\mu - 2\gamma^\nu p_1^\mu}{2\vec p_1\cdot\vec k_2}\right\}u(p_1,s)$$

$p=p_1$
$k=-k_1$
$p'=-p_2$
$k'=k_2$

let's take $$\mathcal{M}^{\mu\nu} = \bar v(p_2,s')\left\{\frac{\gamma^\mu\slashed{k}_1\gamma^\nu - 2\gamma^\mu p_1^\nu}{2\vec p_1\cdot\vec k_1}+\frac{\gamma^\nu\slashed{k}_2\gamma^\mu-2\gamma^\nu p_1^\mu}{2\vec p_1\cdot\vec k_2}\right\}u(p_1,s)$$

$$= \mathcal{M}_1^{\mu\nu} + \mathcal{M}_2^{\mu\nu}$$

$$k_{1\nu}\mathcal{M}_1^{\mu\nu} = \bar v(p_2,s')\,\frac{\gamma^\mu\underbrace{\slashed{k}_1\slashed{k}_1}_{0} - 2\gamma^\mu p_1^\nu k_{1\nu}}{2\vec p_1\cdot\vec k_1}\,u(p_1,s) = \bar v(p_2,s')\cdot\frac{-2\gamma^\mu\vec p_1\cdot\vec k_1}{2\vec p_1\cdot\vec k_1}\,u(p_1,s)$$

$$= -\bar v(p_2,s')\gamma^\mu u(p_1,s).$$

$$k_{2\nu}\mathcal{M}_2^{\mu\nu} = \frac{1}{2\vec p_1\cdot\vec k_2}\cdot\bar v(p_2,s')\left\{\slashed{k}_1\slashed{k}_2\gamma^\mu - 2\slashed{k}_1 p_1^\mu\right\}u(p_1,s)$$

$$= \frac{1}{2\vec p_1\cdot\vec k_2}\cdot\bar v(p_2,s')\left\{(\slashed{p}_1+\slashed{p}_2-\underbrace{\slashed{k}_2)\slashed{k}_2}_{0}\gamma^\mu - 2(\slashed{p}_1+\slashed{p}_2-\slashed{k}_2)p_1^\mu\right\}u(p_1,s)$$

$$= \frac{1}{2\vec p_1\cdot\vec k_2}\,\bar v(p_2,s)\left\{(\slashed{p}_1+\slashed{p}_2)\slashed{k}_2\gamma^\mu - 2(\slashed{p}_1+\slashed{p}_2-\slashed{k}_2)p_1^\mu\right\}u(p_1,s)$$

$$= \frac{1}{2\vec p_1\cdot\vec k_2}\cdot\bar v(p_2,s)\left\{\underbrace{(\slashed{p}_1-m)\slashed{k}_2\gamma^\mu}_{\slashed{p}_1\slashed{k}_2\gamma^\mu - m\slashed{k}_2\gamma^\mu}\ \underbrace{-2(m-m-\slashed{k}_2)p_1^\mu}_{+2\slashed{k}_2 p_1^\mu}\right\}u(p_1,s)$$

$$\slashed{p}_1\slashed{k}_2\gamma^\mu = -\slashed{k}_2\slashed{p}_1\gamma^\mu + 2\vec p_1\cdot\vec k_2\,\gamma^\mu$$

$$= \slashed{k}_2\gamma^\mu\slashed{p}_1 - 2\slashed{k}_2 p_1^\mu + 2\vec p_1\cdot\vec k_2\,\gamma^\mu$$

$$= \frac{1}{2\vec p_1\cdot\vec k_2}\cdot\bar v(p_2,s)\left\{\underbrace{\slashed{k}_2\gamma^\mu\slashed{p}_1}_{m\slashed{k}_2\gamma^\mu} - 2\slashed{k}_2 p_1^\mu + 2(\vec p_1\cdot\vec k_2)\gamma^\mu - m\slashed{k}_2\gamma^\mu + 2\slashed{k}_2 p_1^\mu\right\}u(p_1,s)$$

$$= \bar v(p_2,s)\gamma^\mu u(p_1,s)$$

$\therefore\ k_{2\nu}\mathcal{M}^{\mu\nu} = -\bar v(p_2,s')\gamma^\mu u(p_1,s) + \bar v(p_2,s')\gamma^\mu u(p_1,s) = 0$, this is the prediction of Ward identity!

$\square$

§3. Vertex correction

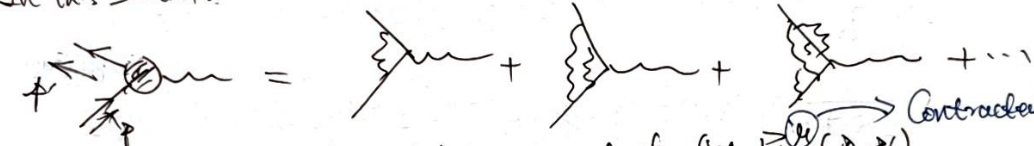
** For QED theory, it's worthy to discuss some higher order diagram, whose phenomenally divergence may seem like break the effectiveness of perturbation, but can be annihilated by some approaches.

** In higher order, we have to deal with the diagram with loops. In this chapter, we will consider three types of typical correction sub-diagram in $e^+e^- \to \mu^+\mu^-$ procs. we can denote them as

Vertex correction. External-leg correction. Vacuum polarization.

For single loop order:

** In this section, we consider the vertex correction, where we have:



Contracted with photon.

Denote this vertex contributes to $i\mu$ as a vertex function $\Gamma^{\mu}(p, p')$

What will be the most general form?; Γ^μ can be rely on $p^\mu, p'^\mu, \gamma^\mu$, and some constant.

Exhaustive choice for any parity-conserving theory!

1. General form of Γ^μ

- Γ^μ transformed as a vector $\Rightarrow$ linear combination of $p^\mu, p'^\mu, \gamma^\mu$

  Denote $\Gamma^\mu = A\gamma^\mu + (p'^\mu + p^\mu)B + (p'^\mu - p^\mu)C$

- Though Γ^μ should be a 4x4 matrix and A, B, C may contains $\not{p}$ or $\not{p}'$, however according to Dirac equation, we can assert A, B, C is only $q^2 = -2p\cdot p' + 2m^2$

- From Ward identity, for a procs $e^- \to e^- + \gamma$; where Γ^μ-vertex (including all levels)

has emerged, we should have $q_\mu \Gamma^\mu \overset{!}{=} 0$ $\quad (q = p' - p)$

which means $A q_\mu \gamma^\mu + (\underbrace{p^2}_{=0} - \underbrace{p'^2}_{=0})B + q^2 C = 0$

while q may not an on-shell value, thus generally $q^2 \neq 0$, then C must be zero.

Thus we have found that $\Gamma^\mu = A(q^2)\gamma^\mu + (p^\mu + p'^\mu)B(q^2)$.

Since we have Gordon identity, then:

$$\bar{u}(p')\Gamma^\mu u(p) = A(q^2)\,\bar{u}(p')\gamma^\mu u(p) + \bar{u}(p')(p^\mu + p'^\mu)B(q^2)u(p)$$

$$= \bar{u}(p')\left\{\frac{p^\mu + p'^\mu}{2m}(A + 2mB) + \frac{i\sigma^{\mu\nu}q_\nu}{2m}A\right\}u(p)$$

thus another more-frequently used general form:

$$\Gamma^{\mu}(p', p) = \gamma^{\mu} F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2m} F_2(q^2)$$

$\longrightarrow$ form factor for vertex correction:

* For tree level, $F_1 = 1$, $F_2 = 0$

Homework III: Consider the scattering of an electron with energy $E \gg m_e$ from a proton initially at rest.

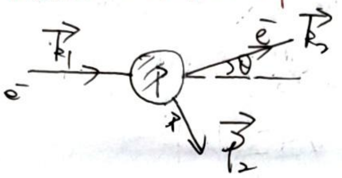
the form of vertex  $= \bar{u}(p')\left[\gamma^{\mu} F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2m} F_2(q^2)\right] u(p) \cdot (-ie)$

will leads to so-called Rosenbluth formula for the elastic scattering cross section.

$$\frac{d\sigma}{d\cos\theta} = \frac{\pi\alpha^2 \left\{\left(F_1^2 - \frac{\vec{q}^2}{4m^2} F_2^2\right)\cos^2\frac{\theta}{2} - \frac{\vec{q}^2}{2m^2}(F_1+F_2)^2 \sin^2\frac{\theta}{2}\right\}}{2E^2\left\{1 + \frac{2E}{m}\sin^2\frac{\theta}{2}\right\}\sin^4\frac{\theta}{2}}$$

θ is the lab-frame scattering angle. When $F_1 = 1$, $F_2 = 0$, and ~~at~~ in the massless limit, the Rosenbluth formula reduces to the Mott formula.

In Lab frame, suppose:



with $\vec{k}_1 = (E, 0, 0, E)$ $\longrightarrow$ ~~here take~~ $\vec{p} = \cancel{p}E = E$ since $E \gg m_e$.

$\vec{p}_1 = (M, 0, 0, 0) \longrightarrow$ mass of proton

$\vec{k}_2 = (E', E'\sin\theta, 0, E'\cos\theta)$

Since $\vec{p}_2$ is on-shell, there should be:

$\vec{p}_2 = \vec{k}_1 + \vec{p}_1 - \vec{k}_2$

$$\vec{p}_2^{\,2} = (\vec{k}_1 + \vec{p}_1 - \vec{k}_2)^2 = \vec{k}_1^2 + \vec{p}_1^2 + \vec{k}_2^2 + 2\vec{k}_1\cdot\vec{p}_1 - 2\vec{k}_1\cdot\vec{k}_2 - 2\vec{p}_1\cdot\vec{k}_2$$

$$= m_e^2 + M^2 + m_e^2 + 2EM - 2EE'(1-\cos\theta) - 2E'M \stackrel{!}{=} M^2$$

$$\therefore\ m_e^2 + (E-E')M \stackrel{!}{=} EE'(1-\cos\theta) \xRightarrow{\text{let } m_e = 0} E' = \frac{ME}{M + 2E\sin^2\frac{\theta}{2}}.$$

$$d\Pi_2 = \frac{d^3\vec{k}_2\, d^3\vec{p}_2}{(2\pi)^2\, 4E' E_{p_2}^{(M)}}\, \delta(E_{p_2} + E' - E - M)\, \delta^{(3)}(\vec{k}_1 + \vec{p}_1 - \vec{k}_2 - \vec{p}_2)$$

$$= \frac{d^3\vec{k}_2}{(2\pi)^2 \cdot 4E' E_{p_2}^{(M)}}\, \delta\left(\sqrt{M^2 + (\vec{k}_1 + \vec{p}_1 - \vec{k}_2)^2} + E' - E - M\right)$$

$$= \frac{k_2^2\, dk_2\, d\Omega}{(2\pi)^2 \cdot 4E' E_{p_2}^{(M)}} \cdot \left|1 + \frac{k_2 - k_1\cos\theta}{E_{p_2}}\right|^{-1} \delta(k_2 - E')$$

$\Rightarrow = \left\{\left[E_{p_2} + \cancel{k_2} E' - E\cos\theta\right] / E_{p_2}\right\}^{-1}$

$= \left\{\left[E(1-\cos\theta) + M\right] / E_{p_2}\right\}^{-1}$

$$= \frac{E'^{\cancel{3}}\, d\Omega}{(2\pi)^2\, 4\cancel{E'}\cancel{E_{p_2}}} \cdot \frac{\cancel{E_{p_2}}}{E(1-\cos\theta) + M} = \frac{E'\, d\Omega}{16\pi^2} \cdot \frac{1}{E(1-\cos\theta) + M} = \frac{E'\, d\cos\theta}{8\pi} \cdot \frac{1}{\cdots}$$

$$\text{thus } \frac{d\sigma}{d\cos\theta} = \frac{1}{4EM}\, d\Pi_2\, |M|^2 = \frac{1}{4EM} \cdot \frac{E'\, d\cos\theta}{8\pi} \cdot \frac{1}{E(1-\cos\theta) + M} \cdot |M|^2 = \frac{E'\, d\cos\theta}{32\pi EM(E(1-\cos\theta) + M)} |M|^2$$

The only mission is $|\mathcal{M}|^2$. In this process.

$$i\mathcal{M} = (-ie)^2\,\bar{u}(p_1)\gamma^\mu u(k_1)\cdot\frac{-ig_{\mu\nu}}{\vec{q}^2+i\varepsilon}\,\bar{U}(p_2)\cdot\Gamma^\nu U(k_2)$$

$$= ie^2\,\bar{u}(p_1)\gamma_\mu u(k_1)\,\frac{1}{\vec{q}^2+i\varepsilon}\,\bar{U}(p_2)\Gamma^\mu U(k_2)$$

$$\frac{1}{4}\sum_{spin}|\mathcal{M}|^2 = \frac{e^2}{4\vec{q}^4}\cdot\left(\bar{u}(p_1)\gamma_\nu u(k_1)\right)\left(\bar{U}(p_2)\Gamma^\mu U(k_2)\right)\cdot\left(\bar{u}(k_1)\gamma_\nu u(p_1)\right)\left(\bar{U}(k_2)\Gamma^\nu U(p_2)\right)$$

$$= \frac{e^2}{4\vec{q}^4}\,\mathrm{Tr}\{\gamma_\mu \not{k}_1\gamma_\nu\not{p}_1\}\cdot\mathrm{Tr}\{\Gamma^\mu(\not{p}_2+M)\Gamma^\nu(\not{k}_2+M)\}$$

After tedious algebra, we will get:

$$\frac{1}{4}\sum_{spin}|\mathcal{M}|^2 = \frac{4e^4M^2}{\vec{q}^4}\left\{\frac{\vec{q}^4}{2M^2}(F_1+F_2)^2+4\left(F_1^2-\frac{\vec{q}^2}{4M^2}F_2^2\right)EE'\cos^2\frac{\theta}{2}\right\}$$

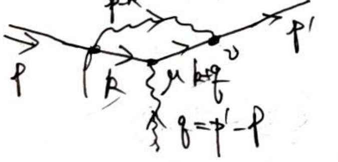
$$= \frac{16e^4M^3}{\vec{q}^4(M+2E\sin^2\frac{\theta}{2})}\left\{\left(F_1^2-\frac{\vec{q}^2}{4M^2}F_2^2\right)\cos^2\frac{\theta}{2}-\frac{\vec{q}^2}{2M^2}(F_1+F_2)^2\sin^2\frac{\theta}{2}\right\}.$$

$$\frac{d\sigma}{d\cos\theta} = \frac{E'\,d\cos\theta}{32\pi\cdot EM(E(1-\cos\theta)+M)}\cdot\frac{1}{4}\sum_{spin}|\mathcal{M}|^2 \qquad E' = \frac{ME}{M+2E\sin^2\frac{\theta}{2}}$$

$$\llcorner\ \frac{1}{(M+2E\sin^2\theta)^2}$$

$$= \frac{\pi\alpha^2}{2E^2\left(1+\frac{2E}{M}\sin^2\frac{\theta}{2}\right)\sin^4\frac{\theta}{2}}\cdot\left\{\left(F_1^2-\frac{\vec{q}^2}{4M^2}F_2^2\right)\cos^2\frac{\theta}{2}-\frac{\vec{q}^2}{2M^2}(F_1+F_2)^2\sin^2\frac{\theta}{2}\right\} \qquad \square$$

2. Single loop computation and Feynman parameter



Let's consider the one loop vertex function:

where $\vec{p}$, $\vec{p}'$ is on-shell

$$\bar{u}(p')\,\delta\Gamma^\mu(p',p)\,u(p) = \bar{u}(p')\cdot\int\frac{d^4k}{(2\pi)^4}\cdot\frac{i(\not{k}+\not{q}+m)}{(\vec{k}+\vec{q})^2-m^2+i\varepsilon}\cdot(-ie\gamma^\mu)\cdot\frac{i(\not{k}+m)}{\vec{k}^2-m^2+i\varepsilon}\cdot(-ie\gamma^\rho)\cdot\frac{-ig_{\nu\rho}}{(\vec{p}-\vec{k})^2}\,u(p)$$

$$(-ie\gamma^\nu)$$

$$= ie^2\,\bar{u}(p')\left\{\int\frac{d^4k}{(2\pi)^4}\cdot\gamma^\nu\frac{\not{k}+\not{q}+m}{(\vec{k}+\vec{q})^2-m^2+i\varepsilon}\cdot\gamma^\mu\cdot\frac{\not{k}+m}{\vec{k}^2-m^2+i\varepsilon}\gamma_\nu\cdot\frac{1}{(\vec{p}-\vec{k})^2}\right\}u(p)$$

$$= ie^2\int\frac{d^4k}{(2\pi)^4}\,\frac{\bar{u}(p')\gamma^\nu(\not{k}+\not{q}+m)\gamma^\mu(\not{k}+m)\gamma_\nu u(p)}{[(\vec{k}+\vec{q})^2-m^2+i\varepsilon][\vec{k}^2-m^2+i\varepsilon][(\vec{p}-\vec{k})^2+i\varepsilon]}$$

$$= 2ie^2 \int \frac{d^4k}{(2\pi)^4} \cdot \frac{\bar u(p')(\not k + m)\gamma^\mu(\not k + \not q + m)u(p)}{[\underbrace{(k+q)}_{=k'}{}^2 - m^2 + i\epsilon][k^2 - m^2 + i\epsilon][(p-k)^2 + i\epsilon]}$$

$$\not k \gamma^\mu \not k + \not k \gamma^\mu (\not q - m) + (\not k - m)\gamma^\mu \not k + m^2 \gamma^\mu (\not q - m)$$

$$\overset{?}{=} 2ie^4 \int \frac{d^4k}{(2\pi)^4} \cdot \frac{\bar u(p')\{\not k \gamma^\mu \not k' + m^2\gamma^\mu - 2m(k^\mu + k'^\mu)\}u(p)}{[\cdots][\cdots][\cdots]} \quad (\#)$$

$$(\not q - m) = \not p' - (\not k + m) \to \not p'$$

$$\therefore \not k \gamma^\mu (\not q - m) = \not k \gamma^\mu \not p'$$

$$= -\not k \not p' \gamma^\mu + 2\not k p'^\mu$$

$$= \not p' \not k \gamma^\mu - 2 p' \cdot k$$

** To integrate the formula above, we should introduce Feynman parameterization technique:

$$\frac{1}{A_1^{m_1}\cdots A_n^{m_n}} = \int_0^1 dx_1\cdots dx_n\, \delta(x_1+\cdots+x_n-1)\cdot\frac{\Gamma(m_1+\cdots+m_n)}{\Gamma(m_1)\cdots\Gamma(m_n)}\cdot\frac{\prod x_i^{m_i-1}}{\left(\sum x_i A_i\right)^{\sum m_i}}$$

Thus in the (#) formula, we have:

$$\frac{1}{[k'^2-m^2+i\epsilon][k^2-m^2+i\epsilon][(k-p)^2+i\epsilon]} = \int_0^1 dx\,dy\, \frac{2}{D^3} \qquad k' = k + q$$

where $D = x(k'^2 - m^2) + y(k^2 - m^2) + (1-x-y)(k-p)^2 + i\epsilon$

$$= x\{\underbrace{(k+q)^2 - m^2}_{k^2+q^2+2k\cdot q - m^2}\} + y(k^2-m^2) + (1-x-y)\underbrace{(k-p)^2}_{k^2+p^2-2k\cdot p} + i\epsilon$$

$$= k^2 + x(q^2 + 2k\cdot q - m^2) - m^2 y + (1-x-y)(p^2 - 2k\cdot p)$$

$$= k^2 + 2k\cdot(x q - \underbrace{(1-x-y)}_{z} p) - xm^2 - m^2 y + \underbrace{(1-x-y)}_{z} p^2 + xq^2$$

$$= (k + xq - zp)^2 - (xq - zp)^2 \underbrace{- xm^2 - m^2 y}_{-(1-z)m^2} + zp^2 + xq^2$$

$$= (\cdots)^2 - x(x-1)q^2 - z(z-1)p^2 + 2xz\, \underbrace{q\cdot p}_{-\frac{q^2}{2}} - (1-z)m^2$$

$$p'^2 = m^2 = (p+q)^2 = \underbrace{p^2}_{m^2} + q^2 + 2p\cdot q$$

$$2p\cdot q = -q^2$$

$$= (\cdots)^2 - [x(x-1) + xz]q^2 + z(z-1)\underbrace{p^2}_{m^2} + (1-z)m^2$$

$$= \underbrace{(\cdots)}_{\ell}{}^2 \underbrace{- xyq^2 + (1-z)^2 m^2}_{=-\Delta}$$

let $D = \ell^2 - \Delta + i\epsilon$. thus we have: $\frac{1}{[\cdots][\cdots][\cdots]} = \int_0^1 dx\,dy \cdot \frac{2}{(\ell^2 + \Delta + i\epsilon)^3}$

and $\bar u(p')\delta\Gamma^\mu(p',p)u(p) = \int_0^1 dx\,dy \int \frac{d^4\ell}{(2\pi)^4}\cdot\frac{\bar u(p')\, f(\ell)\, u(p)}{(\ell^2 - \Delta + i\epsilon)^3}$.

This integral only need to deal with one ℓ^2 on denominator!

$f(\ell) \supseteq \ell^0, \ell^2, \ldots \ell^?$

To further simplify, notice that $f(l) \supset \vec{l}^2$ or l^μ or $l^\mu l^\nu \cdots$

there must be $\int \frac{d^4 l}{(2\pi)^4} \cdot \frac{l^\mu}{D^3} = 0$, and further $\int \frac{d^4 l}{(2\pi)^4} \frac{l^\mu l^\nu}{D^3} = \frac{g^{\mu\nu}}{4} \int \frac{d^4 l}{(2\pi)^4} \cdot \frac{\vec{l}^2}{D^3}$

(From parity symmetry)

$\bar{u}(p') f(l) u(p) = \bar{u}(p') \{ \underbrace{\not{k} \gamma^\mu \not{k}'}_{\mathrm{I}} + m^2 \gamma^\mu \underbrace{- 2m (k^\mu + k'^\mu)}_{\mathrm{II}} \} u(p)$. $k = l - xq + zp$

$k' = k + q = l - (x-1) q + zp$

$\mathrm{I} = \not{k} \gamma^\mu \not{k}' = (\not{l} - x\not{q} + z\not{p}) \gamma^\mu (\not{l} - (x-1)\not{q} + z\not{p})$

$= \underline{\not{l} \gamma^\mu \not{l}} + \underbrace{(-x\not{q} + z\not{p}) \gamma^\mu \not{l} + \not{l} \gamma^\mu (-(x-1)\not{q} + z\not{p})}_{\text{Disappear in the integral.}} + (-x\not{q} + z\not{p}) \gamma^\mu (-(x-1)\not{q} + z\not{p})$

$\downarrow$

$l^\nu \gamma_\nu \gamma^\mu \gamma^\rho l_\rho$

$\rightarrow \frac{1}{4} g^{\nu\rho} \gamma_\nu \gamma^\mu \gamma_\rho \underset{\vec{l}^2}{\vec{l}^2} = \frac{1}{4} \gamma_\nu \gamma^\mu \gamma^\nu \underset{\vec{l}^2}{\vec{l}^2} = -\frac{1}{2} \gamma^\mu \vec{l}^2$

$\therefore \mathrm{I} \rightarrow -\frac{1}{2} \gamma^\mu \vec{l}^2 + (-x\not{q} + z\not{p}) \gamma^\mu (-(x-1)\not{q} + z\not{p})$

$\mathrm{II} = -2m \{ (l^\mu - xq^\mu + zp^\mu) + (l^\mu - (x-1) q^\mu + zp^\mu) \}$

$= -2m \{ \underset{\rightarrow 0}{2l^\mu} - (2x-1) q^\mu + 2zp^\mu \} \rightarrow -2m ((1-2x) q^\mu + 2zp^\mu)$

thus we have $\bar{u}(p') f(l) u(p) \xrightarrow{\text{in} \int d^4 \vec{l}} \bar{u}(p') \{ -\frac{1}{2} \gamma^\mu \vec{l}^2 + \underline{\sum^\mu (q, p; x, z)} \} u(p)$

$\| $

$(-x\not{q} + z\not{p}) \gamma^\mu (-(x-1)\not{q} + z\not{p}) - 2m ((1-2x) q^\mu + 2zp^\mu)$

which means: $\bar{u}(p') \delta\Gamma^\mu (p', p) u(p) = ie^2 \int dx \int \frac{d^4 \vec{l}}{(2\pi)^4} \cdot \frac{-\frac{1}{2} \gamma^\mu \vec{l}^2 + \sum^\mu}{(\vec{l}^2 - \Delta + i\varepsilon)^3}$,

$$= \frac{-ie^2 \gamma^\mu}{2} \int_0^1 dx\, dz \underbrace{\boxed{\int \frac{d^4 \vec{l}}{(2\pi)^4} \frac{\vec{l}^2}{(\vec{l}^2 - \Delta + i\varepsilon)^3}}}_{\equiv I^{(1)}} + ie^2 \int_0^1 dx\, dz\, \sum^\mu \underbrace{\boxed{\int \frac{d^4 \vec{l}}{(2\pi)^4} \cdot \frac{1}{(\vec{l}^2 - \Delta + i\varepsilon)^3}}}_{I^{(0)}}$$

** How to integrate $I^{(n)}$? or more general. $\int \frac{d^4 \vec{l}}{(2\pi)^4} \cdot \frac{\vec{l}^{2n}}{(\vec{l}^2 - \Delta + i\varepsilon)^m} \equiv I^{(n,m)}$?

Let's take Wick Rotation: $l^0 = i l_E$, $l^i = l_E^i$, then $\begin{cases} \vec{l}^2 = l_0^2 - \vec{l}^2 = - l_E^0{}^2 - \vec{l}_E^2 = - \vec{l}_E^2 \\ d^4 l = i d^4 l_E . \end{cases}$

$\therefore I^{(n,m)} = i \int \frac{d^4 l_E}{(2\pi)^4} \cdot \frac{(-\vec{l}_E^2)^n}{(-\vec{l}_E^2 - \Delta + i\varepsilon)^m} = i (-1)^{n+m} \int \frac{d^4 \vec{l}_E}{(2\pi)^4} \cdot \frac{l_E^{2n}}{(l_E^2 + \Delta)^m}$

$= \frac{i (-1)^{n+m}}{(2\pi)^4} \int d\Omega \cdot \int_0^\infty dl_E \cdot \frac{l_E^{2n+3}}{(l_E^2 + \Delta)^m}$

According to the integral formula: $\int_0^\infty dx \cdot \frac{x^{2n-1}}{(x^2+\Delta)^m} = \frac{1}{2}\Delta^{n-m}\cdot\frac{\Gamma(n)\Gamma(m-n)}{\Gamma(m)}$

and the results of $\int d\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$, and for $d=4$, we have $\Omega_4 = 2\pi^2$, there would be:

$$I^{(n,m)} = \frac{i(-1)^{n+m}}{(2\pi)^4}\cdot 2\pi^2 \cdot \frac{1}{2}\Delta^{n-m+2}\cdot\frac{\Gamma(n+2)\Gamma(m-n-2)}{\Gamma(m)} = \frac{i(-1)^{n+m}}{(4\pi)^2}\cdot\Delta^{n-m+2}\cdot\frac{\Gamma(n+2)\Gamma(m-n-2)}{\Gamma(m)}$$

* Is it all correct? $\longrightarrow$ $\boxed{\text{No !!!}}$

Let's re-check integral $I^{(n,m)} \propto \int_0^\infty dl_E \cdot \frac{l_E^{2n+3}}{(l_E^2+\Delta)^m} \xrightarrow{l_E\to\infty} \int_0^\infty dl_E \cdot \frac{l_E^{2n+3}}{l_E^{2m}} = \int_0^\infty l_E^{2(n-m)+3}\, dl_E$

$$= \frac{1}{2(n-m)+3}\cdot l_E^{2(n-m)+4}\Big|_{l_E=0}^{l_E=\infty}$$

If $2(n-m)+4>0$, $\boxed{\text{which means } n>m-2}$, then $l_E^{2(n-m)+4}\Big|_{l_E=\infty} \to \infty$!

In our case (1) is UV-divergent! $\boxed{l_E=\infty} \longrightarrow$ UV divergence !

* How to deal with this divergence? Firstly, we need to take a proper way to describe the divergence.

Approach I (Pauli-Villars regularization).

Introduce a parameter Δ_Λ, then.

$$I^{(n,m)} \propto \int_0^\infty dl_E \cdot \frac{l_E^{2n+3}}{(l_E^2+\Delta)^m} = \lim_{\Delta_\Lambda\to\infty}\int_0^\infty dl_E \cdot \left\{\frac{l_E^{2n+3}}{(l_E^2+\Delta)^m} - \frac{l_E^{2n+3}}{(l_E^2+\Delta_\Lambda)^m}\right\}$$

In our case. $n=1$, $m=3$, then:

$$I^{(1,3)} \propto \lim_{\Delta_\Lambda\to\infty}\int_0^\infty dl_E \left\{\frac{l_E^5}{(l_E^2+\Delta)^3} - \frac{l_E^5}{(l_E^2+\Delta_\Lambda)^3}\right\}$$

$$= \lim_{\Delta_\Lambda\to\infty} \frac{1}{2}\int_0^\infty dl_E \left\{\frac{l_E^2}{(l_E+\Delta)^3} - \frac{l_E^2}{(l_E+\Delta_\Lambda)^3}\right\}$$

$$I^{(1,3)} = \frac{i}{(4\pi)^2}\ln\frac{\Delta_\Lambda}{\Delta}$$

Since $\int_0^\infty dl_E \cdot \frac{l_E^2}{(l_E+\Delta)^3} = \int_\Delta^\infty dt \frac{(t-\Delta)^2}{t^3} = \ln\frac{\infty}{\Delta} - \frac{4}{3}$

thus: $I^{(1,3)} \propto \lim_{\Delta_\Lambda\to\infty} \frac{1}{2}\left\{\ln\frac{\infty}{\Delta} - \ln\frac{\infty}{\Delta_\Lambda}\right\} = \frac{1}{2}\lim_{\Delta_\Lambda\to\infty} \ln\frac{\Delta_\Lambda}{\Delta}$

Approach II (Dimensional regulation)

Let's consider $I^{(n,m)}$ but for d-dim integrals. then:

$$I^{(n,m;d)} = i(-1)^{n+m}\int\frac{d^d l_E}{(2\pi)^4}\cdot\frac{l_E^{2n}}{(l_E^2+\Delta)^m} = i(-1)^{n+m}\int d\Omega_d\int_0^\infty dl_E \cdot \frac{l_E^{2n+d-1}}{(l_E^2+\Delta)^m}$$

Suppose $d=4-2\epsilon$. then:

$$I^{(n,m;4-2\epsilon)} = i(-1)^{n+m}\int d\Omega_{4-2\epsilon}\int_0^\infty dl_E \cdot \frac{l_E^{2n+(4-2\epsilon)-1}}{(l_E^2+\Delta)^m}.$$

Let's check the UV behavior. $\int_\Lambda^\infty dl_E \cdot \frac{l_E^{2n+(4-2\epsilon)-1}}{l_E^{2m}} = \frac{1}{2(n-m-\epsilon)+4}\cdot l_E^{2(n-m-\epsilon)+4}\Big|_{l_E=\infty}$

then it will diverge when $n\geq m+\epsilon-2$, or $\epsilon \leq n-m+2$.

In our case; $n=1$, $m=3$. thus:

$$I^{(1,3;4)} = \lim_{\varepsilon\to 0} I^{(1,3)(4-\varepsilon)} = \frac{i(-1)^{n+m}}{(2\pi)^4}\int d\Omega_{4-2\varepsilon}\int_0^\infty dl_E \cdot \frac{l_E^{2n+(4-2\varepsilon)-1}}{(-l_E^2+\Delta)^m} \qquad \doteq 2N-1 \quad N=n+2-\varepsilon$$

$$= \frac{i(-1)^{n+m}}{(2\pi)^4}\cdot\frac{2\pi^{2-\varepsilon}}{\Gamma(2-\varepsilon)}\cdot\frac{1}{2}\Delta^{n-m+2-\varepsilon}\cdot\frac{\Gamma(n+2-\varepsilon)\,\Gamma(m-n-2+\varepsilon)}{\Gamma(m)}\Bigg|_{n=1,\ m=3}$$

$(2-\varepsilon)(1-\varepsilon)\Gamma(1-\varepsilon)$

$$\therefore = \frac{i}{(2\pi)^4}\frac{2\pi^{2-\varepsilon}}{\Gamma(2-\varepsilon)}\cdot\frac{1}{2}\Delta^{-\varepsilon}\cdot\frac{\Gamma(3-\varepsilon)\,\Gamma(\varepsilon)}{\underbrace{\Gamma(3)}_{=2}} = \frac{i\pi^{-\varepsilon}}{(4\pi)^2}\Gamma(\varepsilon)$$

and we have:

$$\Gamma(\varepsilon) = \frac{1}{\varepsilon} - \gamma + O(\varepsilon)$$

γ ⇓ Euler constant.

$$\therefore I^{(1,3)} \approx \frac{i}{(4\pi)^2}\left(\frac{1}{\varepsilon}-\gamma\right)$$

** Now comes back to vertex function, we have:

$$\bar{u}(p')\,\delta\Gamma^\mu(p',p)\,u(p) = \frac{-ie^2\gamma^\mu}{2}\int_0^1 dxdz\cdot I^{(1,3)} + ie^2\int_0^1 dxdz\cdot\Sigma^\mu\cdot I^{(0)}$$

where $I^{(1,3)} = \frac{i}{(4\pi)^2}\ln\frac{\Delta_\Lambda}{\Delta}$ (Under Pauli-Villars regularization).

$$I^{(0)} = \int\frac{d^4\vec{l}}{(2\pi)^4}\cdot\frac{1}{(\vec{l}^2-\Delta)^3} = i\int\frac{d^4l_E}{(2\pi)^4}\cdot\frac{1}{(l_E^2-\Delta)^3} = \frac{i}{(2\pi)^4}\cdot 2\pi^2\cdot\int_0^\infty dl_E\cdot\frac{1}{(l_E^2-\Delta)^3}$$

$$= \frac{i}{(4\pi)^2}\int_0^\infty dl_E\cdot\frac{1}{(l_E-\Delta)^3} = \frac{i}{(4\pi)^2}\int_\Delta^\infty dx\cdot\frac{1}{x^3} = -\frac{i}{3(4\pi)^2}\cdot\left(-\frac{1}{\Delta^2}\right) = \frac{i}{(4\pi)^2}\cdot\frac{1}{3\Delta^2}$$

$$\therefore \text{(vertex)} = \frac{i}{(4\pi)^2}\cdot(-ie^2\gamma^\mu)\int_0^1 dxdz\ \ln\frac{\Delta_\Lambda}{\Delta} + ie^2\cdot\frac{i}{(4\pi)^2}\cdot\frac{1}{3\Delta^2}\cdot\int_0^1 dxdz\ \Sigma^\mu$$

$$= \frac{\alpha}{4\pi}\gamma^\mu\int_0^1 dxdz\cdot\ln\frac{\Delta_\Lambda}{\Delta} - \frac{\alpha}{4\pi}\cdot\frac{1}{3}\int_0^1\frac{1}{\Delta^2}\Sigma^\mu$$

Finally, when the smoke clears (?):

$$\text{(vertex)} = 2ie^2\int dxdzdy\ \bar{u}(p')\Big\{\gamma^\mu\underbrace{\left(-\frac{1}{2}\ln\frac{z\Lambda^2}{\Delta}+\left[(1-x)(1-y)\vec{q}^2+(1-4z+z^2)m^2\right]\cdot\frac{1}{2\Delta}\right)}_{\text{Contribute to } F_1(\vec{q}^2)}$$

$$+\frac{i\sigma^{\mu\nu}q_\nu}{2m}\cdot\underbrace{\left(2m^2 z(1-z)\frac{1}{2\Delta}\right)}_{\text{Contribute to } F_2(\vec{q}^2)}\Big\}u(p)$$

thus the UV-divergence emerge in:

$$F(\vec{q}^2) = 2ie^2\int dxdydz\cdot\delta(x+y+z-1)\cdot\left(-\frac{1}{2}\ln\frac{z\Lambda^2}{\Delta}+\left[(1-x)(1-y)\vec{q}^2+(1-4z+z^2)m^2\right]\cdot\frac{1}{2\Delta}\right)$$

and it will also emerge for $F(0)$, which should be 1.

⇓

Thus physical part of $\delta F_1(\vec{q}^2)$ is $\delta F_1(\vec{q}^2)-\delta F_1(0)$

⇓

No divergency any more!

§4. External leg correction.

$= 2N-1$

Now let's consider the so-called external-leg correction. If we denote:

$\underset{p}{\rightarrow}$ (1PI) $\underset{p}{\rightarrow} = -i\Sigma(\not p)$ then there will be: → It can be argued that Σ is only rely on $\not p$

—(Σ)— $=$ —(1PI)— $+$ —(1PI)$\xrightarrow{p}$(1PI)— $+$ —(1PI)$\xrightarrow{p}$(1PI)$\xrightarrow{p}$(1PI)— $+ \cdots$

To discuss it easier, let's first consider the Feynman Dygram of correlation Function, then:

$\underset{p}{\rightarrow}$(Σ)$\underset{p}{\rightarrow}$ $=$ $\underset{p}{\rightarrow}$ $+$ $\underset{p}{\rightarrow}$(1PI)$\underset{p}{\rightarrow}$ $+ \cdots$

$$= \frac{i}{\not p - m + i\varepsilon} + \frac{i}{\not p - m + i\varepsilon}\left(-i\Sigma(\not p)\right)\frac{i}{\not p - m + i\varepsilon} + \cdots$$

$$= \frac{i}{\not p - m - \Sigma(\not p) + i\varepsilon} \overset{!}{=} \frac{i\boxed{Z}}{\not p - \boxed{m_{phy}} + i\varepsilon}$$

We have discussed the effect of $\langle \Omega | \hat{T}\{\phi(x)\phi(y)\} | \Omega \rangle$ is the renormalization of field (Z) and mass (m)

① Expect the pole in physical mass:

$$\not p - m - \Sigma(\not p)\Big|_{\not p = m_{phy}} \overset{!}{=} 0 \Rightarrow m_{phy} = m + \Sigma(m_{phy})$$

② Near $m \simeq m_{phy}$. $\Sigma(\not p) = \Sigma(m_{phy}) + \left.\frac{d\Sigma(\not p)}{d\not p}\right|_{\not p = m_{phy}} (\not p - m_{phy})$

$$\therefore \frac{i}{\not p - m - \Sigma(\not p)} = \frac{1}{\underbrace{\not p - m - \Sigma(m_{phy})}_{m_{phy}} - \left.\frac{d\Sigma(\not p)}{d\not p}\right| (\not p - m_{phy})} = \frac{1}{(\not p - m_{phy})\left(1 - \left.\frac{d\Sigma(\not p)}{d\not p}\right|_{\not p = m_{phy}}\right)}$$

$$\therefore Z^{-1} = 1 - \left.\frac{d\Sigma(\not p)}{d\not p}\right|_{\not p = m_{phy}}$$

That's mean: $\underset{p}{\rightarrow}\overset{m}{(\Sigma)}\underset{p}{\rightarrow} = Z \cdot \underset{p}{\xrightarrow{m_{phy}}}$

$(2-(4-2\varepsilon)) = -2(1+\varepsilon)$

Thus our mission is to judge $\underset{p}{\rightarrow}$(Σ)— $\equiv \Sigma(\not p) = ?$

** Single loop calculation: Let's consider $\underset{p}{\rightarrow}\overset{p-k}{\frown}\underset{k}{\rightarrow}\underset{p}{\rightarrow} = -i\Sigma_2(\not p)$

$$-i\Sigma_2(\not p) = (-ie)^2 \int \frac{d^{4-2\varepsilon}\vec{k}}{(2\pi)^{4-2\varepsilon}} \gamma^\mu \frac{i(\not k + m)}{\vec{k}^2 - m^2 + i\varepsilon} \gamma_\mu \frac{-i}{(\vec{p}-\vec{k})^2 - \mu^2 + i\varepsilon}$$

→ this parameter (μ^2) is used to describe the infrared divergence

$$= -e^2 \int \frac{d^{4-2\varepsilon}\vec{k}}{(2\pi)^{4-2\varepsilon}} \frac{-2(1+\varepsilon)\not k + (4-2\varepsilon)m}{[\vec{k}^2 - m^2 + i\varepsilon]\left[(\vec{p}-\vec{k})^2 - \mu^2 + i\varepsilon\right]}$$

$$= -e^2 \int_0^1 dx dy\, \delta(\cdots) \int \frac{d^{4-2\varepsilon}\vec{k}}{(2\pi)^{4-2\varepsilon}} \frac{-2(1+\varepsilon)\not k + (4-2\varepsilon)m}{x(\vec{k}^2 - m^2) + y\left[(\vec{p}-\vec{k})^2 - \mu^2\right] + i\varepsilon}$$

Notice that: Denominator $= x(\vec{k}^2-m^2)+y(\vec{p}-\vec{k})^2-\mu^2 y$

$$= \vec{k}^2 - 2y\vec{p}\cdot\vec{k} - m^2 x + y\vec{p}^2 - \mu^2 y$$

$$= (\vec{k}-y\vec{p})^2 - y^2\vec{p}^2 - m^2 x - \mu^2 y + y\vec{p}^2$$

thus let $\vec{l} = \vec{k} - y\vec{p}$. $\Delta = y^2\vec{p}^2 + m^2 x + \mu^2 y - y\vec{p}^2$

$$= y(y-1)m^2 + m^2 x + \mu^2 y = x(1-y)m^2 + \mu^2 y = x^2 m^2 + y\mu^2$$

then: $$-i\Sigma_2(p) = -e^2\int_0^1 dxdy\,\delta(\cdots)\int\frac{d^4 l}{(2\pi)^4}\cdot\frac{-2(1+\varepsilon)(\not{l}+y\not{p})+(4-2\varepsilon)m}{(l^2-\Delta+i\varepsilon)^2}$$

$$= -ie^2\int_0^1 dxdy\,\delta(\cdots)\int\frac{d^4 l_E}{(2\pi)^4}\cdot\frac{-2y\not{p}+4m+\varepsilon(-2y\not{p}-2m)}{(l_E^2-\Delta)^2}$$

$$= -ie^2\int_0^1 dxdy\,\delta(\cdots)\cdot(-2y\not{p}+4m-2\varepsilon(y\not{p}+m))$$

and there will also be divergence since $2-0=2$!

$$I^{(0,2)} = \lim_{\varepsilon\to 0}\int\frac{d^{4-2\varepsilon}l_E}{(2\pi)^4}\cdot(l_E^2-\Delta)^{-2} \qquad 2(-\varepsilon)+3$$

$$= \lim_{\varepsilon\to 0}\frac{1}{(2\pi)^4}\underbrace{\int d\Omega_{4-2\varepsilon}}_{\frac{2\pi^{2-\varepsilon}}{\Gamma(2-\varepsilon)}}\cdot\underbrace{\int_0^\infty dl_E\,\frac{l_E^{3-2\varepsilon}}{(l_E^2-\Delta)^2}}_{\frac{1}{2}\Delta^{-\varepsilon}\cdot\frac{\Gamma(2-\varepsilon)\Gamma(\varepsilon)}{\Gamma(2)}}$$

$\to \pi^{-\varepsilon} = e^{-\varepsilon\ln\pi}$ $\qquad \to e^{-\varepsilon\ln\Delta} \simeq 1-\varepsilon\ln\Delta$

$$= \lim_{\varepsilon\to 0}\frac{1}{4\pi^2}(1-\varepsilon\ln\Delta)\left(\frac{1}{\varepsilon}+\gamma\right) = \lim_{\varepsilon\to 0}\frac{1}{4\pi^2}\left(\frac{1}{\varepsilon}+\gamma_E-\ln\Delta+\ln 4\pi\right)$$

thus we have: $$-i\Sigma_2(\not{p}) = -ie^2\int_0^1 dxdy\,\delta(\cdots)\left(-2y\not{p}+4m-2\varepsilon(y\not{p}+m)\right)$$

$$-i\Sigma(\not{p}) = \alpha\int_0^1 dxdy\,\delta(\cdots)\left(-2y\not{p}+4m-2\varepsilon(y\not{p}+m)\right)\cdot\left(\frac{1}{\varepsilon}+\gamma_E-\ln\Delta+\ln 4\pi\right)$$

$$= \alpha\int_0^1 dxdy\,\delta(\cdots)\left\{(-2y\not{p}+4m)\left(\frac{1}{\varepsilon}+\gamma_E+\ln 4\pi-\ln(x^2m^2+y\mu^2)\right)-2(y\not{p}+m)\right\}$$

$\hookrightarrow$ Divergent behaviour.

From which we can get:

$$\delta m = m_{phy} - m = \Sigma_2(\not{p}=m) = \alpha m\int_0^1 dxdy\,\delta(\cdots)\left\{(-2y+4)\left(\frac{1}{\varepsilon}+\gamma_E+\ln\frac{4\pi}{x^2m^2+\mu^2 y}\right)-2(y+1)\right\}$$

$\hookrightarrow$ UV-divergence

thus mass is UV-div too.

$$Z_2^{-1}-1 = \left.\frac{d\Sigma_2(\not{p})}{d\not{p}}\right|_{\not{p}=m}$$, and it can be proved that:

$$Z_2^{-1}-1 + \delta F_1(0) = \delta Z_2 + \delta F_1(0) = 0$$

Homework: Here are the optical theorem for a two body process: $\mathrm{Im}\,\mathcal{M}(k\to k) = 2E_{cm}|\vec{p}_1|_{cm}\,\sigma_{tot}(p_1 p_2 \to \text{anything})$

Check it in ϕ^4 theory to order λ^2

In ϕ^4 theory, we have $\sigma_{tot} = \frac{\lambda^2}{16\pi s}$, where $s = E_{cm}^2$. Now consider the imaginary part of $\mathcal{M}$ which only comes from s-channel:

$$i\mathcal{M} = \frac{1}{2}(-i\lambda)^2 \int \frac{d^d \vec{k}}{(2\pi)^d} \frac{i}{\vec{k}^2 - m^2} \cdot \frac{i}{(\vec{k}-\vec{p})^2 - m^2} = -\frac{\lambda^2}{2}\int \frac{d^d \vec{k}}{(2\pi)^d}\int_0^1 dx \cdot \frac{1}{x[\underbrace{(\vec{k}-\vec{p})^2}_{\vec{k}^2 - 2\vec{k}\cdot\vec{p}} - m^2] + y(\vec{k}^2 - m^2)}$$

$$= -\frac{\lambda^2}{2}\int \frac{d^{4-2\varepsilon}\vec{k}}{(2\pi)^{4-2\varepsilon}} \cdot \frac{1}{\vec{k}^2 - 2x\vec{k}\cdot\vec{p} - m^2(1-x)}$$

$$= -\frac{\lambda^2}{2}\int \frac{d^{4-2\varepsilon}\vec{\ell}}{(2\pi)^{4-2\varepsilon}} \cdot \frac{1}{\vec{\ell}^2 - x\vec{p}^2 - m^2(1-x)}$$

$$= \frac{-i\lambda^2}{2(4\pi)^2}\left(\frac{1}{\varepsilon} - \gamma_E + \ln 4\pi - \int_0^1 dx \cdot \ln\{m^2 - x(1-x)s\}\right)$$

thus: $\mathrm{Im}\,\mathcal{M} = -\frac{\lambda^2}{2(4\pi)^2}\int_0^1 dx \cdot \mathrm{Im}\{\ln(m^2 - x(1-x)s^2)\}$

Notice that when $m^2 - x(1-x)s^2 < 0$, which means $\frac{1-\sqrt{1-4m^2/s}}{2} < x < \frac{1+\sqrt{1-4m^2/s}}{2}$,

will gives out non-trivial $\mathrm{Im}\{\ln(\cdots)\} = -\pi$, thus: $\mathrm{Im}\,\mathcal{M} = \frac{\lambda^2}{32\pi}\sqrt{1-4m^2/s} = \frac{\lambda^2 \vec{p}_{cm}}{16\pi E_{cm}}$

$$= 2\vec{p}_1 E_{cm}\sigma_{tot}.$$

Homework: In fact, we have a result between vertex correction and external leg correction, that is $Z_1 = Z_2$, where $Z_1^{-1}\gamma^\mu = \Gamma^\mu(q=0) =$ , and $Z_2^{-1} = 1 - \frac{d\Sigma}{d\not{p}}\Big|_{\not{p}=m}$, where $\Sigma =$ 

This is the results of Ward identity, however it cannot be generally maintained in any regularization approaches:

(1) Recompute δZ_1 and δZ_2 with hard momentum cutoff $\int_0^\Lambda dx$, show that $\delta Z_1 \neq \delta Z_2$.

(2) Recompute δZ_1 and δZ_2 in dimensional-regularization, show that $\delta Z_1 = \delta Z_2$.

In previous section, we have got:

$$\bar{u}(p')\,\delta\Gamma^\mu(p',p)\,u(p) = ie^2\int \frac{d^4\vec{k}}{(2\pi)^4} \frac{\overbrace{\bar{u}(p')\gamma_\nu(\not{k}+\not{q}+m)\gamma^\mu(\not{k}+m)\gamma^\nu u(p)}^{N^\mu}}{[(\vec{k}-\vec{p})^2 - \mu^2][(\vec{k}+\vec{q})^2 - m^2](\vec{k}^2 - m^2)}$$

$$= ie^2\int \frac{d^4\vec{\ell}}{(2\pi)^4}\int dxyz \frac{\bar{u}(p')N^\mu u(p)}{(\vec{\ell}^2 - \Delta)^3} \qquad \begin{cases} \vec{\ell} = \vec{k} + y\vec{q} - x\vec{p} \\ \Delta = xy\vec{q}^2 - (1-z)^2 m^2 + x\mu^2 \end{cases}$$

Now turn the dimension to $4 \to 4-2\varepsilon$, there will be

$$N^\mu = -2\not{k}\gamma^\mu(\not{k}+\not{q}) + 4m(2k^\mu + q^\mu) - (2-2\varepsilon)m^2\gamma^\mu + (2+2\varepsilon)\not{k}\gamma^\mu\not{k}$$

$$= (2-d)\cancel{\not{l}}\gamma^\mu\cancel{\not{l}} - 2(x\not{p} - y\not{q})\gamma^\mu(\cancel{x\not{p}}+(1-y)\not{q}) + 4m\,(2x p^\mu + (1-2y)\,q^\mu) - 2m^2\gamma^\mu$$

and Finally there will be:

$$N^\mu = \left\{ \frac{(2-2\varepsilon)^2}{4-2\varepsilon}\vec{l}^{\,2} - 2(x^2-4x+1)m^2 - 2(x+y)(1-y)\vec{q}^{\,2} \right\} \gamma^\mu - 2x(1-x)\,m\,i\,\sigma^{\mu\nu}q_\nu .$$

~~then of~~ Since $\Gamma^\mu = \gamma^\mu F_1(\vec{q}^{\,2}) + \frac{i\sigma^{\mu\nu}q_\nu}{2m} F_2(\vec{q}^{\,2})$

$$\therefore \delta F_1(0) = -2ie^2 \int_0^1 dx \int_0^{1-x} dy \int \frac{d^{4-2\varepsilon}\vec{l}}{(2\pi)^{4-2\varepsilon}} \cdot \frac{\frac{4(1-\varepsilon)^2}{4-2\varepsilon}\vec{l}^{\,2} - 2(x^2-4x+1)m^2}{(\vec{l}^{\,2}-\Delta)^3}$$

$$= -2ie^2\int_0^1 dx \int_0^{1-x} dy \cdot \left\{ \frac{4(1-\varepsilon)^2}{4-2\varepsilon} \cdot \frac{i}{(2\pi)^{4-2\varepsilon}} \frac{2\pi^{2-\varepsilon}}{\Gamma(2-\varepsilon)} \Delta^{-\varepsilon} \cdot \frac{\Gamma(3-\varepsilon)\Gamma(\varepsilon)}{\Gamma(3)} \right.$$

$$\left. + 2(x^2-4x+1)m^2 \cdot \frac{i}{(2\pi)^{4-2\varepsilon}} \cdot \frac{2\pi^{2-\varepsilon}}{\Gamma(2-\varepsilon)} \cdot \frac{1}{2}\Delta^{-1-\varepsilon} \cdot \frac{\Gamma(2-\varepsilon)\Gamma(\varepsilon+1)}{\Gamma(3)} \right\}$$

$$= \frac{2e^2}{(4\pi)^2} \cdot \int_0^1 dx\,(1-x) \cdot \left\{ \frac{1}{\varepsilon} - \gamma_E + \ln 4\pi - \ln\{(1-x)^2m^2 + x\mu^2\} - 2 + \frac{(x^2-4x+1)m^2}{(1-x)^2m^2 + x\mu^2} \right\}.$$

$$\stackrel{!}{=} -\delta Z_1$$

For Z_2, we have. $-i\Sigma(\not{p}) = -\frac{ie^2}{(4\pi)^{d/2}}\int_0^1 dx \cdot \frac{\Gamma(2-\frac{d}{2})}{\Delta^{2-\frac{d}{2}}} \{(2-d)\,x\not{p} + dm\}$.

Then $$\left.\frac{d\Sigma(\not{p})}{d\not{p}}\right|_{\not{p}=m} = -\frac{e^2}{(4\pi)^2}\int_0^1 dx \left\{ \left(\frac{2}{\varepsilon} + \gamma_E + \ln 4\pi - \ln\{(1-x)^2m^2 + x\mu^2\} \right) - 1 - \frac{2(1-x)(2-x)m^2}{(1-x)^2m^2 + x\mu^2} \right\}.$$

and Now $\delta Z_1 = \delta Z_2$

If we use the hard momentum cutoff.

$$\delta F_1(0) = -2ie^2\int_0^1 dx \int_0^{1-x} dy \int^\Lambda \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2-\Delta)^3}\{k^2 - 2(x^2-4x+1)m^2\}$$

$$= \frac{e^2}{8\pi^2}\int_0^1 dx \cdot (1-x)\left\{ \ln\left(1 + \frac{\Lambda^2}{\Delta}\right) + \frac{(x^2-4x+1)m^2}{\Delta} - \frac{3}{2} \right\}.$$

and $$-i\Sigma(\not{p}) = 2e^2\int^\Lambda \frac{d^4k}{(2\pi)^4}\int_0^1 dx \cdot \frac{x\not{p} - 2m}{(k^2-\Delta)^2} \Rightarrow \delta Z_1 \neq \delta Z_2$$

§5. Vacuum polarization

* Finally, let's talk about the so-called vacuum polarization, which will bring the self-energy of photon

denote: $\mu \overset{q}{\sim} (\text{1PI}) \sim \nu = i\Pi^{\mu\nu}(q)$, and according to $q_\mu \Pi^{\mu\nu}(q) = 0$, there should be:

$$\Pi^{\mu\nu}(q) = \Delta^{\mu\nu}\,\Pi(q^2). \quad \text{where } \Delta^{\mu\nu} = q^2 g^{\mu\nu} - q^\mu q^\nu$$

then $\sim(Q)\sim \; = \; \sim + \sim(\text{1PI})\sim + \sim(\text{1PI})\sim(\text{1PI})\sim \cdots$

$$= \frac{-ig_{\mu\nu}}{q^2} + \frac{-ig_{\mu\rho}}{q^2}\Delta^{\rho\sigma}\Pi(q^2)\,\frac{-ig_{\sigma\nu}}{q^2} + \cdots$$

Notice that $\Delta^\rho_{\ \sigma}\Delta^\sigma_{\ \nu} = \Delta^\rho_{\ \nu}$

$$= \frac{-ig_{\mu\nu}}{q^2} + \frac{-ig_{\mu\rho}}{q^2}\Delta^\rho_{\ \nu}\Pi(q^2) + \frac{-ig_{\mu\rho}}{q^2}\boxed{\Delta^\rho_{\ \sigma}\Delta^\sigma_{\ \nu}}\,\Pi^2(q^2)$$

$\to \Delta = g^{\mu\nu}$

$$\boxed{\Delta^\mu_{\ \nu} = \delta^\mu_\nu - \frac{q^\mu q_\nu}{q^2}}$$

$$= -\frac{ig_{\mu\nu}}{q^2} + \frac{-ig_{\mu\rho}}{q^2}\left\{\delta^\rho_\nu - \frac{q^\rho q_\nu}{q^2}\right\}\left(\Pi(q^2) + \Pi^2(q^2) + \cdots\right)$$

$$= -i\frac{g_{\mu\rho}}{q^2}\left(\delta^\rho_\nu - \delta^\rho_\nu + \frac{q^\rho q_\nu}{q^2}\right) + \frac{-ig_{\mu\rho}}{q^2}\left(\delta^\rho_\nu - \frac{q^\rho q_\nu}{q^2}\right)\sum_{n=0}^{\infty}\Pi^n(q^2)$$

$$= \frac{-i\,q_\mu q_\nu}{q^4} + \frac{-ig_{\mu\rho}}{q^2(1-\Pi(q^2))}\cdot\left(\delta^\rho_\nu - \frac{q^\rho q_\nu}{q^2}\right)$$

In S-matrix, according to Ward identity: $\mu \overset{q}{\sim}(Q)\overset{q}{\sim}\nu \;\to\; \dfrac{-ig_{\mu\nu}}{q^2(1-\Pi(q^2))} \equiv \dfrac{-ig_{\mu\nu}}{q^2}\cdot Z_3$, $\quad Z_3 = \dfrac{1}{1-\Pi(0)}$

In $e^+e^- \to e^+e^-$: $\succ\!\sim(Q)\sim\!\prec \;\; \propto \;\; \dfrac{-ig_{\mu\nu}e^2 Z_3}{q^2} \equiv \dfrac{-ig_{\mu\nu}e_{phy}^2}{q^2} = \;\succ\!\otimes\sim\otimes\!\prec$

$-i\boxed{e_{phy}}\gamma^\mu$ → Charge renormalization $e_{phy} = \sqrt{Z_3}\,e$.

* Now let's compute $\Pi(q^2)$ in the first order.

$$i\Pi_2^{\mu\nu}(q) = \underset{q}{\rightarrow}\!\sim_\mu \bigcirc^{\,k+q}_{\,k} {}_\nu\!\sim\!\overset{q}{\rightarrow} = (-1)(-ie)^2 \int\frac{d^4k}{(2\pi)^4}\mathrm{Tr}\left\{\gamma^\mu\frac{\not k + m}{k^2 - m^2}\cdot\gamma^\nu\cdot\frac{\not k + \not q + m}{(k+q)^2 - m^2}\right\}$$

$$= e^2\int\frac{d^4k}{(2\pi)^4}\frac{\mathrm{Tr}\{\gamma^\mu i(\not k + m)\gamma^\nu i(\not k + \not q + m)\}}{(k^2 - m^2)\,[(k+q)^2 - m^2]}$$

$$= -e^2\int\frac{d^4k}{(2\pi)^4}\int_0^1 dx\,\frac{N^{\mu\nu}}{\{(1-x)(k^2 - m^2) + x[(k+q)^2 - m^2]\}^2}$$

$$= -e^2\int\frac{d^4k}{(2\pi)^4}\int_0^1 dx\cdot\frac{N^{\mu\nu}}{k^2 + \underset{2(1-x)}{\ }q\cdot k + \cancel{x}(q^2 - m^2) - m^2 x}$$

$$= -e^2 \int \frac{d^4\vec{k}}{(2\pi)^4} \int_0^1 dx \cdot \frac{N^{\mu\nu}}{[(\vec{k}+y\vec{q})^2 + x(1-x)\vec{q}^2 - m^2]^2} \qquad y = 1-x$$

$$= -e^2 \int \frac{d^4\vec{\ell}}{(2\pi)^4} \int_0^1 dx \cdot \frac{N^{\mu\nu}}{(\vec{\ell}+\Delta)^2} \qquad \vec{\ell} = \vec{k}+y\vec{q}, \quad \Delta = x(1-x)\vec{q}^2 - m^2.$$

$$N^{\mu\nu} = \mathrm{Tr}\{\gamma^\mu(\not k+m)\gamma^\nu(\not k+\not q+m)\} = \mathrm{Tr}\{\gamma^\mu(\not\ell - y\not q+m)\gamma^\nu(\not\ell+(1-y)\not q+m)\}$$

$$\cancel{= \mathrm{Tr}\{\gamma^\mu \not\ell \gamma^\nu \not\ell\} + \mathrm{Tr}\{\gamma^\mu m \gamma^\nu (\not q + m)\} \quad m^2 g^{\mu\nu}}$$

$$= \mathrm{Tr}\{\gamma^\mu \not\ell \gamma^\nu \not\ell\} + \mathrm{Tr}\{\gamma^\mu(-y\not q+m)\gamma^\nu((1-y)\not q+m)\}$$

Impose dimensional renormalization. trace will not be affected. thus.

$$\mathrm{Tr}\{\gamma^\mu \not\ell \gamma^\nu \not\ell\} = 4\{\ell^\mu\ell^\nu - g^{\mu\nu}\vec{\ell}^2 + \ell^\nu\ell^\mu\} = 4(2\ell^\mu\ell^\nu - g^{\mu\nu}\vec{\ell}^2)$$

$$\mathrm{Tr}\{\gamma^\mu(-y\not q+m)\gamma^\nu[(1-y)\not q+m]\} = -y(1-y)\{\gamma^\mu\not q\gamma^\nu\not q\} + m^2\,\mathrm{Tr}\{\gamma^\mu\gamma^\nu\}$$

$$= -4y(1-y)(2q^\mu q^\nu - g^{\mu\nu}\vec{q}^2) + 4m^2 g^{\mu\nu}$$

$$\therefore N_{\mu\nu} = 4\Big\{\Big(2\ell^\mu\ell^\nu - g^{\mu\nu}\vec{\ell}^2 \underbrace{- 2y(1-y)\,q^\mu q^\nu + g^{\mu\nu}(m^2 + y(1-y)\vec{q}^2)}_{K}\Big)\Big\}$$

$$\to 4\{(\tfrac{2}{d}-1)g^{\mu\nu}\vec{\ell}^2 + K\}$$

which means: $i\Pi_2^{\mu\nu}(\vec{q}^2) = -4e^2\int_0^1 dx \int \frac{d^d\vec{\ell}}{(2\pi)^d} \cdot \frac{(\frac{2}{d}-1)g^{\mu\nu}\vec{\ell}^2 + K}{(\vec{\ell}^2+\Delta)^2}$

$$= -4e^2\int_0^1 dx \cdot \{(\tfrac{2}{d}-1)g^{\mu\nu} I^{(1,2)} + K I^{(0,2)}\}$$

where $I^{(1,2)} = \int \frac{d^d\vec{\ell}}{(2\pi)^d} \frac{\vec{\ell}^2}{(\vec{\ell}^2+\Delta)^2}$, $\quad I^{(0,2)} = \int \frac{d^d\vec{\ell}}{(2\pi)^4} \frac{1}{(\vec{\ell}^2+\Delta)^2}$

(1) $I^{(0,2)} = i\int \frac{d^d\ell_E}{(2\pi)^d} \cdot \frac{1}{(\ell_E^2+\Delta)^2} = \frac{i}{(2\pi)^d} \frac{2\pi^{d/2}}{\Gamma(d/2)} \cdot \int_0^\infty d\ell_E \cdot \frac{\ell_E^{d-1}}{(\ell_E^2+\Delta)^2}$

$d-1 = 2n-1 \to n = \frac{d}{2}$

$$= \frac{i}{(4\pi)^{d/2}} \frac{1}{\Gamma(d/2)} \cancel{\tfrac{1}{2}} \Delta^{\frac{d}{2}-2} \frac{\Gamma(\frac{d}{2})\Gamma(2-\frac{d}{2})}{\Gamma(2)}$$

$$\int_0^\infty dx \cdot \frac{x^{2n-1}}{(x^2+\Delta)^m} = \frac{1}{2}\Delta^{n-m}\frac{\Gamma(n)\Gamma(m-n)}{\Gamma(m)}$$

$$\Downarrow$$

$$\int_0^\infty dx \cdot \frac{x^{d-1}}{(x^2+\Delta)^m} = \frac{1}{2}\Delta^{\frac{d}{2}-m}\frac{\Gamma(d/2)\Gamma(m-\frac{d}{2})}{\Gamma(m)}$$

$$= \frac{i}{(4\pi)^{d/2}} \frac{1}{\Gamma(d/2)} \Delta^{\frac{d}{2}-2} \cdot \frac{\Gamma(\frac{d}{2})\Gamma(2-\frac{d}{2})}{\Gamma(2)}$$

$$= \lim_{\varepsilon\to 0} \frac{i}{(4\pi)^{2-\varepsilon}} \Delta^{-\varepsilon} \frac{\Gamma(2-\frac{d}{2})}{\Gamma(2)}$$

$$= \lim_{\varepsilon\to 0} \frac{i}{(4\pi)^2} \underbrace{\left(\frac{\Delta}{4\pi}\right)^{-\varepsilon}}_{1-\varepsilon\ln(\frac{\Delta}{4\pi})} \underbrace{\Gamma(\varepsilon)}_{(\frac{1}{\varepsilon}-\gamma_E)} = \frac{i}{(4\pi)^2}\left(1-\varepsilon\ln\frac{\Delta}{4\pi}\right)\left(\frac{1}{\varepsilon}-\gamma_E\right)$$

$$\to \frac{i}{(4\pi)^2}\left(\frac{1}{\varepsilon} - \gamma_E - \ln\Delta + \ln 4\pi\right)$$